
UNIT 11 LIMIT AND CONTINUITY

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11.0 INTRODUCTION

In Unit 3 of this course, we have discussed, in detail, the concept of functions and various types of functions. In that unit we have also obtained the value of the function at certain points. That is, the value of a function $f(x)$ has been obtained at certain value of x in its domain.

Here, in this unit, we are going to introduce the concept of limit as well as continuity. That is, we are going to find the limiting value of the function $f(x)$ when x approaches to certain value. That is, we are interested in finding that value to which $f(x)$ approaches to as x approaches to the certain value. Also, this limiting value and the value of the function at certain value of x are compared to define continuity.

11.1 OBJECTIVES

After completing this unit, you should be able to:

- get an idea of limit;
- evaluate the limits of different functions;
- evaluate the infinite limit of some functions;
- check the continuity of a function at a point; and
- check the continuity of a function in its domain.

11.2 CONCEPT OF LIMIT

In Unit 3 of this course, we have discussed concept of function, consider a function

$$y = f(x) = 3x + 2$$

The following table shows the values of y for different values of x which are very close to 2.

x	1.9	1.98	1.998	1.9998	...	1.99999998	...
$y = f(x)$	7.7	7.94	7.994	7.9994	...	7.99999994	...

x	2.1	2.01	2.001	2.0001	...	2.00000001	...
$y = f(x)$	8.3	8.03	8.003	8.0003	...	8.00000003	...

We note that as x approaches to 2 either from left (means x comes nearer and nearer to 2 but remains < 2) or from right (means x comes nearer and nearer to 2 but remains > 2), then $y = f(x)$ approaches to 8 in the same manner.

i.e. as $x \rightarrow 2$ then $f(x) \rightarrow 8$ and we write it as $\lim_{x \rightarrow 2} f(x) = 8$.

In general if $f(x) \rightarrow l$ as $x \rightarrow a$ then we write it as $\lim_{x \rightarrow a} f(x) = l$.

In this unit, we discuss how to evaluate $\lim_{x \rightarrow a} f(x)$ in different situations. In this unit we shall also discuss the concept of infinite limit, some standard limits, left hand limit (L.H.L.) and right hand limit (R.H.L.). Finally we shall conclude the unit after introducing the concept of continuity.

11.3 DIRECT SUBSTITUTION METHOD

Suppose we want to evaluate $\lim_{x \rightarrow a} f(x)$. This method is applied when limiting value of $f(x)$ remains same irrespective of this fact whether x approaches to a from left hand side (L.H.S.) or right hand side (R.H.S.). As the name of this method itself suggests, in this method we directly substitute a in place of x .

Before we take some examples based on the direct substitution method we list some results (algebra of limits) without proof.

If f and g are real valued functions (real valued function means range of the function is subset of $\mathbb{R}$, set of real numbers) defined on the domain D such that

$\lim_{x \rightarrow a} f(x)$, $\lim_{x \rightarrow a} g(x)$ both exist, then the following results hold good.

- $\lim_{x \rightarrow a} (f(x) + g(x)) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$
- $\lim_{x \rightarrow a} (f(x) - g(x)) = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$
- $\lim_{x \rightarrow a} f(x)g(x) = \left(\lim_{x \rightarrow a} f(x) \right) \left(\lim_{x \rightarrow a} g(x) \right)$

4. $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)},$ provided $\lim_{x \rightarrow a} g(x) \neq 0$
5. $\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)}$
6. $\lim_{x \rightarrow a} \frac{1}{g(x)} = \frac{1}{\lim_{x \rightarrow a} g(x)}$ if $g(x) \neq 0$ in a neighbourhood of a and $\lim_{n \rightarrow 0} g(x) \neq 0$
7. $\lim_{x \rightarrow a} \log f(x) = \log \left(\lim_{x \rightarrow a} f(x) \right)$
8. $\lim_{x \rightarrow a} e^{f(x)} = e^{\lim_{x \rightarrow a} f(x)}$
9. $\lim_{x \rightarrow a} f(x)^{g(x)} = \left(\lim_{x \rightarrow a} f(x) \right)^{\lim_{x \rightarrow a} g(x)}$
10. $\lim_{x \rightarrow a} kf(x) = k \lim_{x \rightarrow a} f(x),$ where k is a constant
11. $\lim_{x \rightarrow a} k = k,$ where k is a constant

Remark 1:

- (i) These results are used so frequently; that we have no need to mention these results each time.
- (ii) Hereafter, we will use D.S.M. for Direct Substitution Method.

Now we are in position to discuss some examples based on (D.S.M.).

Example 1: Evaluate the following limits:

$$(i) \lim_{x \rightarrow 3} (x^2 - 2x + 4) \quad (ii) \lim_{x \rightarrow 2} x(x^2 - 4) \quad (iii) \lim_{x \rightarrow -1} (1 + x + x^2 + x^3 + \dots + x^{100})$$

$$(iv) \lim_{x \rightarrow \sqrt{2}} \frac{x^2 + 3}{3 - x^4} \quad (v) \lim_{x \rightarrow 3} \sqrt{25 - x^2}$$

$$(vi) \lim_{x \rightarrow 0} f(x), \text{ where } f(x) = \begin{cases} 2 - x^2, & x \neq 0 \\ 2, & x = 0 \end{cases}$$

Solution:

$$(i) \lim_{x \rightarrow 3} (x^2 - 2x + 4) = (3)^2 - 2 \times 3 + 4 \quad [\text{By D. S. M.}]$$

$$= 9 - 6 + 4 = 7$$

$$(ii) \lim_{x \rightarrow 2} x(x^2 - 4) = 2[(2)^2 - 4] = 2(4 - 4) = 0$$

$$\begin{aligned}
 \text{(iii)} \quad \lim_{x \rightarrow -1} (1 + x + x^2 + x^3 + \dots + x^{100}) &= 1 + (-1)^1 + (-1)^2 + (-1)^3 + \dots + (-1)^{100} \\
 &= 1 + (-1) + (1) + (-1) + (1) + \dots + (-1) + (1) \\
 &= 1 + (50 \text{ terms each containing } 1) \\
 &\quad + (50 \text{ terms each containing } (-1)) \\
 &= 1 + 50 + (-50) = 1
 \end{aligned}$$

$$\text{(iv)} \quad \lim_{x \rightarrow \sqrt{2}} \frac{x^3 + 3}{3 - x^4} = \frac{\lim_{x \rightarrow \sqrt{2}} (x^3 + 3)}{\lim_{x \rightarrow \sqrt{2}} (3 - x^4)} = \frac{(\sqrt{2})^3 + 3}{3 - (\sqrt{2})^4} = \frac{2\sqrt{2} + 3}{3 - 4} = -(2\sqrt{2} + 3)$$

$$\text{(v)} \quad \lim_{x \rightarrow 3} \sqrt{25 - x^2} = \sqrt{\lim_{x \rightarrow 3} (25 - x^2)} = \sqrt{25 - (3)^2} = \sqrt{25 - 9} = \sqrt{16} = 4$$

$$\text{(vi)} \quad f(x) = \begin{cases} 2 - x^2, & x \neq 0 \\ 2, & x = 0 \end{cases}$$

$$\begin{aligned}
 \lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} (2 - x^2) \quad [\because x \rightarrow 0 \Rightarrow x \neq 0, \text{ so } f(x) = 2 - x^2] \\
 &= 2 - (0)^2 = 2 - 0 = 2
 \end{aligned}$$

Remark 2: Limit of polynomial functions is always evaluated by D.S.M.

D.S.M. discussed above does not always work, in some situations it may fail. In next section we shall see when it fails and what are the alternate methods in such situations?

11.4 FAILURE OF DIRECT SUBSTITUTION METHOD

In mathematics following seven forms are known as indeterminate form, i.e. as such these forms are not defined.

$$\text{(i)} \frac{0}{0} \quad \text{(ii)} \frac{\infty}{\infty} \quad \text{(iii)} 0 \times \infty \quad \text{(iv)} \infty - \infty \quad \text{(v)} 0^0 \quad \text{(vi)} 1^\infty \quad \text{(vii)} \infty^0$$

So, if by direct substitution any of the above mentioned forms take place then D.S.M. fails and we need some alternate methods. Some of them are listed below:

I Factorisation Method

II Least Common Multiplier Method

III Rationalisation Method

IV Use of some Standard Results

Let us discuss these methods one by one:

11.4.1 Factorisation Method

This method is useful, when we get $\frac{0}{0}$ form by direct substitution in the given

expression of the type $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$. This will happen if $f(x)$ and $g(x)$ both becomes

zero on direct substitution.

$\Rightarrow$ both have at least one common factor $(x - a)$.

In this case express $f(x) = (x - a)$ (some factor) and $g(x) = (x - a)$ (some factor) either by long division method or by any other method known to you. Then cancel out the common factor and again try D.S.M. If D.S.M. works, we get the required limit.

If D.S.M. fails again, repeat the same procedure. Ultimately, after a finite number of steps, you will get the result as the numerator and denominator both are of finite degrees.

Let us explain the method with the help of the following example.

Example 2: Evaluate the following limits:

$$(i) \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

$$(ii) \lim_{x \rightarrow -1} \frac{x^3 + 1}{x + 1}$$

$$(iii) \lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - 5x + 4}$$

$$(iv) \lim_{x \rightarrow 3} \frac{x^3 - 3x^2 + x - 3}{x^3 - 10x^2 + 27x - 18}$$

Remember, we cannot cancel 0 by 0. But here $x - 2$ is not equal to zero because x is approaching to 2 and not equal to 2 and hence $x - 2$ is approaching to zero and not equal to zero.

Solution:

$$(i) \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} \quad \left[\frac{0}{0} \text{ form, so D.S.M. fails} \right]$$

Using factorisation method, we have

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)}{x - 2} \quad [\because a^2 - b^2 = (a - b)(a + b)]$$

$$= \lim_{x \rightarrow 2} (x + 2) \quad \left[\because x \rightarrow 2 \Rightarrow x - 2 \neq 0, \text{ so dividing numerator and denominator by } x - 2. \right]$$

$$= 2 + 2 = 4 \quad [\text{By D.S.M.}]$$

$$(ii) \lim_{x \rightarrow -1} \frac{x^3 + 1}{x + 1} \quad \left[\frac{0}{0} \text{ form, so D.S.M. fails} \right]$$

Using factorisation method, we have

$$\lim_{x \rightarrow -1} \frac{x^3 + 1}{x + 1} = \lim_{x \rightarrow -1} \frac{x^3 + 1^3}{x + 1}$$

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$$= \lim_{x \rightarrow -1} \frac{(x+1)(x^2 - x + 1)}{x+1} \quad [\because a^3 + b^3 = (a+b)(a^2 - ab + b^2)]$$

$$= \lim_{x \rightarrow -1} (x^2 - x + 1) \quad \left[\begin{array}{l} \because x \rightarrow -1 \Rightarrow x+1 \neq 0, \\ \text{so dividing numerator and} \\ \text{denominator by } x+1. \end{array} \right]$$

$$= (-1)^2 - (-1) + 1 = 1 + 1 + 1 = 3$$

$$(iii) \lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - 5x + 4} \quad \left[\frac{0}{0} \text{ form, so D.S.M. fails} \right]$$

Using factorisation method, we have

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - 5x + 4} &= \lim_{x \rightarrow 1} \frac{x^2 + 2x - x - 2}{x^2 - 4x - x + 4} \\ &= \lim_{x \rightarrow 1} \frac{x(x+2) - 1(x+2)}{x(x-4) - 1(x-4)} = \lim_{x \rightarrow 1} \frac{(x+2)(x-1)}{(x-4)(x-1)} \\ &= \lim_{x \rightarrow 1} \frac{x+2}{x-4} \quad \left[\begin{array}{l} \because x \rightarrow 1 \Rightarrow x-1 \neq 0, \\ \text{so dividing numerator and} \\ \text{denominator by } x-1. \end{array} \right] \\ &= \frac{1+2}{1-4} = \frac{3}{-3} = -1 \quad [\text{By D.S.M.}] \end{aligned}$$

$$(iv) \text{ Let } I = \lim_{x \rightarrow 3} \frac{x^3 - 3x^2 + x - 3}{x^3 - 10x^2 + 27x - 18} \quad \left[\frac{0}{0} \text{ form, so D.S.M. fails} \right]$$

Note: Numerator $\neq 0$ for $x = 3$

As on putting $x = 3$, the numerator and as well denominator both becomes zero, therefore $x - 3$ is a factor of $x^3 - 3x^2 + x$ as well as of

$x^3 - 10x^2 + 27x - 18$. Dividing $x^3 - 3x^2 + x$ by $x - 3$, we get $x^2 + 1$ as the quotient and 0 as the remainder and on dividing $x^3 - 10x^2 + 27x - 18$, we get $x^2 - 7x + 6$ as the quotient and 0 as the remainder.

$$\begin{aligned} \therefore I &= \lim_{x \rightarrow 3} \frac{(x^2 + 1)(x - 3)}{(x^2 - 7x + 6)(x - 3)} = \lim_{x \rightarrow 3} \frac{x^2 + 1}{x^2 - 7x + 6} \quad \left[\begin{array}{l} \text{Cancelling out the} \\ \text{factor } x - 3 \end{array} \right] \\ &= \frac{(3)^2 + 1}{(3)^2 - 7 \times 3 + 6} \quad [\text{By D.S.M.}] \\ &= \frac{9 + 1}{9 - 21 + 6} = \frac{10}{-6} = -\frac{5}{3} \end{aligned}$$

11.4.2 Least Common Multiplier Method

This method is useful in $\infty - \infty$ form.

Procedure: Take L.C.M. of the given expression and simplify it. Most of the times after simplification it reduces to $\frac{0}{0}$ form then solve it as explained in factorisation method.

Let us take an example based on this method.

Example 3: Evaluate $\lim_{x \rightarrow 3} \left(\frac{1}{x-3} - \frac{3}{x^2-3x} \right)$

Solution: $\lim_{x \rightarrow 3} \left(\frac{1}{x-3} - \frac{3}{x^2-3x} \right)$ [$\infty - \infty$ form, so D.S.M. fails]

Using LCM method, we have

$$\lim_{x \rightarrow 3} \left(\frac{1}{x-3} - \frac{3}{x^2-3x} \right) = \lim_{x \rightarrow 3} \left(\frac{1}{x-3} - \frac{3}{x(x-3)} \right) = \lim_{x \rightarrow 3} \left(\frac{x-3}{x(x-3)} \right) = \lim_{x \rightarrow 3} \frac{1}{x} = \frac{1}{3}$$

11.4.3 Rationalisation Method

This method is explained in the following example.

Example 4: Evaluate the following limits:

$$(i) \lim_{x \rightarrow 0} \frac{\sqrt{4+x}-2}{x} \quad (ii) \lim_{x \rightarrow 3} \frac{\sqrt{5x-6}-\sqrt{x+6}}{x^2-9}$$

Solution:

$$(i) \lim_{x \rightarrow 0} \frac{\sqrt{4+x}-2}{x} \quad \left[\frac{0}{0} \text{ form, so D.S.M. fails} \right]$$

Rationalising the numerator, we have

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt{4+x}-2}{x} &= \lim_{x \rightarrow 0} \frac{\sqrt{4+x}-2}{x} \times \frac{\sqrt{4+x}+2}{\sqrt{4+x}+2} \\ &= \lim_{x \rightarrow 0} \frac{(\sqrt{4+x})^2 - (2)^2}{x(\sqrt{4+x}+2)} \quad [\because a^2 - b^2 = (a-b)(a+b)] \\ &= \lim_{x \rightarrow 0} \frac{4+x-4}{x(\sqrt{4+x}+2)} = \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{4+x}+2)} \end{aligned}$$

$$= \lim_{x \rightarrow 0} \frac{1}{\sqrt{4+x}+2} = \frac{1}{\sqrt{4+0}+2} = \frac{1}{2+2} = \frac{1}{4}$$

$$(ii) \lim_{x \rightarrow 3} \frac{\sqrt{5x-6} - \sqrt{x+6}}{x^2-9} \quad \left[\frac{0}{0} \text{ form, so D.S.M. fails} \right]$$

Rationalising the numerator, we have

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{\sqrt{5x-6} - \sqrt{x+6}}{x^2-9} &= \lim_{x \rightarrow 3} \frac{\sqrt{5x-6} - \sqrt{x+6}}{x^2-9} \times \frac{\sqrt{5x-6} + \sqrt{x+6}}{\sqrt{5x-6} + \sqrt{x+6}} \\ &= \lim_{x \rightarrow 3} \frac{(\sqrt{5x-6})^2 - (\sqrt{x+6})^2}{(x^2-9)(\sqrt{5x-6} + \sqrt{x+6})} \\ &= \lim_{x \rightarrow 3} \frac{5x-6 - (x+6)}{(x^2-9)(\sqrt{5x-6} + \sqrt{x+6})} \\ &= \lim_{x \rightarrow 3} \frac{4x-12}{(x^2-9)(\sqrt{5x-6} + \sqrt{x+6})} \\ &= \lim_{x \rightarrow 3} \frac{4(x-3)}{(x-3)(x+3)(\sqrt{5x-6} + \sqrt{x+6})} \\ &= \lim_{x \rightarrow 3} \frac{4}{(x+3)(\sqrt{5x-6} + \sqrt{x+6})} \\ &= \frac{4}{(3+3)(\sqrt{15-6} + \sqrt{3+6})} \\ &= \frac{4}{6(\sqrt{9} + \sqrt{9})} = \frac{4}{6(3+3)} = \frac{4}{36} = \frac{1}{9} \end{aligned}$$

Check Your Progress - 1

1) Evaluate the following limits:

$$(i) \lim_{x \rightarrow 2} (x^2 - 2x + 3)^{x^2+1} \quad (ii) \lim_{x \rightarrow 1} \log (x^4 + x^2 + 1) \quad (iii) \lim_{x \rightarrow 5} 3$$

$$(iv) \lim_{x \rightarrow 3} 4f(x), \quad \text{where } f(x) = (x-5)^2$$

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2) Evaluate the following limits:

$$(i) \lim_{x \rightarrow 2} \frac{x^3 - 7x^2 + 16x - 12}{x^4 - 6x^3 - 3x^2 + 52x - 60} \quad (ii) \lim_{x \rightarrow 2} \frac{x^3 - 4x^2 + 5x - 2}{x^3 - 2x - 4}$$

3) Evaluate $\lim_{x \rightarrow -2} \left(\frac{1}{x+2} - \frac{6}{x^3 + x^2 - 2x} \right)$

4) Evaluate $\lim_{x \rightarrow 2} \frac{\sqrt{3+x} - \sqrt{5}}{x-2}$.

11.4.4 Use of some Standard Results

Here, we list without proof some very useful standard results which hold in limits.

1. $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$ [a and n are any real numbers, provided a^n, a^{n-1} exist]

2. $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = \lim_{\theta \rightarrow 0} \frac{\theta}{\sin \theta} = 1$

3. $\lim_{\theta \rightarrow 0} \cos \theta = 1$

4. $\lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = \lim_{\theta \rightarrow 0} \frac{\theta}{\tan \theta} = 1$

$$5. \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e a, \text{ where } a > 0, a \neq 1 \text{ in particular,}$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \log_e e = 1$$

$$6. \lim_{x \rightarrow 0} \frac{\log_e(1+x)}{x} = 1$$

$$7. \lim_{x \rightarrow 0} (1+x)^{1/x} = e$$

$$8. \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$$

Let us consider an example based on these standard results.

Example 5: Evaluate the following limits:

$$\begin{aligned} \text{(i)} \quad & \lim_{x \rightarrow 3} \frac{x^5 - 243}{x^4 - 81} \quad \text{(ii)} \quad \lim_{x \rightarrow 2} \frac{x^{10/3} - 2^{10/3}}{x^{4/3} - 2^{4/3}} \quad \text{(iii)} \quad \lim_{x \rightarrow 0} \frac{\sin 4x}{3x} \quad \text{(iv)} \quad \lim_{x \rightarrow 0} \cos 5x \\ \text{(v)} \quad & \lim_{x \rightarrow 0} \frac{\tan 3x}{\sin 2x} \quad \text{(vi)} \quad \lim_{x \rightarrow 0} \frac{2^{5x} - 1}{x} \quad \text{(vii)} \quad \lim_{x \rightarrow 0} \frac{e^{ax} - 1}{x} \quad \text{(viii)} \quad \lim_{x \rightarrow 0} \frac{\log(1+5x)}{x} \\ \text{(ix)} \quad & \lim_{x \rightarrow 0} \frac{\log(1+3x)}{e^{2x} - 1} \quad \text{(x)} \quad \lim_{x \rightarrow 0} (1+8x)^{1/x} \end{aligned}$$

Solution:

$$\text{(i)} \quad \text{Let } I = \lim_{x \rightarrow 3} \frac{x^5 - 243}{x^4 - 81} = \lim_{x \rightarrow 3} \frac{x^5 - 3^5}{x^4 - 3^4}$$

Dividing numerator and denominator by $x - 3$, we get

$$\begin{aligned} I &= \lim_{x \rightarrow 3} \frac{\frac{x^5 - 3^5}{x - 3}}{\frac{x^4 - 3^4}{x - 3}} = \frac{\lim_{x \rightarrow 3} \frac{x^5 - 3^5}{x - 3}}{\lim_{x \rightarrow 3} \frac{x^4 - 3^4}{x - 3}} \\ &= \frac{5(3)^{5-1}}{4(3)^{4-1}} = \frac{5}{4} \times \frac{3^4}{3^3} = \frac{5}{4} \times 3 = \frac{15}{4} \quad \left[\text{Using } \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right] \end{aligned}$$

$$\text{(ii)} \quad \lim_{x \rightarrow 2} \frac{x^{10/3} - 2^{10/3}}{x^{4/3} - 2^{4/3}}$$

Dividing numerator and denominator by $x - 2$, we get

$$\begin{aligned}\lim_{x \rightarrow 2} \frac{x^{10/3} - 2^{10/3}}{x^{4/3} - 2^{4/3}} &= \lim_{x \rightarrow 2} \frac{\frac{x^{10/3} - 2^{10/3}}{x - 2}}{\frac{x^{4/3} - 2^{4/3}}{x - 2}} = \frac{\lim_{x \rightarrow 2} \frac{x^{10/3} - 2^{10/3}}{x - 2}}{\lim_{x \rightarrow 2} \frac{x^{4/3} - 2^{4/3}}{x - 2}} \\ &= \frac{\frac{10}{3}(2)^{\frac{10}{3}-1}}{\frac{4}{3}(2)^{\frac{4}{3}-1}} = \frac{10}{3} \times \frac{3}{4} \times \frac{2^{\frac{7}{3}}}{2^{\frac{1}{3}}} = \frac{5}{2} \times 2^{\frac{7-1}{3}} = \frac{5}{2} (2^2) = 10\end{aligned}$$

$$(iii) \lim_{x \rightarrow 0} \frac{\sin 4x}{3x} = \lim_{x \rightarrow 0} \frac{\sin 4x}{4x} \times \frac{4x}{3x} \quad \left[\begin{array}{l} \text{Dividing and multiplying} \\ \text{by } 4x \end{array} \right]$$

$$= \lim_{x \rightarrow 0} \frac{\sin 4x}{4x} \times \frac{4}{3} = \frac{4}{3} \lim_{x \rightarrow 0} \frac{\sin 4x}{4x}$$

$$= \frac{4}{3} \lim_{4x \rightarrow 0} \frac{\sin 4x}{4x} \quad [As x \rightarrow 0 \Rightarrow 4x \rightarrow 0]$$

$$= \frac{4}{3} \times 1 = \frac{4}{3} \quad \left[\because \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1 \right]$$

$$(iv) \lim_{x \rightarrow 0} \cos 5x = \lim_{5x \rightarrow 0} \cos 5x = 1 \quad \left[\begin{array}{l} As x \rightarrow 0 \Rightarrow 5x \rightarrow 0 \text{ and } \\ \lim_{\theta \rightarrow 0} \cos \theta = 1 \end{array} \right]$$

$$(v) \lim_{x \rightarrow 0} \frac{\tan 3x}{\sin 2x} = \lim_{x \rightarrow 0} \frac{\tan 3x}{3x} \times \frac{2x}{\sin 2x} \times \frac{3}{2}$$

$$= \frac{3}{2} \left(\lim_{3x \rightarrow 0} \frac{\tan 3x}{3x} \right) \left(\lim_{2x \rightarrow 0} \frac{2x}{\sin 2x} \right) = \frac{3}{2} (1)(1) = \frac{3}{2}$$

$$\left[\because \lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = 1 \text{ and } \lim_{x \rightarrow 0} \frac{\theta}{\sin \theta} = 1 \right]$$

$$(vi) \lim_{x \rightarrow 0} \frac{2^{5x} - 1}{x} = \lim_{x \rightarrow 0} \frac{2^{5x} - 1}{5x} \times 5 = 5 \lim_{5x \rightarrow 0} \frac{2^{5x} - 1}{5x} \quad [\because x \rightarrow 0 \Rightarrow 5x \rightarrow 0]$$

$$= 5 \log_e 2 \quad \left[\because \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e |a| \right]$$

$$(vii) \lim_{x \rightarrow 0} \frac{e^{ax} - 1}{x} = \lim_{x \rightarrow 0} \frac{e^{ax} - 1}{ax} \times a = a \lim_{ax \rightarrow 0} \frac{e^{ax} - 1}{ax} \quad [\because x \rightarrow 0 \Rightarrow ax \rightarrow 0]$$

$$= a(1) \quad \left[\because \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 \right]$$

$$= a$$

$$(viii) \lim_{x \rightarrow 0} \frac{\log_e (1+5x)}{x} = \lim_{x \rightarrow 0} 5 \frac{\log_e (1+5x)}{5x}$$

$$= 5 \lim_{5x \rightarrow 0} \frac{\log_e (1+5x)}{5x} \quad [\because x \rightarrow 0 \Rightarrow 5x \rightarrow 0]$$

$$= 5(1)$$

$$\left[\because \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1 \right]$$

$$= 5$$

$$(ix) \lim_{x \rightarrow 0} \frac{\log e(1+3x)}{e^{2x}-1} = \lim_{x \rightarrow 0} \frac{\log e(1+3x)}{3x} \times \frac{2x}{e^{2x}-1} \times \frac{3}{2}$$

$$= \frac{3}{2} \left(\lim_{3x \rightarrow 0} \frac{\log(1+3x)}{3x} \right) \left(\lim_{2x \rightarrow 0} \frac{2x}{e^{2x}-1} \right)$$

$$= \frac{3}{2} (1)(1) = \frac{3}{2} \quad \text{as } \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1 \quad \text{and } \lim_{x \rightarrow 0} \frac{x}{e^x-1} = 1$$

$$(x) \lim_{x \rightarrow 0} (1+8x)^{1/x} = \lim_{x \rightarrow 0} \left[(1+8x)^{\frac{1}{8x}} \right]^8 = \left[\lim_{8x \rightarrow 0} (1+8x)^{\frac{1}{8x}} \right]^8 \quad \text{as } x \rightarrow 0 \Rightarrow 8x \rightarrow 0$$

$$= (e)^8 = e^8 \quad \left[\because \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e \right]$$

11.5 CONCEPT OF INFINITE LIMIT

Consider the following cases

$$\frac{1}{10} = 0.1$$

$$\frac{1}{100} = 0.01$$

$$\frac{1}{1000} = 0.001$$

$$\frac{1}{10000} = 0.0001$$

...

...

...

$$\frac{1}{10^n} = \underbrace{0.000\dots 1}_{n \text{ times}}$$

We see that as x (denominator) becomes larger and larger than $\frac{1}{x}$ becomes smaller and smaller and approaches to zero.

$$\therefore \text{ we write } \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

Or $\lim_{x \rightarrow \infty} \frac{1}{x^n} = 0$, where $n > 0$

Remark 3: $x \rightarrow \infty$ means that whatever large real number K (say) we take then $x > K$, i.e. no real number can be greater than x .

Let us consider an example, which involve infinite limit.

Example 6: Evaluate the following limits:

$$(i) \lim_{x \rightarrow \infty} \frac{3x^2 + 5x - 1}{4x^2 - 3x + 9} \quad (ii) \lim_{x \rightarrow \infty} \frac{5x^5 + x + 1}{x^3 + 5} \quad (iii) \lim_{x \rightarrow \infty} \frac{x^5 + 1}{4x^7 + 3x^2 + 7}$$

Solution:

$$(i) \lim_{x \rightarrow \infty} \frac{3x^2 + 5x - 1}{4x^2 - 3x + 9}$$

Here degree of numerator = Degree of denominator = 2

$\therefore$ dividing numerator and denominator by x^2 , we get

$$\lim_{x \rightarrow \infty} \frac{3x^2 + 5x - 1}{4x^2 - 3x + 9} = \lim_{x \rightarrow \infty} \frac{3 + \frac{5}{x} - \frac{1}{x^2}}{4 - \frac{3}{x} + \frac{9}{x^2}} = \frac{3 + 0 - 0}{4 - 0 + 0} = \frac{3}{4} \left[\because \lim_{x \rightarrow \infty} \frac{1}{x^n} = 0 \right. \\ \left. \text{for } n > 0 \right]$$

$$(ii) \lim_{x \rightarrow \infty} \frac{5x^5 + x + 1}{x^3 + 5}$$

Here degree of numerator > degree of denominator.

$\therefore$ dividing numerator and denominator by x^3 [Least of degrees], we get

$$\lim_{x \rightarrow \infty} \frac{5x^5 + x + 1}{x^3 + 5} = \lim_{x \rightarrow \infty} \frac{5x^2 + \frac{1}{x^2} + \frac{1}{x^3}}{1 + \frac{5}{x^3}} = \frac{\infty + 0 + 0}{1 + 0} = \infty$$

$$(iii) \lim_{x \rightarrow \infty} \frac{x^5 + 1}{4x^7 + 3x^2 + 7}$$

Here degree of numerator < degree of denominator.

$\therefore$ dividing numerator and denominator by x^5 [Least of degrees], we get

$$\lim_{x \rightarrow \infty} \frac{x^5 + 1}{4x^7 + 3x^2 + 7} = \lim_{x \rightarrow \infty} \frac{1 + \frac{1}{x^5}}{4x^2 + \frac{3}{x^3} + \frac{7}{x^5}} = \frac{1 + 0}{\infty + 0 + 0} = \frac{1}{\infty} = 0$$

In general, without calculating actual limit we can know the answer in advance of rational functions, in the cases when as $x \rightarrow \infty$ see the following result without proof.

$$\lim_{x \rightarrow \infty} \frac{a_0 x^m + a_1 x^{m-1} + a_2 x^{m-2} + \dots + a_{m-1} x + a_m}{b_0 x^n + b_1 x^{n-1} + b_2 x^{n-2} + \dots + b_{n-1} x + b_n} = \begin{cases} \frac{a_0}{b_0}, & \text{if } m = n \\ 0, & \text{if } m < n \\ \infty, & \text{if } m > n \end{cases}$$

11.6 CONCEPT OF LEFT HAND AND RIGHT HAND LIMITS

We note that on the real line, we can approach any real number 2(say) either from left or from right. Consider the example $y = f(x) = 3x + 2$. We see that as x takes the values 1.9, 1.98, 1.998, 1.9998, ... then corresponding values taken by y are 7.7, 7.94, 7.994, 7.9994, ... respectively as shown below.

x	1.9	1.98	1.998	1.9998	...	1.9999998	...
$y = f(x)$	7.7	7.94	7.994	7.9994	...	7.9999994	...

x	2.1	2.01	2.001	2.0001	...	2.0000001	...
$y = f(x)$	8.3	8.03	8.003	8.0003	...	8.0000003	...

i.e. as x is coming nearer and nearer to 2 from left then y is also coming nearer and nearer to 8 from left. If x approaches like this from left (see Fig. 11.1), then we say that x is approaching from left to 2 and is denoted by putting a -ve sign as a right superscript of 2 i.e. 2^-

i.e. we write the limit of the function as

$$\lim_{x \rightarrow 2^-} f(x) \quad \dots (1)$$

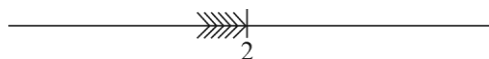


Fig. 11.1

If limit (1) exists, then we call it left hand limit (L.H.L.) of the function $f(x)$ as x tends to 2.

Similarly we see that as x takes the values 2.1, 2.01, 2.001, 2.0001, ... then corresponding values taken by y are 8.3, 8.03, 8.003, 8.0003, ... respectively. i.e. as x is coming nearer and nearer to 2 from right then y is also coming nearer and nearer to 8 from right. If x approaches like this from right (see Fig.

11.2) then we say that x is approaching from right to 2 and is denoted by putting +ve sign as a superscript of 2 i.e. 2^+ and the limit of the function as

$$\lim_{x \rightarrow 2^+} f(x) \quad \dots (2)$$



Fig. 11.2

If limit (2) exists, then we call it right hand limit (R.H.L.) of the function $f(x)$ as x tends to 2.

Remark 4:

- (i) L.H. and R.H. limits are used when functions have different values for $x \rightarrow 2^-$ and $x \rightarrow 2^+$.

For example, in case of

- (a) modulus functions,
 (b) functions having different values just below or above the value to which x is tending, i.e. there is break in function.
- (ii) Limit exists if L.H.L. and R.H.L. both exist and are equal.

Following example illustrates the idea of L.H.L. and R.H.L.

Example 7: Evaluate the following limits:

(i) $\lim_{x \rightarrow 0} |x|$ (ii) $\lim_{x \rightarrow 3} |x - 3|$ (iii) $\lim_{x \rightarrow 1} f(x)$, where $f(x) = \begin{cases} x^2 + 1, & x \leq 1 \\ 1 - x^2, & x > 1 \end{cases}$

(iv) $\lim_{x \rightarrow 4} f(x)$, where $f(x) = \begin{cases} \frac{|x - 4|}{x - 4}, & x \neq 4 \\ 0, & x = 4 \end{cases}$

Solution:

(i) $\lim_{x \rightarrow 0} |x|$

Here we have to use the concept of L.H.L. and R.H.L., because of the presence of the modulus function.

$$\text{L.H.L.} = \lim_{x \rightarrow 0^-} |x|$$

Here, as x is approaching to zero from its left and hence x is having little bit lesser value than 0.

Let us put $x = 0 - h$, where h is +ve real and is very small.

$$\text{As } x \rightarrow 0^- \Rightarrow h \rightarrow 0^+$$

$$\begin{aligned} \text{L.H.L.} &= \lim_{h \rightarrow 0^+} |0 - h| = \lim_{h \rightarrow 0^+} |-h| = \lim_{h \rightarrow 0^+} |-1| \times |h| = \lim_{h \rightarrow 0^+} |h| \text{ as } |-1| = -(-1) = 1 \\ &= \lim_{h \rightarrow 0^+} h \quad [\because h \rightarrow 0^+ \Rightarrow h > 0 \Rightarrow |h| = h] \\ &= 0 \end{aligned} \quad \dots (1)$$

$$\text{R.H.L.} = \lim_{x \rightarrow 0^+} |x|$$

Here, as x is approaching to zero from its right and hence x is having slightly greater value than 0.

Let us put $x = 0 + h$, where h is +ve real and is very small.

$$\text{As } x \rightarrow 0^+ \Rightarrow h \rightarrow 0^+$$

$$\therefore \text{R.H.L.} = \lim_{h \rightarrow 0^+} |0 + h| = \lim_{h \rightarrow 0^+} |h| = \lim_{h \rightarrow 0^+} h = 0 \quad \dots (2)$$

From (1) and (2)

$$\text{L.H.L.} = \text{R.H.L.}$$

$$\therefore \lim_{x \rightarrow 0} |x| \text{ exists and equal to 0.}$$

$$(ii) \lim_{x \rightarrow 3} |x - 3|$$

$$\text{L.H.L.} = \lim_{x \rightarrow 3^-} |x - 3|$$

Putting $x = 3 - h$, where h is +ve real and very small.

$$\text{As } x \rightarrow 3^- \Rightarrow h \rightarrow 0^+$$

$$\text{L.H.L.} = \lim_{h \rightarrow 0^+} |3 - h - 3| = \lim_{h \rightarrow 0^+} |-h| = \lim_{h \rightarrow 0^+} |h| = \lim_{h \rightarrow 0^+} h = 0 \quad \dots (1)$$

$$\text{R.H.L.} = \lim_{x \rightarrow 3^+} |x - 3|$$

Putting $x = 3 + h$ as $x \rightarrow 3^+ \Rightarrow h \rightarrow 0^+$

$$\text{R.H.L.} = \lim_{h \rightarrow 0^+} |3 + h - 3| = \lim_{h \rightarrow 0^+} |h| = \lim_{h \rightarrow 0^+} h = 0 \quad \dots (2)$$

From (1) and (2)

$$\text{L.H.L.} = \text{R.H.L.}$$

$$\therefore \lim_{x \rightarrow 3} |x - 3| \text{ exists and equal to 0.}$$

$$(iii) \lim_{x \rightarrow 1} f(x), \text{ where } f(x) = \begin{cases} x^2 + 1, & x \leq 1 \\ 1 - x^2, & x > 1 \end{cases}$$

$$\text{L.H.L.} = \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (x^2 + 1) \left[\begin{array}{l} \because x \rightarrow 1^- \text{ means } x \text{ is slightly less than } 1 \\ \text{and hence in this case } f(x) = x^2 + 1 \end{array} \right]$$

$$= (1)^2 + 1 = 1 + 1 = 2 \quad \dots (1)$$

$$\text{R.H.L.} = \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (1 - x^2) \left[\begin{array}{l} \because x \rightarrow 1^+ \text{ means } x \text{ is slightly greater than } 1 \\ \text{and hence in this case } f(x) = 1 - x^2 \end{array} \right]$$

$$= 1 - (1)^2 = 1 - 1 = 0 \quad \dots (2)$$

From (1) and (2)

L.H.L. $\neq$ R.H.L.

$\therefore \lim_{x \rightarrow 1} f(x)$ does not exist.

$$(iv) \lim_{x \rightarrow 4} f(x), \text{ where } f(x) = \begin{cases} \frac{|x-4|}{x-4}, & x \neq 4 \\ 0, & x = 4 \end{cases}$$

$$\text{L.H.L.} = \lim_{x \rightarrow 4^-} f(x) = \lim_{x \rightarrow 4^-} \frac{|x-4|}{x-4} \left[\begin{array}{l} \because x \rightarrow 4^- \Rightarrow x \text{ is slightly less than } 4 \\ \text{i.e } x \neq 4, \text{ so in this case } f(x) = \frac{|x-4|}{x-4} \end{array} \right]$$

Putting $x = 4 - h$, where h is +ve real and very small.

As $x \rightarrow 4^- \Rightarrow h \rightarrow 0^+$

$$\begin{aligned} \text{L.H.L.} &= \lim_{h \rightarrow 0^+} \frac{|4-h-4|}{4-h-4} = \lim_{h \rightarrow 0^+} \frac{|-h|}{-h} = \lim_{h \rightarrow 0^+} \frac{|-1| \times |h|}{-h} \\ &= \lim_{h \rightarrow 0^+} \frac{(1)(h)}{-h} = \lim_{h \rightarrow 0^+} (-1) = -1 \quad \dots (1) \end{aligned}$$

$$\text{R.H.L.} = \lim_{x \rightarrow 4^+} f(x) = \lim_{x \rightarrow 4^+} \frac{|x-4|}{x-4}$$

Putting $x = 4 + h$ as $x \rightarrow 4^+ \Rightarrow h \rightarrow 0^+$

$$\therefore \text{R.H.L.} = \lim_{h \rightarrow 0^+} \frac{|4+h-4|}{4+h-4} = \lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = \lim_{h \rightarrow 0^+} 1 = 1 \quad \dots (2)$$

From (1) and (2)

L.H.L. $\neq$ R.H.L.

$\therefore \lim_{x \rightarrow 4} f(x)$ does not exist.

Example 8: If $\lim_{x \rightarrow 0} f(x)$ exists, then find the value of k for

$$f(x) = \begin{cases} x - |x|, & x \leq 0 \\ k, & x > 0 \end{cases}$$

Solution: $f(x) = \begin{cases} x - |x|, & x \leq 0 \\ k, & x > 0 \end{cases}$

$$\text{L.H.L.} = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (x - |x|)$$

Putting $x = 0 - h$ as $x \rightarrow 0^- \Rightarrow h \rightarrow 0^+$

$$\begin{aligned} \text{L.H.L.} &= \lim_{h \rightarrow 0^+} (0 - h - |0 - h|) = \lim_{h \rightarrow 0^+} (-h - |-h|) \\ &= \lim_{h \rightarrow 0^+} (-h - |1| \times |h|) = \lim_{h \rightarrow 0^+} (-h - |h|) = \lim_{h \rightarrow 0^+} (-h - h) \\ &= \lim_{h \rightarrow 0^+} (-2h) = -2(0) = 0 \end{aligned} \quad \dots (1)$$

$$\text{R.H.L.} = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} k = k \quad \dots (2)$$

Since, it is given that $\lim_{x \rightarrow 0} f(x)$ exists.

$\therefore$ we must have

$$\text{L.H.L.} = \text{R.H.L.} \Rightarrow 0 = k \text{ or } k = 0$$

11.7 CONTINUITY OF A FUNCTION AT A POINT

In Sec. 11.6, we have discussed the concept of L.H.L. and R.H.L. Adding one more-step, we can define continuity at a point.

A function $f(x)$ is said to be continuous at $x = a$ if

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a), \text{ i.e. for continuity at a point } x = a, \text{ we must have}$$

$$\text{i.e. } \text{L.H.L.}_{\text{at } x=a} = \text{R.H.L.}_{\text{at } x=a} = \text{value of the function at } x = a$$

Diagrammatically, continuity at $x = a$ means graph of the function $f(x)$ from a value slightly less than 'a' to a value slightly greater than 'a' has no gap, i.e. if we draw the graph with pencil then we don't have to lift the pencil as we cross the point where $x = a$. Look at the Fig. 11.3 to 11.5.

In Fig. 11.3 $f(x)$ is not continuous at $x = a$.

In Fig. 11.4 $f(x)$ is not continuous at $x = a$.

In Fig. 11.5 $f(x)$ is continuous at $x = a$.

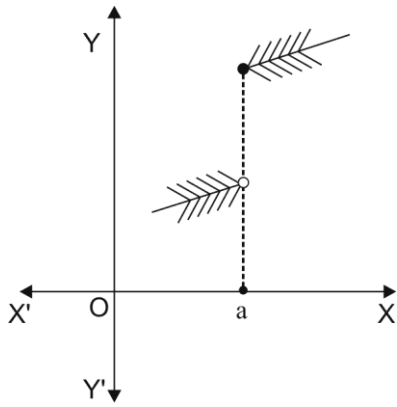


Fig. 11.3

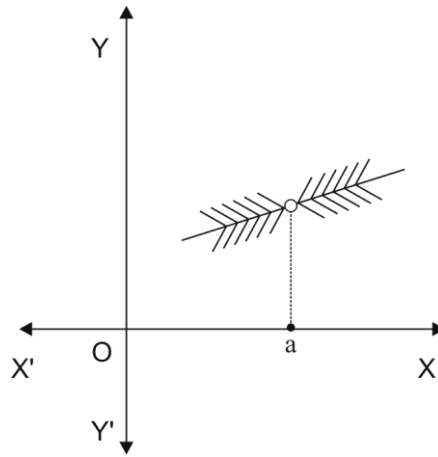


Fig. 11.4

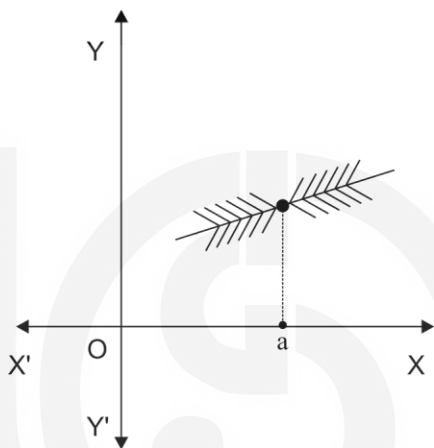


Fig. 11.5

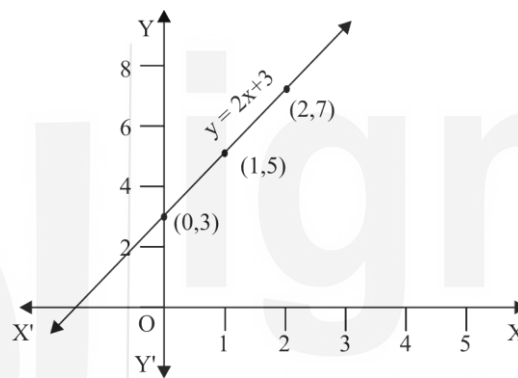


Fig. 11.6

Functions whose graphs are given in Fig. 11.6 and Fig. 11.7 are discussed below.

- (i) Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = 2x + 3$$

x	0	1	2
y	3	5	7

See the graph of this function in Fig. 11.6. We note that this function is continuous at all points of its domain as there is no gap at any point in its graph.

- (ii) Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1, & x \geq 1 \\ 2, & x < 1 \end{cases}$$

See the graph of this function in Fig. 11.7.

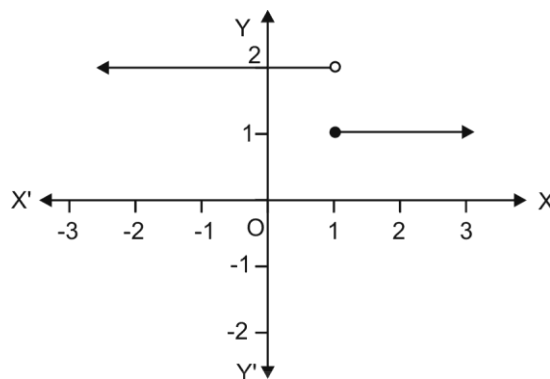


Fig.11.7

We note that, if we draw the graph of this function with pencil, then we will have to lift the pencil as we cross the point where $x = 1$. Therefore this function is not continuous at $x = 1$. Also this function is continuous at all points of its domain except at $x = 1$.

Now, let us consider some examples of continuity at a point.

Example 9: Discuss the continuity of the following functions at given point:

(i) $f(x) = |x|$ at $x = 0$

(ii) $f(x) = |x - 3|$ at $x = 3$

(iii) $f(x) = \begin{cases} x^2 + 1, & x \leq 1 \\ 1 - x^2, & x > 1 \end{cases}$ at $x = 1$

(iv) $f(x) = \begin{cases} \frac{|x - 4|}{x - 4}, & x \neq 4 \\ 0, & x = 4 \end{cases}$ at $x = 4$

(v) $f(x) = \frac{x}{|x|}$ at $x = 0$

Solution:

(i) $f(x) = |x|$ at $x = 0$

$$\text{L.H.L.} = 0, \quad \text{R.H.L.} = 0 \quad \left[\begin{array}{l} \text{Already calculated in} \\ \text{Example 7 of this unit} \end{array} \right]$$

$$\text{Also, at } x = 0, f(x) = |0| = 0$$

$$\therefore \text{L.H.L.}_{\text{at } x=0} = \text{R.H.L.}_{\text{at } x=0} = f(0)$$

$$\Rightarrow f(x) \text{ is continuous at } x = 0$$

$$(ii) f(x) = |x - 3|, \quad \text{at } x = 3$$

$$\text{L.H.L.} = 0, \quad \text{R.H.L.} = 0 \quad \left[\begin{array}{l} \text{Already calculated in} \\ \text{Example 7 of this unit} \end{array} \right]$$

$$\text{Also, } f(3) = |3 - 3| = |0| = 0$$

$$\therefore \text{L.H.L.}_{\text{at } x=3} = \text{R.H.L.}_{\text{at } x=3} = f(0)$$

$$\Rightarrow f(x) \text{ is continuous at } x = 3$$

$$(iii) f(x) = \begin{cases} x^2 + 1, & x \leq 1 \\ 1 - x^2, & x > 1 \end{cases} \quad \text{at } x = 1$$

$$\text{L.H.L.} = 2, \text{ but } \text{R.H.L.} = 0$$

$$\text{As } \text{L.H.L.}_{\text{at } x=1} \neq \text{R.H.L.}_{\text{at } x=1}$$

$$\therefore f \text{ is not continuous at } x = 1$$

$$(iv) f(x) = \begin{cases} \frac{|x-4|}{x-4}, & x \neq 4 \\ 0, & x = 4 \end{cases} \quad \text{at } x = 4$$

$$\therefore \text{L.H.L.}_{\text{at } x=4} = -1, \text{ but } \text{R.H.L.}_{\text{at } x=4} = 1 \quad \left[\begin{array}{l} \text{Already calculated in} \\ \text{Example 7 of this unit} \end{array} \right]$$

$$\text{As } \text{L.H.L.}_{\text{at } x=4} \neq \text{R.H.L.}_{\text{at } x=4}$$

$$\therefore f(x) \text{ is not continuous at } x = 4.$$

$$(v) f(x) = \frac{x}{|x|}, \quad \text{at } x = 0$$

$$\text{L.H.L.} = \lim_{x \rightarrow 0^-} \frac{x}{|x|}$$

$$\text{Putting } x = 0 - h \text{ as } x \rightarrow 0^- \Rightarrow h \rightarrow 0^+$$

$$\text{L.H.L.} = \lim_{h \rightarrow 0^+} \frac{0 - h}{|0 - h|} = \lim_{h \rightarrow 0^+} \frac{-h}{|-h|} = \lim_{h \rightarrow 0^+} \frac{-h}{|-1| \times |h|}$$

$$= \lim_{h \rightarrow 0^+} \frac{-h}{h} = \lim_{h \rightarrow 0^+} (-1) = -1 \quad \dots (1)$$

$$\text{R.H.L.} = \lim_{x \rightarrow 0^+} \frac{x}{|x|} = \lim_{x \rightarrow 0^+} \frac{x}{x} = \lim_{x \rightarrow 0^+} 1 = 1 \quad \dots (2)$$

From (1) and (2)

$$\text{L.H.L.} \neq \text{R.H.L.}$$

$$\therefore \lim_{x \rightarrow 0} \frac{x}{|x|} \text{ does not exist.}$$

Hence f is not continuous at $x = 0$.

Example 10: Find the values of a and b , if the function f given below is continuous at $x = 2$

$$f(x) = \begin{cases} 7, & x < 2 \\ ax + b, & x > 2 \\ a + 5, & x = 2 \end{cases}$$

$$\text{Solution: L.H.L.} = \lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} 7 = 7$$

$$\text{R.H.L.} = \lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (ax + b) = 2a + b$$

$$\text{Also } f(2) = a + 5$$

Since, f is given to be continuous at $x = 2$, therefore we must have

$$\text{L.H.L.}_{\text{at } x=2} = \text{R.H.L.}_{\text{at } x=2} = f(2)$$

$$\Rightarrow 7 = 2a + b = a + 5$$

$$\text{I} \quad \text{II} \quad \text{III}$$

$$\text{I \& III} \Rightarrow 7 = a + 5 \Rightarrow a = 2$$

$$\text{I \& II} \Rightarrow 7 - 2a = b \Rightarrow b = 7 - 4 = 3$$

$$\therefore a = 2, b = 3$$

Check Your Progress – 2

1) Evaluate the following limits:

$$(i) \lim_{x \rightarrow \sqrt{2}} \frac{x^{10} - 32}{x - \sqrt{2}} \quad (ii) \lim_{x \rightarrow 0} \frac{(ab)^{3x} - 1}{x} \quad (iii) \lim_{x \rightarrow 0} \frac{e^{\sin x} - 1}{\tan x}$$

$$(iv) \lim_{x \rightarrow 0} \frac{\log(1 + 8x^2)}{e^{x^2} - 1} \quad (v) \lim_{x \rightarrow 0} \frac{a^x - e^x}{x} \quad (vi) \lim_{x \rightarrow 0} \frac{e^x - (1 + 2x)}{2^x - 1}$$

$$(vii) \lim_{x \rightarrow 0} \frac{x(1 + 2x)^{1/x} - (e^{2x} - 1)}{x}$$

2) Evaluate $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 + 3} + \sqrt[4]{x^4 + 2}}{\sqrt[7]{x^7 + x^2} - \sqrt[3]{x^2 + 5}}.$

3) Evaluate the following limits:

(i) $\lim_{x \rightarrow 0^-} \frac{5x + |x|}{3|x| - 7x}$ (ii) $\lim_{x \rightarrow 5^+} (3 - |x|)$ (iii) $\lim_{x \rightarrow 0} \frac{x}{|x|}$

4) If $\lim_{x \rightarrow 3} f(x)$ exists then find a, for $f(x) = \begin{cases} ax + 3, & x \leq 3 \\ 2(x + 1), & x > 3 \end{cases}$

5) Find the relation between a and b if the function f is given to be

continuous at $x = 0$, where

$$f(x) = \begin{cases} 2x - a, & x \geq 0 \\ ax + b + 3, & x < 0 \end{cases}$$

11.8 CONTINUOUS FUNCTION

In section 11.7, we have discussed the continuity of a function at a point. In this section, we define what we mean by continuous function.

Continuous Function: A function f is said to be continuous if it is continuous at each point of its domain.

For example, function $y = f(x) = 2x + 3$ whose graph is given in Fig. 11.6 is a continuous function as we have already discussed that it is continuous at all points of its domain.

Algebra of continuous functions:

If f and g are two continuous functions on a common domain then

- (1) $f + g$ is continuous
- (2) $f - g$ is continuous
- (3) fg is continuous
- (4) f/g is continuous, provided $g(x) \neq 0 \forall$ points x of its domain.

11.9 SUMMARY

This unit introduces the foundational concept of a limit in calculus, which describes the behaviour of a function as it approaches a specific value. This unit details various evaluation techniques, starting with the direct substitution method. Further, this unit highlights the importance of direction through Left-Hand Limits (L.H.L.) and Right-Hand Limits (R.H.L.), noting that a limit exists only if both sides converge to the same value, and also explores infinite limits in which a function grows without bound. Finally, these concepts ended in the study of continuity, defining a function as continuous at a point if its limit equals the function value at that point, and extending this definition to describe a **continuous function** as one that remains unbroken across its entire domain.

11.10 SOLUTIONS/ANSWERS TO CHECK YOUR PROGRESS

Check Your Progress – 1

$$1) (i) \lim_{x \rightarrow 2} (x^2 - 2x + 3)^{x^2+1} = (2^2 - 2 \times 2 + 3)^{2^2+1} = 3^5 = 243$$

$$(ii) \lim_{x \rightarrow 1} \log(x^4 + x^2 + 1) = \log\left(\lim_{x \rightarrow 1} (x^4 + x^2 + 1)\right) \\ = \log(1^4 + 1^2 + 1) = \log 3$$

$$(iii) \lim_{x \rightarrow 5} 3 = 3 \quad \text{as } f(x) = 3 \text{ is a constant function}$$

$$(iv) \lim_{x \rightarrow 3} 4f(x) = 4 \lim_{x \rightarrow 3} f(x) = 4 \lim_{x \rightarrow 3} (x - 5)^2 \\ = 4(3 - 5)^2 = 4(-2)^2 = 4 \times 4 = 16$$

$$2) (i) \text{ Let } I = \lim_{x \rightarrow 2} \frac{x^3 - 7x^2 + 16x - 12}{x^4 - 6x^3 - 3x^2 + 52x - 60} \quad \left[\frac{0}{0} \text{ form, so D.S.M. fails} \right]$$

Using factorisation method, we have

$$I = \lim_{x \rightarrow 2} \frac{(x - 2)(x^2 - 5x + 6)}{(x - 2)(x^3 - 4x^2 - 11x + 30)}$$

Cancelling the common factor $x - 2 \neq 0$, we get

$$I = \lim_{x \rightarrow 2} \frac{x^2 - 5x + 6}{x^3 - 4x^2 - 11x + 30} \quad \left[\text{Again } \frac{0}{0} \text{ form} \right]$$

Again using factorisation method, we have

$$I = \lim_{x \rightarrow 2} \frac{(x - 2)(x - 3)}{(x - 2)(x^2 - 2x - 15)} = \lim_{x \rightarrow 2} \frac{x - 3}{x^2 - 2x - 15} \\ = \frac{2 - 3}{4 - 4 - 15} \quad [\text{By D.S.M.}] \\ = \frac{-1}{-15} = \frac{1}{15}$$

$$(ii) \lim_{x \rightarrow 2} \frac{x^3 - 4x^2 + 5x - 2}{x^3 - 2x - 4} \quad \left[\frac{0}{0} \text{ form, so D.S.M. fails} \right]$$

Using factorisation method, we have

Limit and Continuity

$$\begin{aligned}\lim_{x \rightarrow 2} \frac{x^3 - 4x^2 + 5x - 2}{x^3 - 2x - 4} &= \lim_{x \rightarrow 2} \frac{(x-2)(x^2 - 2x + 1)}{(x-2)(x^2 + 2x + 2)} \\ &= \lim_{x \rightarrow 2} \frac{x^2 - 2x + 1}{x^2 + 2x + 2} = \frac{4 - 4 + 1}{4 + 4 + 2} = \frac{1}{10}\end{aligned}$$

$$3) \lim_{x \rightarrow -2} \left(\frac{1}{x+2} - \frac{6}{x^3 + x^2 - 2x} \right) \quad [\infty - \infty \text{ form, so D.S.M. fails}]$$

Using LCM method, we have

$$\begin{aligned}\lim_{x \rightarrow -2} \left(\frac{1}{x+2} - \frac{6}{x^3 + x^2 - 2x} \right) &= \lim_{x \rightarrow -2} \left(\frac{1}{x+2} - \frac{6}{x(x^2 + x - 2)} \right) \\ &= \lim_{x \rightarrow -2} \left(\frac{1}{x+2} - \frac{6}{x(x+2)(x-1)} \right) \\ &= \lim_{x \rightarrow -2} \left(\frac{x(x-1) - 6}{x(x+2)(x-1)} \right) = \lim_{x \rightarrow -2} \frac{x^2 - x - 6}{x(x+2)(x-1)} \\ &= \lim_{x \rightarrow -2} \frac{(x-3)(x+2)}{x(x+2)(x-1)} \\ &= \lim_{x \rightarrow -2} \frac{x-3}{x(x-1)} \quad \left[\begin{array}{l} \because x \rightarrow -2 \Rightarrow x \neq -2 \\ \Rightarrow x+2 \neq 0 \text{ so, cancelling} \\ \text{out } x+2 \end{array} \right] \\ &= \frac{-2-3}{-2(-2-1)} = \frac{-5}{6} = -\frac{5}{6}\end{aligned}$$

$$4) \lim_{x \rightarrow 2} \frac{\sqrt{3+x} - \sqrt{5}}{x-2}$$

Rationalising the numerator, we have

$$\begin{aligned}\lim_{x \rightarrow 2} \frac{\sqrt{3+x} - \sqrt{5}}{x-2} &= \lim_{x \rightarrow 2} \frac{\sqrt{3+x} - \sqrt{5}}{x-2} \times \frac{\sqrt{3+x} + \sqrt{5}}{\sqrt{3+x} + \sqrt{5}} \\ &= \lim_{x \rightarrow 2} \frac{(\sqrt{3+x})^2 - (\sqrt{5})^2}{(x-2)(\sqrt{3+x} + \sqrt{5})} \quad [\because (a-b)(a+b) = a^2 - b^2] \\ &= \lim_{x \rightarrow 2} \frac{3+x-5}{(x-2)(\sqrt{3+x} + \sqrt{5})} = \lim_{x \rightarrow 2} \frac{x-2}{(x-2)(\sqrt{3+x} + \sqrt{5})} \\ &= \lim_{x \rightarrow 2} \frac{1}{\sqrt{3+x} + \sqrt{5}} \quad \left[\begin{array}{l} \text{Canceling out the common} \\ \text{factor } x-2 \neq 0 \end{array} \right]\end{aligned}$$

$$= \frac{1}{\sqrt{3+2} + \sqrt{5}} = \frac{1}{\sqrt{5} + \sqrt{5}} = \frac{1}{2\sqrt{5}}$$

$$= \frac{1}{2\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{\sqrt{5}}{10} \quad \left[\begin{array}{l} \text{Rationalising the} \\ \text{denominator} \end{array} \right]$$

Check Your Progress – 2

$$1) (i) \lim_{x \rightarrow \sqrt{2}} \frac{x^{10} - 32}{x - \sqrt{2}} = \lim_{x \rightarrow \sqrt{2}} \frac{x^{10} - (\sqrt{2})^{10}}{x - \sqrt{2}}$$

$$= 10 \left((\sqrt{2})^{10-1} \right) \quad \left[\because \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right]$$

$$= 10(\sqrt{2})^9 = 10(\sqrt{2})^8(\sqrt{2}) = 10 \times 16 \times \sqrt{2} = 160\sqrt{2}$$

$$(ii) \lim_{x \rightarrow 0} \frac{(ab)^{3x} - 1}{x} = \lim_{x \rightarrow 0} \frac{(ab)^{3x} - 1}{3x} \times 3$$

$$= 3 \lim_{3x \rightarrow 0} \frac{(ab)^{3x} - 1}{3x} \quad \text{as } x \rightarrow 0 \Rightarrow 3x \rightarrow 0$$

$$= 3 \log|ab| \quad \left[\because \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log|a| \right]$$

$$(iii) \lim_{x \rightarrow 0} \frac{e^{\sin x} - 1}{\tan x} = \lim_{x \rightarrow 0} \frac{e^{\sin x} - 1}{\sin x} \times \frac{\sin x}{x} \times \frac{x}{\tan x}$$

$$= \left(\lim_{\sin x \rightarrow 0} \frac{e^{\sin x} - 1}{\sin x} \right) \left(\lim_{x \rightarrow 0} \frac{\sin x}{x} \right) \left(\lim_{x \rightarrow 0} \frac{x}{\tan x} \right)$$

$$\text{as } x \rightarrow 0 \Rightarrow \sin x \rightarrow 0$$

$$= (1)(1)(1) = 1 \quad \left[\begin{array}{l} \because \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1, \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1 \\ \text{and } \lim_{\theta \rightarrow 0} \frac{\theta}{\tan \theta} = 1 \end{array} \right]$$

$$(iv) \lim_{x \rightarrow 0} \frac{\log(1+8x^2)}{e^{x^2} - 1} = \lim_{x \rightarrow 0} \frac{\log(1+8x^2)}{8x^2} \times \frac{x^2}{e^{x^2} - 1} \times 8$$

$$= 8 \left(\lim_{8x^2 \rightarrow 0} \frac{\log(1+8x^2)}{8x^2} \right) \left(\lim_{x^2 \rightarrow 0} \frac{x^2}{e^{x^2} - 1} \right)$$

$$\left[\because \text{as } x \rightarrow 0 \Rightarrow 8x^2 \rightarrow 0 \text{ and } x^2 \rightarrow 0 \right]$$

$$= 8(1)(1) = 8 \quad \left[\begin{array}{l} \because \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1 \text{ and } \\ \lim_{x \rightarrow 0} \frac{x}{e^x - 1} = 1 \end{array} \right]$$

$$\begin{aligned} \text{(v)} \quad \lim_{x \rightarrow 0} \frac{a^x - e^x}{x} &= \lim_{x \rightarrow 0} \frac{a^x - 1 - (e^x - 1)}{x} = \lim_{x \rightarrow 0} \left(\frac{a^x - 1}{x} - \frac{e^x - 1}{x} \right) \\ &= \left(\lim_{x \rightarrow 0} \frac{a^x - 1}{x} \right) - \left(\lim_{x \rightarrow 0} \frac{e^x - 1}{x} \right) \end{aligned}$$

$$= \log|a| - \log|e| \quad \left[\begin{array}{l} \because \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log|a| \text{ and } \\ \lim_{x \rightarrow 0} \frac{e^x - 1}{1} = \log|e| \end{array} \right]$$

$$= \log \left| \frac{a}{e} \right| \quad \text{as } \log \frac{m}{n} = \log m - \log n$$

$$\text{or } \lim_{x \rightarrow 0} \frac{a^x - e^x}{x} = \log|a| - 1 \quad \text{as } \log e = 1$$

$$\text{(vi)} \quad \text{Let } I = \lim_{x \rightarrow 0} \frac{e^x - (1 + 2x)}{2^x - 1} = \lim_{x \rightarrow 0} \frac{(e^x - 1) - 2x}{2^x - 1}$$

Dividing numerator and denominator by x , we get

$$I = \lim_{x \rightarrow 0} \frac{\frac{e^x - 1}{x} - 2}{\frac{2^x - 1}{x}} = \frac{\lim_{x \rightarrow 0} \frac{e^x - 1}{x} - 2}{\lim_{x \rightarrow 0} \frac{2^x - 1}{x}} = \frac{1 - 2}{\log 2} = \frac{-1}{\log 2}$$

$$\begin{aligned} \text{(vii)} \quad \lim_{x \rightarrow 0} \frac{x(1 + 2x)^{1/x} - (e^{2x} - 1)}{x} &= \lim_{x \rightarrow 0} \left((1 + 2x)^{1/x} - \frac{e^{2x} - 1}{x} \right) \\ &= \lim_{x \rightarrow 0} (1 + 2x)^{1/x} - \lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x} \\ &= \lim_{x \rightarrow 0} \left[(1 + 2x)^{\frac{1}{2x}} \right]^2 - \lim_{x \rightarrow 0} \frac{e^{2x} - 1}{2x} \times 2 \\ &= \left[\lim_{2x \rightarrow 0} (1 + 2x)^{\frac{1}{2x}} \right]^2 - 2 \lim_{2x \rightarrow 0} \frac{e^{2x} - 1}{2x} \\ &\quad \text{as } x \rightarrow 0 \Rightarrow 2x \rightarrow 0 \end{aligned}$$

$$= (e)^2 - 2 \log e \left[\begin{array}{l} \because \lim_{x \rightarrow 0} (1+x)^{1/x} = e \\ \text{and } \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \log e \end{array} \right]$$

$$= e^2 - 2 \quad \text{as } \log e = 1$$

$$2) \lim_{x \rightarrow \infty} \frac{\sqrt{x^2 + 3} + \sqrt[4]{x^4 + 2}}{\sqrt[7]{x^7 + x^2} - \sqrt[3]{x^2 + 5}} = \lim_{x \rightarrow \infty} \frac{x \left(\sqrt{1 + \frac{3}{x^2}} \right) + x \left(\sqrt[4]{1 + \frac{2}{x^4}} \right)}{x \left(\sqrt[7]{1 + \frac{1}{x^5}} \right) - x \left(\sqrt[3]{\frac{1}{x} + \frac{5}{x^3}} \right)}$$

$$= \lim_{x \rightarrow \infty} \frac{\sqrt{1 + \frac{3}{x^2}} + \sqrt[4]{1 + \frac{2}{x^4}}}{\sqrt[7]{1 + \frac{1}{x^5}} - \sqrt[3]{\frac{1}{x} + \frac{5}{x^3}}}$$

$$= \frac{\sqrt{1+0} + \sqrt[4]{1+0}}{\sqrt[7]{1+0} - \sqrt[3]{0+0}} = \frac{1+1}{1-0} = \frac{2}{1} = 2$$

Rule to be Remembered:

$$\sqrt[n]{f(x)} = [f(x)]^{1/n}$$

$$3) (i) \lim_{x \rightarrow 0^-} \frac{5x + |x|}{3|x| - 7x}$$

Putting $x = 0 - h$ as $x \rightarrow 0^- \Rightarrow h \rightarrow 0^+$

$$\lim_{x \rightarrow 0^-} \frac{5x + |x|}{3|x| - 7x} = \lim_{h \rightarrow 0^+} \frac{5(0-h) + |0-h|}{3|0-h| - 7(0-h)} = \lim_{h \rightarrow 0^+} \frac{-5h + |-h|}{3|-h| + 7h}$$

$$= \lim_{h \rightarrow 0^+} \frac{-5h + |-1||h|}{3|-1||h| + 7h} = \lim_{h \rightarrow 0^+} \frac{-5h + h}{3h + 7h} = \lim_{h \rightarrow 0^+} \frac{-4h}{10h}$$

$$= \lim_{h \rightarrow 0^+} \frac{-2}{5} = -\frac{2}{5}$$

$$(ii) \lim_{x \rightarrow 5^+} (3 - |x|)$$

Putting $x = 5 + h$ as $x \rightarrow 5^+ \Rightarrow h \rightarrow 0^+$

$$\therefore \lim_{x \rightarrow 5^+} (3 - |x|) = \lim_{h \rightarrow 0^+} (3 - |5 + h|)$$

$$= \lim_{h \rightarrow 0^+} (3 - (5 + h)) \left[\begin{array}{l} \because \text{as } h \rightarrow 0^+ \Rightarrow 5 + h > 5 > 0 \\ \Rightarrow |5 + h| = 5 + h \end{array} \right]$$

$$= \lim_{h \rightarrow 0^+} (-2 - h) = -2 - 0 = -2$$

$$(iii) \lim_{x \rightarrow 0} \frac{x}{|x|}$$

$$L.H.L. = \lim_{x \rightarrow 0^-} \frac{x}{|x|}$$

$$\text{Putting } x = 0 - h \text{ as } x \rightarrow 0^- \Rightarrow h \rightarrow 0^+$$

$$L.H.L. = \lim_{h \rightarrow 0^+} \frac{0 - h}{|0 - h|} = \lim_{h \rightarrow 0^+} \frac{-h}{|-h|} = \lim_{h \rightarrow 0^+} \frac{-h}{-1|h|}$$

$$= \lim_{h \rightarrow 0^+} \frac{-h}{h} = \lim_{h \rightarrow 0^+} (-1) = -1 \quad \dots (1)$$

$$R.H.L. = \lim_{x \rightarrow 0^+} \frac{x}{|x|} = \lim_{x \rightarrow 0^+} \frac{x}{x} = \lim_{x \rightarrow 0^+} 1 = 1 \quad \dots (2)$$

$$\left[\begin{array}{l} \because x \rightarrow 0^+ \Rightarrow x \text{ is slightly greater} \\ \text{than } 0, \text{ so } |x| = x \end{array} \right]$$

From (1) and (2)

$$L.H.L. \neq R.H.L.$$

$$\therefore \lim_{x \rightarrow 0} \frac{x}{|x|} \text{ does not exist.}$$

$$4) f(x) = \begin{cases} ax + 3, & x \leq 3 \\ 2(x + 1), & x > 3 \end{cases}$$

$$L.H.L. = \lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} (ax + 3) = 3a + 3 \quad \dots (1)$$

$$R.H.L. = \lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} 2(x + 1) = 2(3 + 1) = 8 \quad \dots (2)$$

Since, it is given that $\lim_{x \rightarrow 3} f(x)$ exists.

$\therefore$ we must have

$$L.H.L. = R.H.L.$$

$$\Rightarrow 3a + 3 = 8 \Rightarrow 3a = 5 \Rightarrow a = 5/3$$

$$5) f(x) = \begin{cases} 2x - a, & x \geq 0 \\ ax + b + 3, & x < 0 \end{cases} \quad \text{at } x = 0$$

$$L.H.L. = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (ax + b + 3)$$

$$\text{Putting } x = 0 - h \text{ as } x \rightarrow 0^- \Rightarrow h \rightarrow 0^+$$

$$\text{L.H.L.} = \lim_{h \rightarrow 0^+} (a(0-h) + b + 3) = \lim_{h \rightarrow 0^+} (-ah + b + 3) = b + 3$$

$$\text{R.H.L.} = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (2x - a) = 2(0) - a = -a$$

$$\text{Also } f(0) = 2(0) - a = 0 - a = -a$$

Since f is given to be continuous at $x = 0$, so we must have

$$\text{L.H.L.} = \text{R.H.L.} = f(0)$$

$$\Rightarrow b + 3 = -a \Rightarrow a + b + 3 = 0$$

Which is the required relation between a and b .

11.11 REFERENCES/FURTHER READINGS

1. Mathematics Class XI By RD Sharma , Dhanpat Rai Publication , 2011
2. Mathematics Class XII Part – 1 By Ravi Prakash , GRB Books, Second Edition 2025
3. BCS-012 : Basic Mathematics :
https://onlinecourses.swayam2.ac.in/nou26_ma07/preview (January 2026 session)

UNIT 12 DIFFERENTIATION

Structure

- 12.0 Introduction
- 12.1 Objectives
- 12.2 Definition of Derivative, its Meaning and Geometrical Interpretation
- 12.3 Derivative at a Point
- 12.4 Derivative by First Principle
- 12.5 Chain Rule
- 12.6 Derivatives of Exponential, Logarithmic, Parametric and Implicit Functions
- 12.7 Derivatives of Higher Orders
- 12.8 Concept of Maxima and Minima
- 12.9 Summary
- 12.10 Solutions/Answers to Check Your Progress
- 12.11 References/Further Readings

12.0 INTRODUCTION

In the preceding unit, we have discussed concept of limit and continuity. In fact, the definition of derivative involves these concepts. So, learner must go through the previous unit before starting this unit. Derivatives have large number of applications in the fields of mathematics, statistics, economics, insurance, industrial, health sector, etc.

In this unit, we will present this concept from a very simple and elementary point of view, keeping in mind that learner knows nothing about derivatives. In this unit, we have discussed some examples basically based on the formulae for derivatives of a constant, polynomial, exponential, logarithmic, parametric and implicit functions. Product rule, quotient rule, chain rule has also been discussed. Finally, we close this unit by giving a touch to higher order derivatives and maxima and minima of functions.

12.1 OBJECTIVES

After completing this unit, you should be able to:

- find derivative of a function at a particular point and at a general point;
- find derivative by first principle;
- find derivative of some commonly used functions;
- apply the chain rule;
- find derivative of exponential, logarithmic, parametric and implicit functions;
- find higher order derivatives; and
- find maxima and minima of a function.

12.2 DEFINITION OF DERIVATIVE, ITS MEANING AND GEOMETRICAL INTERPRETATION

Definition

Let $f : D \rightarrow \mathbb{R}$ be a function, where $D \subseteq \mathbb{R}$,

i.e. f is a real valued function defined on D .

Let $a \in D$ then derivative of f at $x = a$ is denoted by $f'(a)$ and is defined as

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}, \text{ provided limit exists} \quad \dots (1)$$

From definition (1), we see that $f'(a)$ measures the rate at which the function $f(x)$ changes at $x = a$. This is clear from the figure 12.1 given below.

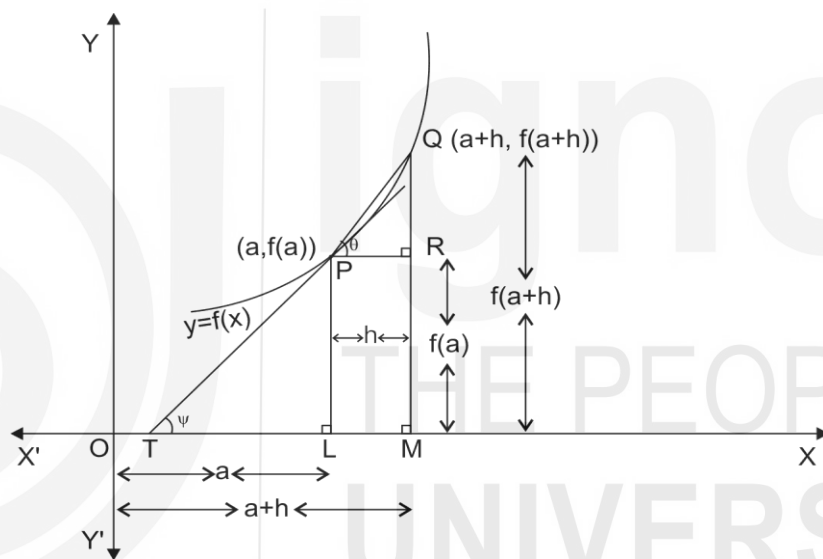


Fig. 12.1: Geometric Interpretation

Let PT be the tangent at point P of the curve of the function $y = f(x)$.

Draw $PL \perp OX$, $QM \perp OX$, $PR \perp QM$

Let $OL = a$, $OM = a + h$

$$\therefore PR = LM = OM - OL = a + h - a = h$$

$$\text{and } RQ = MQ - MR = MQ - LP = f(a+h) - f(a)$$

$$\therefore (1) \Rightarrow$$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{RQ}{PR}$$

$$= \lim_{h \rightarrow 0} \tan \theta \quad \left[\because \text{in } \Delta PQR, \tan \theta = \frac{\text{perpendicular}}{\text{Base}} = \frac{RQ}{PR} \right]$$

Now as $h \rightarrow 0$, chord PQ tends to coincide with the tangent at point P ,

i.e. as $h \rightarrow 0 \Rightarrow \theta \rightarrow \psi$

$$\therefore f'(a) = \lim_{h \rightarrow 0} \tan \theta = \tan \psi$$

$$\text{i.e. } f'(a) = \tan \psi$$

i.e. (derivative at point $x = a$) = (tangent of the angle which the tangent line at $x = a$ makes with +ve direction of x-axis)

In fact, if a line makes an angle θ with position direction of x-axis, then value of $\tan \theta$ is known as slope of the line.

Thus in mathematical language we can say

(Derivative at a point) $x=a$ is the slope of the tangent at that point. ... (2)

i.e. we can say that derivative measures the rate at which the tangent to the curve at point $x = a$ is changing

Meaning

Rewriting (1)

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \quad \dots (3)$$

From the knowledge of previous unit, we know that limit in R.H.S. of (1) or (3) exists if

$$\lim_{h \rightarrow 0^-} \frac{f(a+h) - f(a)}{h} \text{ and } \lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h} \text{ both exist and are equal.}$$

In view of (2), we have, limit in (1) exists if

$$\left(\begin{array}{l} \text{Slope of the tangent to the left} \\ \text{of the point } x = a \end{array} \right) = \left(\begin{array}{l} \text{Slope of the tangent to the right} \\ \text{of the point } x = a \end{array} \right)$$

i.e. limit in (1) exists if $x = a$ is not a corner point.

i.e. $f'(a)$ does not exist at corner points. ... (4)

For example, consider the function

$$f(x) = |x|$$

See the graph of this function in Fig. 12.2. We observe that $x = 0$ is a corner point in its graph.

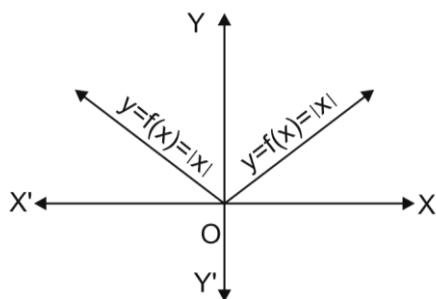


Fig . 12.2: Graph of Function $f(x) = |x|$

So $f(x) = |x|$ is not differentiable at $x = 0$. but there is no other corner point in its graph, so it is differentiable at all points of the domain except $x = 0$.

Remark1:

- (i) The last paragraph is very useful to understand the concept of derivative for those learners, who do not have mathematical background.
- (ii) However, the units have been written keeping in mind that the learner has no mathematical background after 10th standard.
- (iii) In the definition of derivative of a function at a point given by (1), we see that in order to find the derivative at the said point, we have to evaluate the limit in R.H.S. But sometimes functions may have different values for $h \rightarrow 0^-$ and $h \rightarrow 0^+$. In such cases like modulus function or where there is break for function in order to evaluate the limit we have to find out the L.H.L. and R.H.L.as we have done in the previous unit. But in the definition of derivative these L.H.L. and R.H.L. are known as left hand derivative (L.H.D.) and right hand derivative (R.H.D.) respectively.

A function is said to have derivative at a point if L.H.D. and R.H.D. both exist and are equal at that point, i.e.L.H.D. = R.H.D.

We denote L.H.D. of the function $f(x)$ at $x = a$ by $L(f'(a))$ and R.H.D. of the function $f(x)$ at $x = a$ by $R(f'(a))$. See Example 2 of this unit for more clarity.

12.3 DERIVATIVE AT A POINT

Here, we give some examples which will illustrate the idea as to how we calculate derivative of a function at a point.

Example 1: Find the derivative of the following functions at the indicated points:

- (i) $f(x) = a$, at $x = 5$, where a is a real constant
- (ii) $f(x) = ax + b$, at $x = 2$, $a \neq 0$
- (iii) $f(x) = ax^2 + bx + c$, at $x = 3$, $a \neq 0$

$$(iv) f(x) = \frac{1}{x}, \quad \text{at } x = 1$$

Solution:

$$(i) f(x) = a, \quad \text{where } a \text{ is real constant}$$

By definition

$$f'(5) = \lim_{h \rightarrow 0} \frac{f(5+h) - f(5)}{h} = \lim_{h \rightarrow 0} \frac{a - a}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = \lim_{h \rightarrow 0} 0 = 0$$

$$(ii) f(x) = ax + b, \quad a \neq 0$$

By definition

$$\begin{aligned} f'(2) &= \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{a(2+h) + b - (2a + b)}{h} = \lim_{h \rightarrow 0} \frac{ah}{h} = \lim_{h \rightarrow 0} a = a \end{aligned}$$

$$(iii) f(x) = ax^2 + bx + c, \quad a \neq 0$$

By definition

$$\begin{aligned} f'(3) &= \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} = \lim_{h \rightarrow 0} \frac{a(3+h)^2 + b(3+h) + c - (9a + 3b + c)}{h} \\ &= \lim_{h \rightarrow 0} \frac{ah^2 + 6ah + bh}{h} = \lim_{h \rightarrow 0} (ah + 6a + b) = 6a + b \end{aligned}$$

$$(iv) f(x) = \frac{1}{x}$$

By definition

$$\begin{aligned} f'(1) &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{\frac{1}{1+h} - \frac{1}{1}}{h} \\ &= \lim_{h \rightarrow 0} \frac{1 - (1+h)}{h(1+h)} = \lim_{h \rightarrow 0} \frac{-h}{h(1+h)} = -\lim_{h \rightarrow 0} \frac{1}{1+h} = -\frac{1}{1+0} = -1 \end{aligned}$$

Example 2: Find the derivative (if exists) of the following functions at the indicated points.

$$(i) f(x) = |x| \quad \text{at } x = 0$$

$$(ii) f(x) = \begin{cases} 5 + 2x, & x \geq 1 \\ 9 - 2x, & x < 1 \end{cases} \quad \text{at } x = 1$$

Solution:

(i) By definition

Differentiation

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{|0+h| - |0|}{h} = \lim_{h \rightarrow 0} \frac{|h| - 0}{h} = \lim_{h \rightarrow 0} \frac{|h|}{h}$$

We note that to deal with $|h|$ we must know the sign of h in advance. So

We must have to calculate L.H.D. and R.H.D. separately.

$$\text{L.H.D.} = \lim_{h \rightarrow 0^-} \frac{|h|}{h}$$

Putting $h = 0 - k$ as $h \rightarrow 0^- \Rightarrow k \rightarrow 0^+$

$$\begin{aligned} \text{L.H.D.} &= \lim_{k \rightarrow 0^+} \frac{|0-k|}{0-k} = \lim_{k \rightarrow 0^+} \frac{|-k|}{-k} = \lim_{k \rightarrow 0^+} \frac{|-1| \times |k|}{-k} = \lim_{k \rightarrow 0^+} \frac{k}{-k} \\ &= \lim_{k \rightarrow 0^+} (-1) = -1 \end{aligned} \quad \dots (1)$$

$$\text{R.H.D.} = \lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = \lim_{h \rightarrow 0^+} (1) = 1 \quad \dots (2)$$

From (1) and (2), we have

$$\text{L.H.D.} \neq \text{R.H.D.}$$

$\Rightarrow f'(0)$ does not exist.

(ii) By definition

$$f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{f(1+h) - (5 + 2 \times 1)}{h} = \lim_{h \rightarrow 0} \frac{f(1+h) - 7}{h}$$

We note that function has different values for $x < 1$ and $x > 1$, so we must have to calculate L.H.D. and R.H.D. separately.

$$\begin{aligned} L(f'(1)) &= \text{L.H.D.} = \lim_{h \rightarrow 0^-} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0^-} \frac{9 - 2(1+h) - 7}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{-2h}{h} = \lim_{h \rightarrow 0^-} (-2) = -2 \end{aligned} \quad \dots (1)$$

$$\begin{aligned} R(f'(1)) &= \text{R.H.D.} = \lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0^+} \frac{5 + 2(1+h) - 7}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{2h}{h} = \lim_{h \rightarrow 0^+} (2) = 2 \end{aligned} \quad \dots (2)$$

From (1) and (2)

$$L(f'(1)) \neq R(f'(1))$$

$\Rightarrow f'(1)$ does not exist.

Remark 2: In part (i) $x = 0$ is a corner point (see Fig. 12.2) that is why its derivative did not exist at this point, which was indicated in equation (4) in Sec. 12.2. Same is the case in part (ii).

12.4 DERIVATIVE BY FIRST PRINCIPLE

In section 12.3 of this unit, we have discussed as to how we calculate the derivative of a function at a given point $x = a$ (say). Suppose we want to calculate the derivative at 10 points, then using the definition 10 times is a very time consuming and lengthy procedure. To get rid of this difficulty, we will introduce a procedure in this section which will provide us the derivative of the function at a general point. After calculating the derivative at the general point we can replace this point by any number of points very quickly (provided derivative at the required point exists). Let us first describe the procedure as to how we calculate the derivative at a general point. After this we shall give some results to get a good understanding of the procedure. This process of finding derivative is known as derivative by **first principle** or by **definition** or by **delta** method or **ab-intio**.

Let us explain the procedure of first principle for the function

$$y = f(x) \quad \dots (1)$$

in the following steps.

Step I: Let δx be the small increment (+ve or -ve) in the value of x and δy be the corresponding increment in the value of y .

$\therefore$ (1) becomes

$$y + \delta y = f(x + \delta x) \quad \dots (2)$$

Step II: (2) – (1) gives

$$\delta y = f(x + \delta x) - f(x) \quad \dots (3)$$

Step III: First we simplify the expression in the R.H.S. of (3). After simplifying the expression, we divide both sides by δx and get

$$\frac{\delta y}{\delta x} = \frac{f(x + \delta x) - f(x)}{\delta x}$$

Step IV: Proceeding limit as $\delta x \rightarrow 0$ on both sides

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x} \quad \dots (4)$$

Step V: The term in L.H.S. of (4) is denoted by $\frac{dy}{dx}$ and limit in R.H.S. of (4)

is evaluated using suitable formula discussed in the previous unit

$$\text{i.e. } \frac{dy}{dx} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x} \quad \dots (5)$$

The expression obtained after simplification of the R.H.S. of (5), is derivative of y w.r.t. x at a general point x .

If we want the derivative of the function $y = f(x)$ at a particular point $x = a$ (say), then replace x by a in the result.

Some Results:

Result 1: Find the derivative of the constant function given by $f(x) = k$, where k is a real constant **by using first principle.**

Solution: Let $y = f(x) = k$... (1)

Step I: Let δx be the small increment in the value of x and δy be the corresponding increment in the value of y .

$\therefore$ (1) becomes

$$y + \delta y = k \quad \dots (2)$$

Step II: (2) – (1) gives

$$y + \delta y - y = k - k$$

$$\text{Or } \delta y = 0 \quad \dots (3)$$

Step III: Dividing on both sides of (3) by δx

$$\frac{\delta y}{\delta x} = \frac{0}{\delta x} = 0 \Rightarrow \frac{\delta y}{\delta x} = 0$$

Step IV: Proceeding limit as $\delta x \rightarrow 0$, we get

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} 0$$

$$\text{Step V: } \frac{dy}{dx} = 0$$

$$\text{i.e. } \frac{d}{dx}(k) = 0.$$

So, the derivative of a constant function is zero.

Result 2: Find the derivatives of the functions given by

$$(i) f(x) = x^2 \quad (ii) f(x) = x^3$$

by using first principle.

Solution:(i) Let $y = f(x) = x^2$... (1)

Step I: Let δx be the small increment in the value of x and δy be the corresponding increment in the value of y .

$\therefore$ (1) becomes

$$y + \delta y = (x + \delta x)^2 = x^2 + (\delta x)^2 + 2x \times \delta x \quad \dots (2)$$

Step II: (2) – (1) gives

$$\delta y = x^2 + (\delta x)^2 + 2x \times \delta x - x^2 = (\delta x)^2 + 2x \times \delta x$$

Step III: Dividing on both sides by δx , we get

$$\frac{\delta y}{\delta x} = \frac{(\delta x)^2 + 2x \times \delta x}{\delta x} = \delta x + 2x$$

Step IV: Proceeding limit as $\delta x \rightarrow 0$, we get

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} (\delta x + 2x)$$

Step V: $\frac{dy}{dx} = \lim_{\delta x \rightarrow 0} \delta x + \lim_{\delta x \rightarrow 0} 2x = 0 + 2x = 2x$

$$\text{i.e. } \frac{d}{dx}(x^2) = 2x = 2x^{2-1}$$

Second Method:

$$\text{Let } y = f(x) = x^2 \quad \dots (1)$$

Step I: Let δx be the small increment in the value of x and δy be the corresponding increment in the value of y .

$\therefore$ (1) becomes

$$y + \delta y = (x + \delta x)^2 \quad \dots (2)$$

Step II: (2) – (1) gives

$$\delta y = (x + \delta x)^2 - x^2$$

Step III: Dividing on both sides by δx , we get

$$\frac{\delta y}{\delta x} = \frac{(x + \delta x)^2 - x^2}{\delta x} = \frac{(x + \delta x)^2 - x^2}{(x + \delta x) - x}$$

Step IV: Proceeding limit as $\delta x \rightarrow 0$, we get

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{(x + \delta x)^2 - x^2}{(x + \delta x) - x}$$

Step V: $\frac{dy}{dx} = \lim_{x+\delta x \rightarrow x} \frac{(x + \delta x)^2 - x^2}{(x + \delta x) - x}$ as $\delta x \rightarrow 0 \Rightarrow x + \delta x \rightarrow x$

$$\text{i.e. } \frac{d}{dx}(x^2) = 2x^{2-1} = 2x \quad \left[\because \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right]$$

(iii) Let $y = f(x) = x^3 \quad \dots (1)$

Step I: Let δx be the small increment in the value of x and δy be the corresponding increment in the value of y .

$\therefore (1)$ becomes

$$y + \delta y = (x + \delta x)^3 = x^3 + 3x^2 \times \delta x + 3x \times (\delta x)^2 + (\delta x)^3 \quad \dots (2)$$

Step II: $(2) - (1)$ gives

$$\begin{aligned} \delta y &= x^3 + 3x^2 \times \delta x + 3x \times (\delta x)^2 + (\delta x)^3 - x^3 \\ &= 3x^2 \times \delta x + 3x \times (\delta x)^2 + (\delta x)^3 \end{aligned}$$

Step III: Dividing on both sides by δx , we get

$$\frac{\delta y}{\delta x} = \frac{3x^2 \times \delta x + 3x \times (\delta x)^2 + (\delta x)^3}{\delta x} = 3x^2 + 3x \times \delta x + (\delta x)^2$$

Step IV: Proceeding limit as $\delta x \rightarrow 0$, we get

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} (3x^2 + 3x \times \delta x + (\delta x)^2)$$

Step V: $\frac{dy}{dx} = \lim_{\delta x \rightarrow 0} 3x^2 + \lim_{\delta x \rightarrow 0} 3x \times \delta x + \lim_{\delta x \rightarrow 0} (\delta x)^2 = 3x^2 + 0 + 0 = 3x^2$

i.e. $\frac{d}{dx}(x^3) = 3x^2 = 3x^{3-1}$

Second Method:

Let $y = f(x) = x^3 \quad \dots (1)$

Step I: Let δx be the small increment in the value of x and δy be the corresponding increment in the value of y .

$\therefore (1)$ becomes

$$y + \delta y = (x + \delta x)^3 \quad \dots (2)$$

Step II: $(2) - (1)$ gives

$$\delta y = (x + \delta x)^3 - x^3$$

Step III: Dividing both the sides by δx , we get

$$\frac{\delta y}{\delta x} = \frac{(x + \delta x)^3 - x^3}{\delta x} = \frac{(x + \delta x)^3 - x^3}{(x + \delta x) - x}$$

Step IV: Proceeding limit as $\delta x \rightarrow 0$, we get

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{(x + \delta x)^3 - x^3}{(x + \delta x) - x}$$

Step V: $\frac{dy}{dx} = \lim_{x+\delta x \rightarrow x} \frac{(x + \delta x)^3 - x^3}{(x + \delta x) - x}$ as $\delta x \rightarrow 0 \Rightarrow x + \delta x \rightarrow x$

i.e. $\frac{d}{dx}(x^3) = 3x^{3-1}$ $\left[\because \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right]$

Similarly, $\frac{d}{dx}(x^n) = nx^{n-1}$

Result 3: Find the derivative of the function given by

$$f(x) = (ax + b)^2, \quad \text{where } a, b \text{ are real constants and } a \neq 0$$

by using first principle.

Solution: Let us use second method here.

Let $y = f(x) = (ax + b)^2$... (1)

Step I: Let δx be the small increment in the value of x and δy be the corresponding increment in the value of y .

$\therefore$ (1) becomes

$$y + \delta y = [a(x + \delta x) + b]^2 = (ax + a\delta x + b)^2 \dots (2)$$

Step II: (2) – (1) gives

$$\delta y = (ax + a\delta x + b)^2 - (ax + b)^2$$

Step III: Dividing both the sides by δx , we get

$$\begin{aligned} \frac{\delta y}{\delta x} &= \frac{(ax + a\delta x + b)^2 - (ax + b)^2}{\delta x} = a \left[\frac{(ax + a\delta x + b)^2 - (ax + b)^2}{a\delta x} \right] \\ &= a \left[\frac{(ax + a\delta x + b)^2 - (ax + b)^2}{(ax + a\delta x + b) - (ax + b)} \right] \end{aligned}$$

(Note: $(ax + b + a\delta x)^2 = (ax + b)^2 + 2a\delta x(ax + b) + a^2(\delta x)^2$ and use it.

Step IV: Proceeding limit as $\delta x \rightarrow 0$, we get

Differentiation

$$\begin{aligned}\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} &= a \lim_{\delta x \rightarrow 0} \left[\frac{(ax + a\delta x + b)^2 - (ax + b)^2}{(ax + a\delta x + b) - (ax + b)} \right] \\ &= a \lim_{a\delta x \rightarrow 0} \left[\frac{(ax + a\delta x + b)^2 - (ax + b)^2}{(ax + a\delta x + b) - (ax + b)} \right] \text{ as } \delta x \rightarrow 0 \Rightarrow a\delta x \rightarrow 0\end{aligned}$$

$$\begin{aligned}\text{Step V: } \frac{dy}{dx} &= a \lim_{ax+a\delta x+b \rightarrow ax+b} \left[\frac{(ax + a\delta x + b)^2 - (ax + b)^2}{(ax + a\delta x + b) - (ax + b)} \right] \\ &= a \times 2(ax + b)^{2-1} \quad \left[\because \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right]\end{aligned}$$

$$\text{i.e. } \frac{d}{dx}(ax + b)^2 = 2a(ax + b)^{2-1}$$

Similarly, we can easily obtain

$$\frac{d}{dx}(ax + b)^n = na(ax + b)^{n-1}$$

Result 4: Find the derivative of the exponential function $f(x) = e^{ax}$ by using first principle.

Solution: Let $y = f(x) = e^{ax}$... (1)

Step I: Let δx be the small increment in the value of x and δy be the corresponding increment in the value of y .

$\therefore$ (1) becomes

$$y + \delta y = e^{a(x+\delta x)} = e^{ax+a\delta x} \quad \dots (2)$$

Step II: (2) – (1) gives

$$\begin{aligned}\delta y &= e^{ax+a\delta x} - e^{ax} = e^{ax} e^{a\delta x} - e^{ax} \quad [\because a^{m+n} = a^m a^n] \\ &= e^{ax}(e^{a\delta x} - 1)\end{aligned}$$

Step III: Dividing both the sides by δx , we get

$$\frac{\delta y}{\delta x} = e^{ax} \frac{(e^{a\delta x} - 1)}{\delta x} = ae^{ax} \left(\frac{e^{a\delta x} - 1}{a\delta x} \right)$$

Step IV: Proceeding limit as $\delta x \rightarrow 0$, we get

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} ae^{ax} \left(\frac{e^{a\delta x} - 1}{a\delta x} \right)$$

Step V: $\frac{dy}{dx} = ae^{ax} \lim_{a\delta x \rightarrow 0} \frac{e^{a\delta x} - 1}{a\delta x} \text{ as } \delta x \rightarrow 0 \Rightarrow a\delta x \rightarrow 0$

$$= ae^{ax} (1) \quad \left[\because \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 \right]$$

$$\text{i.e. } \frac{d}{dx}(e^{ax}) = ae^{ax}$$

Result 5: Find the derivative of the logarithm function $f(x) = \log_a x$, where $a > 0, a \neq 1$.

by using first principle.

Solution: Let $y = f(x) = \log_a x$, where $x > 0$... (1)

Step I: Let δx be the small increment in the values of x

and δy be the corresponding increment in the value of y

$\therefore$ (1) becomes

$$y + \delta y = \log_a (x + \delta x) \quad \dots (2)$$

Step II: (2) – (1) gives

$$\begin{aligned} \delta y &= \log_a (x + \delta x) - \log_a x = \log_a \left(\frac{x + \delta x}{x} \right) \quad \left[\because \log m - \log n = \log \frac{m}{n} \right] \\ &= \log_a \left(1 + \frac{\delta x}{x} \right) \end{aligned}$$

Step III: Dividing both the sides by δx , we get

$$\begin{aligned} \frac{\delta y}{\delta x} &= \frac{\log_a \left(1 + \frac{\delta x}{x} \right)}{\delta x} = \frac{1}{x} \cdot \frac{x}{\delta x} \log_a \left(1 + \frac{\delta x}{x} \right) \\ &= \frac{1}{x} \log_a \left(1 + \frac{\delta x}{x} \right)^{\frac{x}{\delta x}} \quad [\because n \log m = \log m^n] \end{aligned}$$

Step IV: Proceeding limit as $\delta x \rightarrow 0$, we get

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{1}{x} \log_a \left(1 + \frac{\delta x}{x} \right)^{\frac{x}{\delta x}}$$

Step V: $\frac{dy}{dx} = \frac{1}{x} \lim_{\frac{\delta x}{x} \rightarrow 0} \log_a \left(1 + \frac{\delta x}{x} \right)^{\frac{x}{\delta x}} \text{ as } \delta x \rightarrow 0 \Rightarrow \frac{\delta x}{x} \rightarrow 0$

If a function is **continuous** then it **respects limit** i.e. if a function f is continuous and a is a point of its domain, then $\lim_{x \rightarrow a} f(x) = f(\lim_{x \rightarrow a} x) = f(a)$ i.e. limit can be taken inside the function.
i.e. Role of limit and function can be interchanged.

Differentiation

$$= \frac{1}{x} \log_a \left[\lim_{\frac{\delta x}{x} \rightarrow 0} \left(1 + \frac{\delta x}{x} \right)^{\frac{x}{\delta x}} \right] \left[\begin{array}{l} \because \log \text{ arithm is a continuous} \\ \text{function on its domain} \end{array} \right] = \frac{1}{x} \log_a e$$

$$\left[\because \lim_{x \rightarrow 0} (1+x)^{1/x} = e \right]$$

i.e.

$$\frac{d}{dx}(\log_a x) = \frac{1}{x} \log_a e$$

In particular, if base of the logarithmic is e in place of a, then

$\frac{d}{dx}(\log_e x) = \frac{1}{x} \log_e e = \frac{1}{x}$

as $\log_e e = 1$

Remark 3 : Keep all these formulae put in the rectangular boxes always in mind, as we will use these formulae later on as standard results.

Some more Formulae of Finding Derivatives:

If u and v are functions of x, then

(i) $\frac{d}{dx}(cu) = c \frac{du}{dx}$, where c is a real constant

(ii) $\frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$

(iii) $\frac{d}{dx}(u.v) = u \frac{dv}{dx} + v \frac{du}{dx}$ (Known as **Product Rule**)

(iv) $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$ (Known as **Quotient Rule**)

Remark 4: Aim of this unit from learners point of view is not to focus on the derivations of the formulae. But main aim of this unit is able to make the learners user friendly as to how these results can be used whenever we encounter a situation where derivative is involved. That is, why we will not provide the derivations of more formulae.

Various formulae which are used in many practical situations are listed below:

S. No	Function	Derivative of the function
1	k (constant)	$\frac{d}{dx}(k) = 0$

2	x^n	$\frac{d}{dx}(x^n) = nx^{n-1}$
3	$(ax + b)^n$	$\frac{d}{dx}(ax + b)^n = na(ax + b)^{n-1}$
4	Exponential function (i) a^{bx} (ii) e^{bx}	(i) $\frac{d}{dx}(a^{bx}) = ba^{bx} \log a$ (ii) $\frac{d}{dx}(e^{bx}) = be^{bx}$
5	Logarithmic function (i) $\log_a x$ (ii) $\log_e x$	(i) $\frac{d}{dx}(\log_a x) = \frac{1}{x} \log_a e$ (ii) $\frac{d}{dx}(\log_e x) = \frac{1}{x}$
6	cu, where c is constant and u is a function of x	$\frac{d}{dx}(cu) = c \frac{d}{dx}(u)$
7	(i) $u \pm v$ (ii) uv (iii) $\frac{u}{v}$ where, u, v are functions of x.	(i) $\frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$ (ii) $\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$ (Product Rule) (iii) $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$ (Quotient Rule)
8	$[f(x)]^n$, n is +ve or -ve realnumber	$\frac{d}{dx}[f(x)]^n = n[f(x)]^{n-1} \frac{d}{dx}(f(x))$
9	$\frac{1}{f(x)}$	$\frac{d}{dx}\left(\frac{1}{f(x)}\right) = \frac{d}{dx}(f(x))^{-1} = \frac{-1}{[f(x)]^2} \frac{d}{dx}(f(x))$
10	$y = f(u)$ $u = g(w)$ $w = h(x)$	$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dw} \frac{dw}{dx}$ (Chain Rule)
11	Parametric functions $x = f(t)$ $y = g(t)$	$\frac{dy}{dx} = \frac{dy}{dt} \bigg/ \frac{dx}{dt}$
12	Polynomial function $f(x) = a_0x^n + a_1x^{n-1} + \dots$ $+ a_{n-1}x + a_n$	$\frac{d}{dx}(f(x)) = na_0x^{n-1} + (n-1)a_1x^{n-2} + \dots + a_{n-1}$

Now we take some examples. We will write “Diff. w.r.t x” in place of “differentiating with respect to x”.

Example 3: Find derivative of the following functions:

(i) 5 (ii) $\alpha + \beta$ (iii) $\frac{1}{\sqrt{11}}$ (iv) $\frac{-\pi}{\sqrt{17}}$ (v) x^{11}

(vi) $x^{5/2}$ (vii) $\frac{1}{\sqrt[3]{x}}$ (viii) $\frac{1}{x^7}$ (ix) $\frac{1}{x}(x)\sqrt{x}$

(xi) $(2x+5)^3$ (xii) $(4-3x)^8$ (xiii) $\left(5-\frac{3}{2}x\right)^{4/9}$

(xiv) $\sqrt{2+3x}$ (xv) $x^5 + x^2 + 1$

(xvi) $\frac{x^3 + x^2 + 1}{x}$ (xvii) $\left(x^2 + \frac{1}{x^2}\right)^2$

(xviii) $\left(x + \frac{1}{x}\right)^{100}$ (xix) $(x^2+1)(x-1)$

(xx) $x^{-3}(1+x^2+x^5+x^8)$ (xxi) $(x^2+1)(x^3+x^2+1)$

(xxii) $(4x+1)^3(7x+1)^4$ (xxiii) $(x+2)^2(x+3)^4(x+1)^5$

Solution:

(i) Let $y = 5$

Diff. w.r.t.x

$$\frac{dy}{dx} = 0$$

$$\left[\because 5 \text{ is a constant and derivative of a constant function is zero. } \right]$$

(ii) Let $y = \alpha + \beta$

Diff. w.r.t.x

$$\frac{dy}{dx} = 0$$

$$\left[\because \alpha \text{ and } \beta \text{ both are constants } \Rightarrow \alpha + \beta \text{ is constant and derivative of a constant function is zero. } \right]$$

(iii) Let $y = \frac{1}{\sqrt{11}}$

Diff. w.r.t.x

$$\frac{dy}{dx} = 0$$

$$\left[\because \frac{1}{\sqrt{11}} \text{ is a constant and derivative of a constant function is zero. } \right]$$

(iv) Let $y = \frac{-\pi}{\sqrt{17}}$

Diff. w.r.t.x

$$\frac{dy}{dx} = 0$$

$$\left[\because \frac{-\pi}{\sqrt{17}} \text{ is a constant and derivative of a constant function is zero.} \right]$$

(v) Let $y = x^{11}$

Diff. w.r.t.x

$$\frac{dy}{dx} = 11x^{11-1} = 11x^{10}$$

$$\left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right]$$

(vi) Let $y = x^{5/2}$

Diff. w.r.t.x

$$\frac{dy}{dx} = \frac{5}{2}x^{\frac{5}{2}-1} = \frac{5}{2}x^{\frac{3}{2}}$$

$$\left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right]$$

(vii) Let $y = \frac{1}{\sqrt[3]{x}} = \frac{1}{x^{1/3}} = x^{-1/3}$

Diff. w.r.t.x

$$\frac{dy}{dx} = -\frac{1}{3}x^{-4/3}$$

$$\left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right]$$

(viii) Let $y = \frac{1}{x^7} = x^{-7}$

Diff. w.r.t.x

$$\frac{dy}{dx} = -7x^{-8} = -\frac{7}{x^8}$$

$$\left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right]$$

(ix) Let $y = \frac{1}{x} = x^{-1}$

Diff. w.r.t.x

$$\frac{dy}{dx} = (-1)x^{-2} = -\frac{1}{x^2}$$

$$\left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right]$$

(x) Let $y = \sqrt{x} = x^{1/2}$

Diff. w.r.t.x

$$\frac{dy}{dx} = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$$

$$\left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right]$$

(xi) Let $y = (2x+5)^3$

Diff. w.r.t.x

Differentiation

$$\frac{dy}{dx} = 3(2x+5)^2(2) = 6(2x+5)^2 \quad \left[\because \frac{d}{dx}((ax+b)^n) = n(ax+b)^{n-1}a \right]$$

(xii) Let $y = (4-3x)^8$

Diff. w.r.t.x

$$\frac{dy}{dx} = 8(4-3x)^7(-3) = -24(4-3x)^7 \quad \left[\because \frac{d}{dx}((ax+b)^n) = n(ax+b)^{n-1}a \right]$$

(xiii) Let $y = \left(5 - \frac{3}{2}x\right)^{4/9}$

Diff. w.r.t.x

$$\begin{aligned} \frac{dy}{dx} &= \frac{4}{9} \left(5 - \frac{3}{2}x\right)^{-5/9} \left(-\frac{3}{2}\right) \quad \left[\because \frac{d}{dx}((ax+b)^n) = n(ax+b)^{n-1}a \right] \\ &= -\frac{2}{3} \left(5 - \frac{3}{2}x\right)^{-5/9} \end{aligned}$$

(xiv) Let $y = \sqrt{2+3x} = (2+3x)^{1/2}$

Diff. w.r.t.x

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{2} (2+3x)^{-1/2} (3) \quad \left[\because \frac{d}{dx}((ax+b)^n) = n(ax+b)^{n-1}a \right] \\ &= \frac{3}{2\sqrt{2+3x}} \end{aligned}$$

(xv) Let $y = x^5 + x^2 + 1$

Diff. w.r.t.x

$$\frac{dy}{dx} = 5x^4 + 2x + 0 = 5x^4 + 2x \quad \left[\text{Using formula written at serial number 12 of the table of formulae} \right]$$

(xvi) Let $y = \frac{x^3 + x^2 + 1}{x} = \frac{x^3}{x} + \frac{x^2}{x} + \frac{1}{x} = x^2 + x + \frac{1}{x}$

$$y = x^2 + x + \frac{1}{x}$$

Diff. w.r.t.x

$$\frac{dy}{dx} = \frac{d}{dx}(x^2) + \frac{d}{dx}(x) + \frac{d}{dx}\left(\frac{1}{x}\right) = 2x + 1 - \frac{1}{x^2} \quad \left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right]$$

(xvii) Let $y = \left(x^2 + \frac{1}{x^2}\right)^2$

Diff. w.r.t.x

$$\frac{dy}{dx} = 2\left(x^2 + \frac{1}{x^2}\right) \cdot \frac{d}{dx}\left(x^2 + \frac{1}{x^2}\right) \quad \left[\because \frac{d}{dx}(f(x))^n = n(f(x))^{n-1} \frac{d}{dx}(f(x)) \right]$$

$$= 2\left(x^2 + \frac{1}{x^2}\right)\left(2x - \frac{2}{x^3}\right) = 4\left(x^3 - \frac{1}{x} + \frac{1}{x} - \frac{1}{x^5}\right) = 4\left(x^3 - \frac{1}{x^5}\right)$$

(xviii) Let $y = \left(x + \frac{1}{x}\right)^{100}$

Diff. w.r.t.x

$$\frac{dy}{dx} = 100\left(x + \frac{1}{x}\right)^{99} \frac{d}{dx}\left(x + \frac{1}{x}\right) \quad [\text{Same reason as given in (xvii)}]$$

$$= 100\left(x + \frac{1}{x}\right)^{99} \left(1 - \frac{1}{x^2}\right)$$

(xix) Let $y = (x^2 + 1)(x - 1) = x^3 - x^2 + x - 1$

Diff. w.r.t.x

$$\frac{dy}{dx} = 3x^2 - 2x + 1 - 0 \quad \left[\text{Using formula written at serial number 12 of the table of formulae.} \right]$$

$$= 3x^2 - 2x + 1$$

Alternatively: Using Product Rule

$$\frac{dy}{dx} = (x^2 + 1) \frac{d}{dx}(x - 1) + (x - 1) \frac{d}{dx}(x^2 + 1)$$

$$= (x^2 + 1)(1 - 0) + (x - 1)(2x + 0)$$

$$= x^2 + 1 + 2x^2 - 2x = 3x^2 - 2x + 1$$

(xx) Let $y = x^{-3}(1 + x^2 + x^5 + x^8)$

$$y = x^{-3} + x^{-1} + x^2 + x^5$$

Diff. w.r.t.x

$$\frac{dy}{dx} = -3x^{-4} - 1x^{-2} + 2x + 5x^4 \quad \left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right]$$

$$= -3x^{-4} - x^{-2} + 2x + 5x^4$$

(xxi) Let $y = (x^2 + 1)(x^3 + x^2 + 1)$

Diff. w.r.t.x

[Using Product Rule]

$$\frac{dy}{dx} = (x^2 + 1) \frac{d}{dx}(x^3 + x^2 + 1) + (x^3 + x^2 + 1) \frac{d}{dx}(x^2 + 1)$$

$$= (x^2 + 1)(3x^2 + 2x) + (x^3 + x^2 + 1)(2x)$$

$$= 3x^4 + 2x^3 + 3x^2 + 2x + 2x^4 + 2x^3 + 2x$$

Differentiation

$$= 5x^4 + 4x^3 + 3x^2 + 4x$$

(xxii) Let $y = (4x + 1)^3 (7x + 1)^4$

Diff. w.r.t.x [Using Product rule]

$$\begin{aligned}\frac{dy}{dx} &= (4x + 1)^3 \frac{d}{dx} (7x + 1)^4 + (7x + 1)^4 \frac{d}{dx} (4x + 1)^3 \\&= (4x + 1)^3 4(7x + 1)^3 7 + (7x + 1)^4 3(4x + 1)^2 4 \\&= 4(4x + 1)^2 (7x + 1)^3 [7(4x + 1) + 3(7x + 1)] \\&= 4(4x + 1)^2 (7x + 1)^3 (49x + 10)\end{aligned}$$

(xxiii) Let $y = (x + 2)^2 (x + 3)^4 (x + 1)^5$

Diff.w.r.t.x

$$\begin{aligned}\frac{dy}{dx} &= (x + 2)^2 (x + 3)^4 \frac{d}{dx} (x + 1)^5 + (x + 2)^2 (x + 1)^5 \frac{d}{dx} (x + 3)^4 \\&\quad + (x + 3)^4 (x + 1)^5 \frac{d}{dx} (x + 2)^2 \\&\quad \left[\because \text{if } u, v, w, \text{ are functions of } x, \text{ then} \right. \\&\quad \left. \left[\frac{d}{dx} (uvw) = uv \frac{d}{dx} (w) + uw \frac{d}{dx} (v) + vw \frac{d}{dx} (u) \right] \right] \\&= (x + 2)^2 (x + 3)^4 5(x + 1)^4 + (x + 2)^2 (x + 1)^5 4(x + 3)^3 \\&\quad + (x + 3)^4 (x + 1)^5 2(x + 2) \\&= (x + 2)(x + 3)^3 (x + 1)^4 [5(x + 2)(x + 3) + 4(x + 2)(x + 1) + 2(x + 3)(x + 1)] \\&= (x + 2)(x + 3)^3 (x + 1)^4 [5(x^2 + 5x + 6) + 4(x^2 + 3x + 2) + 2(x^2 + 4x + 3)] \\&= (x + 2)(x + 3)^2 (x + 1)^4 (11x^2 + 45x + 44)\end{aligned}$$

Example 4: Find the derivative of the following functions:

(i) $\frac{x+1}{x-1}$ (ii) $\frac{8x+3}{6-5x}$ (iii) $\frac{a^2}{x^2+a^2}$ (iv) $\sqrt{\frac{x^2+1}{x+1}}$

Solution:

(i) Let $y = \frac{x+1}{x-1}$

Diff. w.r.t.x

$$\frac{dy}{dx} = \frac{(x-1) \frac{d}{dx} (x+1) - (x+1) \frac{d}{dx} (x-1)}{(x-1)^2} \quad [\text{Using Quotient Rule}]$$

$$= \frac{(x-1).1 - (x+1).1}{(x-1)^2} = \frac{x-1-x-1}{(x-1)^2} = \frac{-2}{(x-1)^2}$$

(ii) Let $y = \frac{8x+3}{6-5x}$

Diff. w.r.t.x

$$\frac{dy}{dx} = \frac{(6-5x) \frac{d}{dx}(8x+3) - (8x+3) \frac{d}{dx}(6-5x)}{(6-5x)^2} \quad [\text{Using Quotient Rule}]$$

$$= \frac{(6-5x).8 - (8x+3)(-5)}{(6-5x)^2} = \frac{63}{(6-5x)^2}$$

(iii) Let $y = \frac{a^2}{x^2 + a^2} = a^2(x^2 + a^2)^{-1}$

Diff. w.r.t.x

$$\frac{dy}{dx} = a^2(-1)(x^2 + a^2)^{-2} \frac{d}{dx}(x^2 + a^2)$$

[Don't use quotient rule here because there is no function of x in numerator.]

$$= -\frac{a^2}{(x^2 + a^2)^2} \cdot 2x = \frac{-2a^2x}{(x^2 + a^2)^2}$$

(iv) Let $y = \sqrt{\frac{x^2+1}{x+1}}$

Diff. w.r.t.x

$$\frac{dy}{dx} = \frac{1}{2\sqrt{\frac{x^2+1}{x+1}}} \frac{d}{dx} \left(\frac{x^2+1}{x+1} \right) \left[\because \frac{d}{dx} (f(x))^n = n(f(x))^{n-1} \frac{d}{dx} (f(x)) \right]$$

$$= \frac{1}{2\sqrt{\frac{x^2+1}{x+1}}} \left[\frac{(x+1) \frac{d}{dx}(x^2+1) - (x^2+1) \frac{d}{dx}(x+1)}{(x+1)^2} \right] \quad [\text{Using Quotient Rule}]$$

$$= \frac{(x+1)(2x-1) - (x^2+1)(1)}{2\sqrt{x^2+1}(x+1)^{3/2}} = \frac{2x^2+2x-x^2-1}{2\sqrt{x^2+1}(x+1)^{3/2}}$$

$$= \frac{x^2+2x-1}{2(x+1)^{3/2} \times \sqrt{x^2+1}} = \frac{(x-1)^2}{2(x+1)^{3/2} \sqrt{x^2+1}}$$

Check Your Progress – 1

1) Find the derivative of the following functions at the indicated points

Differentiation

(i) $f(x) = x^3 + x + 1$, at $x = -1$

(ii) $f(x) = 2 - 3x^2$, at $x = 1/2$

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2) Find the value of a, if $f'(-2) = 3$, where $f(x) = 2x^2 - 3ax + 5$

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3) Differentiate the following functions w.r.t. x

(i) πe (ii) $\frac{1}{x^7}$ (iii) $\sqrt{x}$ (iv) $(4 - 3x)^8$ (v) $\left(x^3 + \frac{1}{x^3}\right)^2$

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4) Find the derivative of the following functions:

(i) $\left(x + \frac{1}{x}\right)\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)$ (ii) $\left(x^3 + \frac{1}{x^3}\right)\left(x + \frac{1}{x}\right)$ (iii) $\frac{x^2}{x^3 + 1}$ (iv)

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12.5 CHAIN RULE

Sometimes variables y and x are connected by the relations of the form

$$y = f(u), u = g(w), w = h(x)$$

and we want to differentiate y w.r.t.x. then chain rule is used, which gives

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dw} \frac{dw}{dx}$$

Following example will illustrate the rule more clearly.

Example 5: Find $\frac{dy}{dx}$ in the following cases.

(i) $y = 3u$, $u = v^2$, $v = 4x^2 + 5$

(ii) $y = u^2$, $u = 3v$, $v = \frac{x}{x+1}$

Solution:

(i) $y = 3u$, $u = v^2$, $v = 4x^2 + 5$

Diff. w.r.t. u Diff. w.r.t. v Diff. w.r.t. x

$$\frac{dy}{du} = 3 \qquad \frac{du}{dv} = 2v \qquad \frac{dv}{dx} = 8x$$

$\therefore$ by chain rule

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dv} \frac{dv}{dx} = 3(2v)(8x)$$

$$= 48xv = 48x(4x^2 + 5) \text{ [Replacing the value of } v \text{ in terms of } x]$$

(ii) $y = u^2$, $u = 3v$, $v = \frac{x}{x+1}$

Diff. w.r.t. u Diff. w.r.t. v Diff. w.r.t. x

$$\frac{dy}{du} = 2u \qquad \frac{du}{dv} = 3 \qquad \frac{dv}{dx} = \frac{(x+1) \cdot 1 - x \cdot (1)}{(x+1)^2} = \frac{1}{(x+1)^2}$$

$\therefore$ by chain rule

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dv} \frac{dv}{dx} = (2u)(3) \left(\frac{1}{(1+x)^2} \right)$$

$$= \frac{6u}{(1+x)^2} = \frac{6(3v)}{(1+x)^2} = \frac{18v}{(1+x)^2} = \frac{18x}{(1+x)^3}$$

12.6 DERIVATIVES OF EXPONENTIAL, LOGRITHMIC, PARAMETRIC AND IMPLICIT FUNCTIONS

Let us first take up some examples on derivatives of exponential and logarithmic functions.

Example 6: Find the derivative of the following functions:

(i) 2^x (ii) 5^{ax} (iii) e^{3x} (iv) e^{9x^2} (v) $\log(1+x^2)$

$$(vi) \log_x x^2 \quad (vii) \log_2 x \quad (viii) \log_x 2 \quad (ix) \frac{1}{\log(1+x)}$$

Solution:

(i) Let $y = 2^x$

Diff. w.r.t.x

$$\frac{dy}{dx} = 2^x \log 2 \quad \left[\because \frac{d}{dx}(a^x) = a^x \log a \right]$$

(ii) Let $y = 5^{ax}$

Diff. w.r.t.x

$$\begin{aligned} \frac{dy}{dx} &= 5^{ax} \log 5 \frac{d}{dx}(ax) \quad \left[\because \frac{d}{dx}(a^{bx}) = ba^{bx} \log a \right] \\ &= 5^{ax} a \log 5 \end{aligned}$$

(iii) Let $y = e^{3x}$

Diff. w.r.t.x

$$\frac{dy}{dx} = e^{3x} (3) = 3e^{3x} \quad \left[\because \frac{d}{dx}(e^{ax}) = ae^{ax} \right]$$

(iv) Let $y = e^{9x^2}$

Diff. w.r.t.x

$$\begin{aligned} \frac{dy}{dx} &= e^{9x^2} \frac{d}{dx}(9x^2) = e^{9x^2} (18x) \left[\text{Using } \frac{d}{dx}(e^{f(x)}) = e^{f(x)} \frac{d}{dx}(f(x)) \right] \\ &= 18x e^{9x^2} \end{aligned}$$

(v) Let $y = \log(1+x^2)$

Diff. w.r.t.x

$$\frac{dy}{dx} = \frac{1}{1+x^2} \frac{d}{dx}(1+x^2) = \frac{2x}{1+x^2} \quad \left[\because \frac{d}{dx}(\log f(x)) = \frac{1}{f(x)} \frac{d}{dx}(f(x)) \right]$$

(vi) Let $y = \log_x x^2 = \frac{\log x^2}{\log x}$ [Using base change formula]

$$= \frac{2 \log_x x}{\log_x x} = 2 \quad \left[\because \log m^n = n \log m \right]$$

Diff. w.r.t.x

$$\frac{dy}{dx} = 0$$

(vii) Let $y = \log_2 x$

Diff. w.r.t. x

$$\frac{dy}{dx} = \frac{1}{x} \log_2 e \quad \left[\because \frac{d}{dx}(\log_a x) = \frac{1}{x} \log_a e \right]$$

(viii) Let $y = \log_x 2 = \frac{\log 2}{\log x}$ [Using base change formula]

$$y = (\log 2)(\log x)^{-1}$$

Diff. w.r.t. x

$$\frac{dy}{dx} = -(\log 2)(\log x)^{-2} \frac{d}{dx}(\log x) = -\frac{(\log 2)}{(\log x)^2} \times \frac{1}{x} = -\frac{\log 2}{x(\log x)^2}$$

(ix) Let $y = \frac{1}{\log(1+x)}$ $y = [\log(1+x)]^{-1}$

Diff. w.r.t. x

$$\begin{aligned} \frac{dy}{dx} &= -1[\log(1+x)]^{-2} \frac{d}{dx}(\log(1+x)) \\ &= \frac{-1}{[\log(1+x)]^2} \times \frac{1}{1+x} = -\frac{1}{(1+x)[\log(1+x)]^2} \end{aligned}$$

Following is an example on derivatives of parametric functions.

Example 7: Find $\frac{dy}{dx}$ for the following parametric functions:

(i) $x = 1 + t$, $y = 2 + t^2$ (ii) $x = \frac{1+3t^2}{1-t}$, $y = \frac{1+t^2}{1-t}$

Solution:

(i) $x = 1 + t$, $y = 2 + t^2$

Diff. w.r.t. t Diff. w.r.t. t

$$\frac{dx}{dt} = 1, \quad \frac{dy}{dt} = 2t$$

$$\therefore \frac{dy}{dx} = \frac{dy}{dt} \bigg/ \frac{dx}{dt} = \frac{2t}{1} = 2t$$

(ii) $x = \frac{1+3t^2}{1-t}$

Diff. w.r.t. t

$$\frac{dx}{dt} = \frac{(1-t) \frac{d}{dt}(1+3t^2) - (1+3t^2) \frac{d}{dt}(1-t)}{(1-t)^2}$$

Differentiation

$$= \frac{(1-t)(6t) - (1+3t^2)(-1)}{(1-6^2)} = \frac{6t - 6t^2 + 1 + 3t^2}{(1-t)^2} = \frac{1+6t-3t^2}{(1-t)^2}$$

$$\text{Now, } y = \frac{1+t^2}{1-t}$$

Diff. w.r.t.t

$$\begin{aligned}\frac{dy}{dt} &= \frac{(1-t)\frac{d}{dt}(1+t^2) - (1+t^2)\frac{d}{dt}(1-t)}{(1-t)^2} = \frac{(1-t)(2t) - (1+t^2)(-1)}{(1-t)^2} \\ &= \frac{2t - 2t^2 + 1 + t^2}{(1-t)^2} = \frac{1+2t-t^2}{(1-t)^2}\end{aligned}$$

$$\therefore \frac{dy}{dx} = \frac{dy}{dt} \bigg/ \frac{dx}{dt} = \frac{1+2t-t^2}{1+6t-3t^2}$$

Check Your Progress – 2

1) Find the derivative of the following functions:

(i) $a^{\log_a 2^x}$ (ii) $3^{\log_3 x^2}$

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2) Find $\frac{dy}{dx}$ if $x = 2 + 4t^2$, $y = 9t^2 + 3t + 1$.

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Implicit Function

A function defined by $y = f(x)$ is known as explicit function. But sometimes y cannot be easily expressed in terms of x . A function of the form

$$f(x, y) = c, \text{ where } c \text{ is a constant}$$

is known as implicit function.

Procedure:

In case of implicit function, differentiate the given relation w.r.t. x and collect

all the terms of $\frac{dy}{dx}$ to the left hand side and finally dividing both sides by a

term attached with $\frac{dy}{dx}$, we get the value of $\frac{dy}{dx}$.

Following example will explain the procedure more clearly:

Example 8: Find $\frac{dy}{dx}$ in the following cases:

(i) $x^2 + y^2 = c^2$ (ii) $(x - a)^2 + (y - b)^2 = r^2$ (iii) $x^3 + y^3 + xy = 5$

Solution:

(i) $x^2 + y^2 = c^2$

Diff. w.r.t. x

$$2x + 2y \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x}{y}$$

(ii) $(x - a)^2 + (y - b)^2 = r^2$

Diff. w.r.t. x

$$2(x - a) + 2(y - b) \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x - a}{y - b}$$

(iii) $x^3 + y^3 + xy = 5$

Diff. w.r.t. x

$$3x^2 + 3y^2 \frac{dy}{dx} + x \frac{dy}{dx} + y \cdot 1 = 0 \quad [\text{Using Product Rule in the term } xy]$$

$$\Rightarrow (3y^2 + x) \frac{dy}{dx} = -(y + 3x^2)$$

$$\Rightarrow \frac{dy}{dx} = -\frac{3x^2 + y}{x + 3y^2}$$

12.7 DERIVATIVE OF HIGHER ORDERS

Sometimes we are to differentiate the function more than once.

Derivative of y w.r.t x twice is denoted by $\frac{d^2y}{dx^2}$,

Derivative of y w.r.t x thrice is denoted by $\frac{d^3y}{dx^3}$,

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n times differentiation of y w.r.t x is denoted by $\frac{d^n y}{dx^n}$.

Following example illustrate the idea of higher order derivatives.

Example 9: Find the indicated derivatives for the following functions:

(i) $\frac{d^4 y}{dx^4}$ for $y = \frac{1}{x}$ (ii) $\frac{d^3 y}{dx^3}$ for $y = e^{ax}$

(iii) $\frac{d^3 y}{dx^3}$ for $y = (ax + b)^7$ (iv) $\frac{d^2 y}{dx^2}, \frac{d^3 y}{dx^3}, \frac{d^4 y}{dx^4}$ for $y = x^2 + x + 1$

Solution:

(i) $y = \frac{1}{x}$

Diff. w.r.t.x

$$\frac{dy}{dx} = \frac{-1}{x^2}$$

Again diff. w.r.t.x

$$\frac{d^2 y}{dx^2} = \frac{(-1)(-2)}{x^3} = \frac{2}{x^3}$$

Again diff. w.r.t.x

$$\frac{d^3 y}{dx^3} = \frac{2(-3)}{x^4} = -\frac{6}{x^4}$$

Again diff. w.r.t.x

$$\frac{d^4 y}{dx^4} = \frac{-6(-4)}{x^5} = \frac{24}{x^5}$$

(ii) $y = e^{ax}$

Diff. w.r.t.x

$$\frac{dy}{dx} = e^{ax} \frac{d}{dx}(ax) \quad \left[\because \frac{d}{dx}(e^{f(x)}) = e^{f(x)} \frac{d}{dx}(f(x)) \right]$$

$$= e^{ax}(a) = ae^{ax}$$

Again diff. w.r.t.x

$$\frac{d^2 y}{dx^2} = a^2 e^{ax}$$

Again diff. w.r.t.x

$$\frac{d^3y}{dx^3} = a^3 e^{ax}$$

$$(iii) \ y = (ax + b)^7$$

Diff. w.r.t.x

$$\frac{dy}{dx} = 7(ax + b)^6 a \quad \left[\because \frac{d}{dx} ((ax + b)^n) = n(ax + b)^{n-1} a \right]$$

Again diff. w.r.t.x

$$\frac{d^2y}{dx^2} = 42(ax + b)^5 a^2$$

Again diff. w.r.t.x

$$\frac{d^3y}{dx^3} = 210(ax + b)^4 a^3$$

$$(iv) \ y = x^2 + x + 1$$

Diff. w.r.t.x

$$\frac{dy}{dx} = 2x + 1 \quad \left[\text{Using formula written at serial number 12 of the table of formulae} \right]$$

Again diff. w.r.t.x

$$\frac{d^2y}{dx^2} = 2$$

Again diff. w.r.t.x

$$\frac{d^3y}{dx^3} = 0$$

Again diff. w.r.t.x

$$\frac{d^4y}{dx^4} = 0$$

12.8 CONCEPT OF MAXIMA AND MINIMA

Differentiation has a large number of applications in different fields such as mathematics, statistics, economics, actuarial science, etc. Concept of differentiation is also useful to obtain maxima or minima point(s) and their corresponding value(s) of a given function. Actually sometimes we are interested only to find maximum or minimum value(s) of a function. The aim of this section is to meet this interest. Without going into theoretical details, let us discuss the concept geometrically.

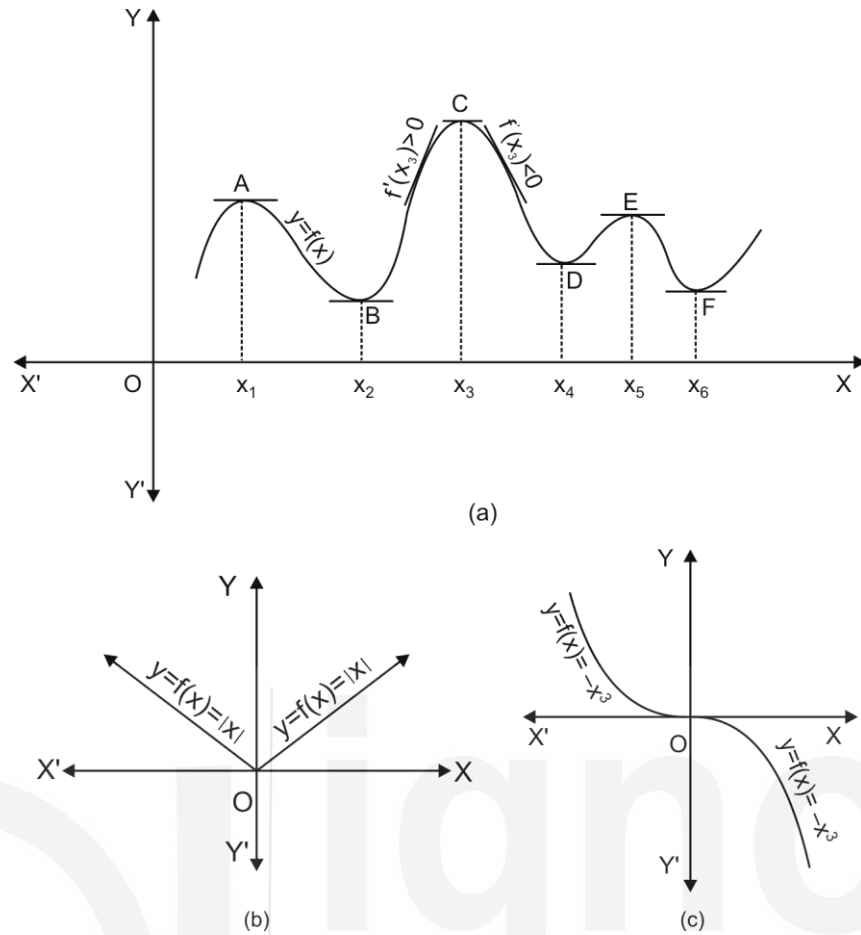


Fig. 12.3: Diagrammatic representation of Differentiation

In equation (2) of Sec. 12.2 of this unit we have seen that
(Derivative at a point) $_{x=a}$ is the slope of the tangent at that point. ... (1)

In Fig 12.3a we see that x_1 is a point where the function $y = f(x)$ takes maximum value (local) compare to all points which are very close to x_1
i.e. $f(x_1) > f(x)$... (2)

for all points x which are very close to x_1

The point x_1 is known as local maxima point of the function $y = f(x)$ (local maxima means that it satisfy the equation (2) as given above, i.e. there may be points in the domain of the function where value of the function f is greater than $f(x_1)$)

Similarly, points x_3 and x_5 in the same figure are points of maxima (local). On the other hand in the same figure x_2 is a point where the function $y = f(x)$ takes minimum value (local) compare to all points which are very close to x_2
i.e. $f(x_2) < f(x)$

for all points x which are very close to x_2

The point x_2 is known as local minima point of the function $y = f(x)$.

Similarly x_4 and x_6 are points of local minima.

But one interesting point to be noted here is that tangent lines at all the points whether it is point of maxima or minima is parallel to x -axis.

i.e. slope of the tangent at point of maxima or minima = $\tan \theta = 0 \quad \dots (3)$

$$\left[\begin{array}{l} \because \text{if a line makes an angle } \theta \text{ with +ve direction of} \\ x\text{-axis then slope of the line is defined as } \tan \theta, \text{ in} \\ \text{this case } \theta = 0 \text{ as tangent line is parallel to } x\text{-axis.} \end{array} \right]$$

In view of equation (1) and (3), we have

Derivative of the function at a point of maxima or minima = 0 $\dots (4)$

, provided that derivative at that point exists

Also note that converse of (4) does not hold.

For example, take graph of the function $y = f(x) = -x^3$ shown in Fig. 12.3 c.

we see that

$$\left. \frac{dy}{dx} \right|_{x=0} = f'(0) = -3(0)^2 = 0 \quad \left[\because \frac{dy}{dx} = f'(x) = -3x^2 \right]$$

But $x = 0$ is neither point of maxima nor minima.

Now consider the function $y = f(x) = |x|$, whose graph is shown in Fig. 12.3 b.

We see that $x = 0$ is a point of minima, in fact absolute minima or global minima (absolute minima or global minima means that function assumes smallest value at $x = 0$ in whole domain of the function not only at those points which are very close to $x = 0$)

But we have seen in part (i) of Example 2 of this unit that derivative of the function $y = f(x) = |x|$ does not exist at $x = 0$.

Equation (4) implies points obtained by putting first derivative equal to zero may be points of maxima or minima. Second derivative test differentiate between points of maxima and minima which is stated below:

Second Derivative Test: It states that if the function f is twice differentiable at a point ' c ', where c is point of the domain of the function f , then

(i) c is point of local minima if $f'(c) = 0$ and $f''(c) > 0$.

(ii) c is point of local maxima if $f'(c) = 0$ and $f''(c) < 0$.

(iii) test fails if $f''(c) = 0$. In this case we use first order derivative test, which can be concluded as:

- if $f'(c)$ changes its sign from positive to negative as we cross the point $x = c$, then $x = c$ is point of maxima (see Fig. 12.3 a at point x_3)
- if $f'(c)$ changes its sign from negative to positive as we cross the point $x = c$, then $x = c$ is point of minima. This can be noted at points x_2, x_4, x_6) in Fig. 12.3 a.
- if $f'(c)$ does not change its sign as we cross the point $x = c$, then $x = c$ is neither point of minima nor maxima. See Fig. 12.3 c in which $x = 0$ is such a point. Point of this nature is called point of inflection. Normal curve has two such points at $x = \mu + \sigma$ and $x = \mu - \sigma$. You can observe it by differentiating normal density twice and putting double derivative equal to zero.

Example 10: Find local maximum and minimum values of the function

$$f(x) = 2x^3 - 15x^2 + 36x + 9.$$

Solution: Given function is

$$f(x) = 2x^3 - 15x^2 + 36x + 9$$

Dif. w.r.t. x

$$f'(x) = 6x^2 - 30x + 36 \quad \dots (1)$$

For maxima or minima

$$f'(x) = 0$$

$$\Rightarrow 6x^2 - 30x + 36 = 0$$

$$\Rightarrow x^2 - 5x + 6 = 0$$

$$\Rightarrow (x - 2)(x - 3) = 0$$

$$\Rightarrow x = 2, 3$$

Dif. (1) w.r.t. x

$$f''(x) = 12x - 30$$

$$\text{At } x = 2, f''(2) = 24 - 30 = -6 < 0$$

$\therefore$ by second order derivative test, $x = 2$ is point of maxima and maximum value is given by

$$f(2) = 2(2)^3 - 15(2)^2 + 36(2) + 9 = 16 - 60 + 72 + 9 = 27$$

$$\text{At } x = 3, f''(3) = 36 - 30 = 6 > 0$$

$\therefore$ by second order derivative test, $x = 3$ is point of minima and minimum value of the function is given by

$$f(3) = 2(3)^3 - 15(3)^2 + 36(3) + 9 = 54 - 135 + 108 + 9 = 36$$

Example 11: Find local maximum and minimum values of the function

$$f(x) = \frac{x^4}{4} + \frac{x^3}{3} - 2x^2 - 4x + 5.$$

Solution: Given function is

$$f(x) = \frac{x^4}{4} + \frac{x^3}{3} - 2x^2 - 4x + 5$$

Diff. w.r.t. x

$$f'(x) = \frac{4x^3}{4} + \frac{3x^2}{3} - 4x - 4 = x^3 + x^2 - 4x - 4 \quad \dots (1)$$

For maxima or minima

$$f'(x) = 0$$

$$\Rightarrow x^3 + x^2 - 4x - 4 = 0 \quad \dots (2)$$

By inspection $x = -1$ is a root of equation (2)

$$\Rightarrow (x + 1) \text{ is a factor of } x^3 + x^2 - 4x - 4$$

$\therefore$ (2) can be written as

$$(x + 1)(x^2 - 4) = 0$$

$$\Rightarrow (x + 1)(x - 2)(x + 2) = 0$$

$$\Rightarrow x = -1, 2, -2$$

Diff. (1) w.r.t. x

$$f''(x) = 3x^2 + 2x - 4$$

$$\text{At } x = -1, f''(-1) = 3(-1)^2 + 2(-1) - 4 = 3 - 2 - 4 = -3 < 0$$

$$\text{At } x = 2, f''(2) = 3(2)^2 + 2(2) - 4 = 12 > 0 \text{ and}$$

$$\text{At } x = -2, f''(-2) = 3(-2)^2 + 2(-2) - 4 = 4 > 0$$

$\therefore$ by second order derivative test $x = 2, -2$ are points of minima and $x = -1$ is point of maxima.

Local minimum value at $x = 2$ is $f(2) = \frac{(2)^4}{4} + \frac{(2)^3}{3} - 2(2)^2 - 4(2) + 5 = -\frac{13}{3}$ and

Local minimum value at $x = -2$ is given by

$$f(-2) = \frac{(-2)^4}{4} + \frac{(-2)^3}{3} - 2(-2)^2 - 4(-2) + 5 = \frac{19}{3}$$

Local maximum value at $x = -1$ is given by

$$f(-1) = \frac{(-1)^4}{4} + \frac{(-1)^3}{3} - 2(-1)^2 - 4(-1) + 5 = \frac{83}{12}$$

Check Your Progress – 3

1) Find $\frac{dy}{dx}$, if $xy^3 + xe^x + xe^{-y} = 3$

2) Find local maximum and minimum values of the function

$$f(x) = 4x^3 - 21x^2 + 18x + 9.$$

12.9 SUMMARY

As this unit transitions from the study of limits (previous unit) to the derivative, this unit begins by defining it fundamentally as the instantaneous rate of change and geometrically as the slope of the tangent line to a curve at a given point. It introduces the First Principle of Derivatives—the formal limit-based method for finding a derivative—before moving on to efficient differentiation techniques like the Chain Rule for composite functions. Further, in the unit, we discussed the derivatives of specialised functions, including exponential, logarithmic, parametric, and implicit forms. This unit also explored higher-order derivatives, in which a function is differentiated multiple times. Finally, this unit describes the concepts and tools to identify the maxima and minima of functions, using the first- and second-derivative tests to locate the "peaks" and "valleys" of a graph where the rate of change is zero.

12.10 SOLUTIONS/ANSWERS TO CHECK YOUR PROGRESS

Check Your Progress – 1

1) (i) $f(x) = x^3 + x + 1$, at $x = -1$

By definition

$$\begin{aligned} f'(-1) &= \lim_{h \rightarrow 0} \frac{f(-1+h) - f(-1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(-1+h)^3 + (-1+h) + 1 - [(-1)^3 + (-1) + 1]}{h} \\ &= \lim_{h \rightarrow 0} \frac{-1 + h^3 + 3h - 3h^2 - 1 + h + 1 - (-1 - 1 + 1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^3 - 3h^2 + 4h}{h} = \lim_{h \rightarrow 0} (h^2 - 3h + 4) = 0 - 0 + 4 = 4 \end{aligned}$$

(ii) By definition

$$\begin{aligned} f'\left(\frac{1}{2}\right) &= \lim_{h \rightarrow 0} \frac{f\left(\frac{1}{2}+h\right) - f\left(\frac{1}{2}\right)}{h} \\ &= \lim_{h \rightarrow 0} \frac{2 - 3\left(\frac{1}{2}+h\right)^2 - \left[2 - 3\left(\frac{1}{2}\right)^2\right]}{h} \\ &= \lim_{h \rightarrow 0} \frac{2 - 3\left(\frac{1}{4} + h + h^2\right) - \left(2 - \frac{3}{4}\right)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-3h - 3h^2}{h} = \lim_{h \rightarrow 0} (-3 - 3h) = -3 - 0 = -3 \end{aligned}$$

2) $f(x) = 2x^2 - 3ax + 5$

By definition

$$\begin{aligned} f'(-2) &= \lim_{h \rightarrow 0} \frac{f(-2+h) - f(-2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{2(-2+h)^2 - 3a(-2+h) + 5 - [2(-2)^2 - 3a(-2) + 5]}{h} \\ &= \lim_{h \rightarrow 0} \frac{2(4 + h^2 - 4h) + 6a - 3ah + 5 - (8 + 6a + 5)}{h} \end{aligned}$$

$$= \lim_{h \rightarrow 0} \frac{2h^2 - 8h - 3ah}{h} = \lim_{h \rightarrow 0} (2h - 8 - 3a) = -8 - 3a$$

But according to problem

$$f'(-2) = 3 \Rightarrow -8 - 3a = 3 \Rightarrow -3a = 11 \Rightarrow a = -\frac{11}{3}$$

3)(i) Let $y = \pi e$

Diff. w.r.t. x

$$\frac{dy}{dx} = 0 \quad \left[\because \pi \text{ and } e \text{ both are constant} \Rightarrow \pi e \text{ is a constant} \right. \\ \left. \text{and derivative of a constant function is zero.} \right]$$

(ii) Let $y = \frac{1}{x^7} = x^{-7}$

Diff. w.r.t. x

$$\frac{dy}{dx} = -7x^{-8} = -\frac{7}{x^8} \left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right]$$

(iii) Let $y = \sqrt{x} = x^{1/2}$

Diff. w.r.t. x

$$\frac{dy}{dx} = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}} \left[\because \frac{d}{dx}(x^n) = nx^{n-1} \right]$$

(iv) Let $y = (4 - 3x)^8$

Diff. w.r.t. x

$$\frac{dy}{dx} = 8(4 - 3x)^7(-3) \left[\because \frac{d}{dx}((ax + b)^n) = n(ax + b)^{n-1}a \right] \\ = -24(4 - 3x)^7$$

(v) Let $y = \left(x^3 + \frac{1}{x^3}\right)^2$

Diff. w.r.t. x

$$\frac{dy}{dx} = 2\left(x^3 + \frac{1}{x^3}\right) \frac{d}{dx}\left(x^3 + \frac{1}{x^3}\right) \left[\because \frac{d}{dx}(f(x))^n = n(f(x))^{n-1} \frac{d}{dx}(f(x)) \right] \\ = 2\left(x^3 + \frac{1}{x^3}\right) \left(3x^2 - \frac{3}{x^4}\right) \\ = 6\left(x^5 - \frac{1}{x} + \frac{1}{x} - \frac{1}{x^7}\right) = 4\left(x^5 - \frac{1}{x^7}\right)$$

4) (i) $y = \left(x + \frac{1}{x}\right) \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right) = x^{3/2} + x^{1/2} + \frac{1}{x^{1/2}} + \frac{1}{x^{3/2}}$

$$y = x^{3/2} + x^{1/2} + x^{-1/2} + x^{-3/2}$$

Diff. w.r.t.x

$$\frac{dy}{dx} = \frac{3}{2}x^{1/2} + \frac{1}{2}x^{-1/2} - \frac{1}{2}x^{-3/2} - \frac{3}{2}x^{-5/2}$$

$$= \frac{3}{2}\sqrt{x} + \frac{1}{2\sqrt{x}} - \frac{1}{2x^{3/2}} - \frac{3}{2x^{5/2}}$$

(ii) Let $y = \left(x^3 + \frac{1}{x^3}\right) \left(x + \frac{1}{x}\right) = x^4 + x^2 + \frac{1}{x^2} + \frac{1}{x^4}$

Diff. w.r.t.x

$$\frac{dy}{dx} = 4x^3 + 2x - \frac{2}{x^3} - \frac{4}{x^5}$$

(iii) Let $y = \frac{x^2}{x^3 + 1}$

Diff. w.r.t.x

$$\frac{dy}{dx} = \frac{(x^3 + 1) \frac{d}{dx}(x^2) - x^2 \frac{d}{dx}(x^3 + 1)}{(x^3 + 1)^2} \quad \text{[Using Quotient Rule]}$$

$$= \frac{(x^3 + 1)(2x) - x^2(3x^2)}{(x^3 + 1)^2}$$

$$= \frac{2x^4 + 2x - 3x^4}{(x^3 + 1)^2} = \frac{2x - x^4}{(x^3 + 1)^2} = \frac{x(2 - x^3)}{(x^3 + 1)^2}$$

(iv) Let $y = \frac{x^2 + x}{a} = \frac{1}{a}(x^2 + x)$

Diff. w.r.t.x

$$\frac{dy}{dx} = \frac{1}{a} \frac{d}{dx}(x^2 + x) = \frac{1}{a}(2x + 1) \left[\begin{array}{l} \text{Do not use quotient rule} \\ \text{because in the denominator} \\ \text{there is no function of } x. \end{array} \right]$$

Check Your Progress – 2

1)(i) Let $y = a^{\log_a 2^x}$ $y = 2^x$ $[\because a^{\log_a f(x)} = f(x)]$

Diff. w.r.t.x

Differentiation

$$\frac{dy}{dx} = 2^x \log 2 \left[\because \frac{d}{dx} (a^x) = a^x \log a \right]$$

(ii) Let $y = 3^{\log_3 x^2} = x^2$ $[\because a^{\log_a f(x)} = f(x)]$

Diff. w.r.t.x

$$\frac{dy}{dx} = 2x \left[\because \frac{d}{dx} (x^n) = nx^{n-1} \right]$$

2) $x = 2 + 4t^2$, $y = 9t^2 + 3t + 1$

Diff. w.r.t.t

Diff. w.r.t.t

$$\frac{dx}{dt} = 8t$$

$$\frac{dy}{dt} = 18t + 3$$

$$\therefore \frac{dy}{dx} = \frac{dy}{dt} \bigg/ \frac{dx}{dt} = \frac{18t + 3}{8t}$$

Check Your Progress – 3

1) $xy^3 + xe^x + xe^{-y} = 3$

Diff. w.r.t.x

$$x \left(3y^2 \frac{dy}{dx} \right) + (1)y^3 + xe^x + (1)e^x + xe^{-y} \left(-\frac{dy}{dx} \right) + (1)e^{-y} = 0$$

$$\Rightarrow (3xy^2 - xe^{-y}) \frac{dy}{dx} = -y^3 - xe^x - e^x - e^{-y}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{y^3 + xe^x + e^x + e^{-y}}{3xy^2 - xe^{-y}} = \frac{y^3 + xe^x + e^x + e^{-y}}{xe^{-y} - 3xy^2}$$

2) Given function is

$$f(x) = 4x^3 - 21x^2 + 18x + 9$$

Diff. w.r.t. x

$$f'(x) = 12x^2 - 42x + 18 \quad \dots (1)$$

For maxima or minima

$$f'(x) = 0$$

$$\Rightarrow 12x^2 - 42x + 18 = 0$$

$$\Rightarrow 2x^2 - 7x + 3 = 0$$

$$\Rightarrow 2x^2 - 6x - x + 3 = 0$$

$$\Rightarrow 2x(x - 3) - 1(x - 3) = 0$$

$$\Rightarrow (x - 3)(2x - 1) = 0$$

$$\Rightarrow x = 3, 1/2$$

Diff. (1) w.r.t. x

$$f''(x) = 24x - 42$$

$$\text{At } x = 3, f''(3) = 24 \times 3 - 42 = 72 - 42 = 30 > 0$$

$$\text{At } x = 1/2, f''(1/2) = 24 \times \frac{1}{2} - 42 = 12 - 42 = -30 < 0$$

$\therefore$ by second order derivative test $x = 3$ is point of minima and $x = 1/2$ is point of maxima.

Local minimum value at $x = 3$ is given by

$$f(3) = 4(3)^3 - 21(3)^2 + 18(3) + 9 = 108 - 189 + 54 + 9 = -18$$

Local maximum value at $x = 1/2$ is given by

$$\begin{aligned} f\left(\frac{1}{2}\right) &= 4\left(\frac{1}{2}\right)^3 - 21\left(\frac{1}{2}\right)^2 + 18\left(\frac{1}{2}\right) + 9 \\ &= \frac{1}{2} - \frac{21}{4} + 9 + 9 = \frac{21 - 21 + 72}{4} = \frac{53}{4} \end{aligned}$$

12.11 REFERENCES/FURTHER READINGS

1. Mathematics Class XI By RD Sharma , Dhanpat Rai Publication , 2011
2. Mathematics Class XII Part – 1 By Ravi Prakash , GRB Books, Second Edition 2025
3. BCS-012 : Basic Mathematics :
https://onlinecourses.swayam2.ac.in/nou26_ma07/preview (January 2026 session)

UNIT 13 INDEFINITE INTEGRATION

Structure

- 13.0 Introduction
- 13.1 Objectives
- 13.2 Meaning and Terminology used.
- 13.3 Integration of some Particular Functions
- 13.4 Integration by Substitution
- 13.5 Integration using Partial Fractions
- 13.6 Integration by Parts
- 13.7 Summary
- 13.8 Solutions/Answers to Check Your Progress
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13.0 INTRODUCTION

In the previous unit, we have studied the differentiation of some functions. Here, in this unit we are going to discuss the reverse process of differentiation known as integration.

In this unit, we will study the integration of some commonly used functions in section 13.3, integration by substitutions in section 13.4, integration by using partial fractions in section 13.5 and integration by parts in section 13.6.

13.1 OBJECTIVES

After completing this unit, you should be able to:

- evaluate the integration of some commonly used functions;
- evaluate the integration by substitution method;
- evaluate the integration using partial fractions; and
- evaluate the integration by parts.

13.2 MEANING AND TERMINOLOGY USED

Notations

In fact, summation is convenient way to represent the sum of discrete values only. If the variable is continuous, then the summation cannot be used in the way it is used for discrete values. Summation is obtained by the process of integration in case of continuous variable.

Origin of integration lies in the process of summation. In mathematics, the words “Summation” and “Integration” are used for the words “to unite”.

Indefinite Integration

In previous unit we have studied differentiation. The integration is just the reverse process of differentiation. Actually, it is an antiderivative of a function. That is, if $f'(x)$ is derivative of $f(x)$ and hence $f'(x)$ is derivative of $f(x) + c$ ($\because$ derivative of constant is zero). And therefore, $f(x) + c$ is the integration of $f'(x)$.

Integral of a function $f(x)$ w.r.t. x is denoted by $\int f(x)dx$

,where $f(x)$ is known as integrand, dx reflects the message that integrand is to be integrated w.r.t. to the variable x and the entire process of finding the integral of integrand is known as integration. The symbol $\int$ has its origin from the letter S , which was used for summation.

Let us consider a simple example first and then give a list of the formulae.

We know that the function x is the differentiation of $\frac{x^2}{2} + c$ w.r.t. x .

$\therefore \frac{x^2}{2} + c$ is integration of x .

i.e. $\int x dx = \frac{x^2}{2} + c$, where c is known as constant of integration.

Similarly, integration of other functions can be obtained. Integrations of some commonly used functions are listed in the following table.

List of Formulae of Integration

S. No.	Function $f(x)$	$\int f(x)dx$
1	k (constant function)	$kx + c$, where c is constant of integration
2	x^n	$\frac{x^{n+1}}{n+1} + c$, $n \neq -1$
3	$\frac{1}{x}$	$\log x + c$
4	$(ax + b)^n$	$\frac{(ax + b)^{n+1}}{a(n+1)} + c$, $n \neq -1$
5	$\frac{1}{ax + b}$	$\frac{1}{a} \log ax + b + c$
6	Exponential functions (i) a^{mx} (ii) a^{mx+n}	(i) $\frac{a^{mx}}{m \log_e a } + c$ (ii) $\frac{a^{mx+n}}{m \log a } + c$

	(iii) e^{ax}	(iii) $\frac{e^{ax}}{a} + c$
	(iv) e^{ax+b}	(iv) $\frac{e^{ax+b}}{a} + c$

Remark 1:

If f, g are integral functions such that $f + g, f - g$, are defined and a, b are real constants, then

$$(i) \int a(f(x))dx = a \int f(x)dx$$

$$(ii) \int (af(x) \pm bg(x))dx = a \int f(x)dx \pm b \int g(x)dx$$

13.3 INTEGRATION OF SOME PARTICULAR FUNCTIONS

In this section, we learn how the formulae mentioned in the table on previous page are used.

Example 1: Evaluate the following integrals:

$$(i) \int \sqrt{5}dx \quad (ii) \int 0dx \quad (iii) \int \pi dx \quad (iv) \int x^3 dx$$

$$(v) \int x^{7/2}dx \quad (vi) \int \frac{1}{x^5} dx \quad (vii) \int \frac{1}{x^{7/2}} dx \quad (viii) \int \frac{1}{\sqrt[3]{x}} dx$$

$$(ix) \int (x^3 + x + 5)dx \quad (x) \int (x^2 + 1)(x - 1)dx \quad (xi) \int \frac{x^6 - x^4 + 1}{x^2} dx$$

$$(xii) \int \frac{x^4 + x^3 + 3}{\sqrt{x}} dx \quad (xiii) \int \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right) dx \quad (xiv) \int \left(x + \frac{1}{x} \right)^2 dx$$

$$(xv) \int \left(8x^3 + 2x - 3 + \frac{1}{x^2} + \frac{2}{x^3} \right) dx$$

Solution:

$$(i) \int \sqrt{5}dx = \sqrt{5}x + c \quad \left[\begin{array}{l} \because \sqrt{5} \text{ is a constant and if } k \text{ is} \\ \text{constant then } \int kdx = kx + c \end{array} \right]$$

where c is constant of integration.

Indefinite Integration

Note: Constant of integration c is added everywhere, so in future we will not write 'where c is constant of integration'.

$$(ii) \quad \int 0 dx = 0x + c = c \quad \text{as } 0 \text{ is constant}$$

$$(iii) \quad \int \pi dx = \pi x + c \quad \text{as } \pi \text{ is constant}$$

$$(iv) \quad \int x^3 dx = \frac{x^{3+1}}{3+1} + c = \frac{x^4}{4} + c \quad \left[\because \int x^n dx = \frac{x^{n+1}}{n+1} + c \right]$$

$$(v) \quad \int x^{7/2} dx = \frac{x^{\frac{7}{2}+1}}{\frac{7}{2}+1} + c = \frac{2}{9} x^{\frac{9}{2}} + c \quad \left[\because \int x^n dx = \frac{x^{n+1}}{n+1} + c \right]$$

$$(vi) \quad \int \frac{1}{x^5} dx = \int x^{-5} dx = \frac{x^{-5+1}}{-5+1} + c = -\frac{1}{4} x^{-4} + c$$

$$\left[\because \int x^n dx = \frac{x^{n+1}}{n+1} + c \right]$$

$$(vii) \quad \int \frac{1}{x^{7/2}} dx = \int x^{-7/2} dx = \frac{x^{-5/2}}{-5/2} + c = -\frac{2}{5} x^{-5/2} + c$$

$$\left[\because \int x^n dx = \frac{x^{n+1}}{n+1} + c \right]$$

$$(viii) \quad \int \frac{1}{\sqrt[3]{x}} dx = \int \frac{1}{x^{1/3}} dx = \int x^{-1/3} dx = \frac{x^{2/3}}{2/3} + c = \frac{3}{2} x^{2/3} + c$$

$$\left[\because \int x^n dx = \frac{x^{n+1}}{n+1} + c \right]$$

$$(ix) \quad \int (x^3 + x + 5) dx = \int x^3 dx + \int x dx + \int 5 dx = \frac{x^4}{4} + \frac{x^2}{2} + 5x + c$$

$$(x) \quad \int (x^2 + 1)(x - 1) dx = \int (x^3 - x^2 + x - 1) dx$$

$$= \frac{x^4}{4} - \frac{x^3}{3} + \frac{x^2}{2} - x + c \quad \left[\begin{array}{l} \because \int x^n dx = \frac{x^{n+1}}{n+1} + c \\ \text{and } \int k dx = kx + c \end{array} \right]$$

$$(xi) \quad \int \frac{x^6 - x^4 + 1}{x^2} dx = \int \left(\frac{x^6}{x^2} - \frac{x^4}{x^2} + \frac{1}{x^2} \right) dx = \int (x^4 - x^2 + x^{-2}) dx$$

$$= \frac{x^5}{5} - \frac{x^3}{3} + \frac{x^{-1}}{-1} + c = \frac{x^5}{5} - \frac{x^3}{3} - \frac{1}{x} + c$$

$$\begin{aligned}
 \text{(xii)} \quad \int \frac{x^4 + x^3 + 3}{\sqrt{x}} dx &= \int \left(\frac{x^4}{\sqrt{x}} + \frac{x^3}{\sqrt{x}} + \frac{3}{\sqrt{x}} \right) dx = \int (x^{7/2} + x^{5/2} + 3x^{-1/2}) dx \\
 &= \frac{x^{9/2}}{9/2} + \frac{x^{7/2}}{7/2} + 3 \frac{x^{1/2}}{1/2} + c = \frac{2}{9} x^{9/2} + \frac{2}{7} x^{7/2} + 6\sqrt{x} + c
 \end{aligned}$$

$$\begin{aligned}
 \text{(xiii)} \quad \int \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right) dx &= \int (x^{1/2} + x^{-1/2}) dx = \frac{x^{3/2}}{3/2} + \frac{x^{1/2}}{1/2} + c \\
 &= \frac{2}{3} x^{3/2} + 2\sqrt{x} + c
 \end{aligned}$$

$$\begin{aligned}
 \text{(xiv)} \quad \int \left(x + \frac{1}{x} \right)^2 dx &= \int \left(x^2 + \frac{1}{x^2} + 2.x \cdot \frac{1}{x} \right) dx \\
 &= \int (x^2 + x^{-2} + 2) dx = \frac{x^3}{3} + \frac{x^{-1}}{-1} + 2x + c = \frac{x^3}{3} - \frac{1}{x} + 2x + c
 \end{aligned}$$

$$\begin{aligned}
 \text{(xv)} \quad \int \left(8x^3 + 2x - 3 + \frac{1}{x^2} + \frac{2}{x^3} \right) dx &= \int (8x^3 + 2x - 3 + x^{-2} + 2x^{-3}) dx \\
 &= \frac{8x^4}{4} + \frac{2x^2}{2} - 3x + \frac{x^{-1}}{-1} + \frac{2x^{-2}}{-2} + c \\
 &= 2x^4 + x^2 - 3x - \frac{1}{x} - \frac{1}{x^2} + c
 \end{aligned}$$

Example 2: Evaluate the following integrals:

$$\text{(i)} \quad \int (2x + 3)^5 dx \quad \text{(ii)} \quad \int (5 - 9x)^6 dx \quad \text{(iii)} \quad \int (9x + 5)^{3/2} dx$$

$$\text{(iv)} \quad \int \sqrt[5]{8 - 3x} dx \quad \text{(v)} \quad \int \frac{(3 + 2x)^{7/2}}{\sqrt{3 + 2x}} dx \quad \text{(vi)} \quad \int \frac{1}{(7x + 2)^3} dx$$

$$\text{(vii)} \quad \int \frac{1}{\sqrt{3x + 5}} dx \quad \text{(viii)} \quad \int e^{5 \log \sqrt{3x - 5}} dx \quad \text{(ix)} \quad \int a^{\log_a \sqrt{4x + 5}} dx$$

$$\text{(x)} \quad \int a^{3x} dx \quad \text{(xi)} \quad \int e^{3x} dx \quad \text{(xii)} \quad \int e^{5x+7} dx$$

$$\text{(xiii)} \quad \int a^{3-2x} dx \quad \text{(xiv)} \quad \int (5e^{7x} + x^2) dx \quad \text{(xv)} \quad \int (a^x + e^x + a^a + e^a + a^e) dx$$

$$\text{(xvi)} \quad \int 5^x 2^x dx \quad \text{(xvii)} \quad \int \frac{2^x}{3^x} dx \quad \text{(xviii)} \quad \int \frac{(a^x - b^x)^2}{a^x b^x} dx$$

$$\text{(xix)} \quad \int (e^{a \log x} + a^{x \log_a a} + a^{m \log_a a}) dx$$

$$\text{(xx)} \quad \int \left(\frac{x}{3} + (5x - 3)^3 + x\sqrt{x} + \frac{1}{\sqrt{5 + 2x}} + a^a \right) dx$$

Solution:

$$(i) \int (2x+3)^5 dx = \frac{(2x+3)^{5+1}}{2(5+1)} + c \quad \left[\begin{array}{l} \because \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c \\ \text{Here } a=2, n=5 \end{array} \right]$$

$$= \frac{(2x+3)^6}{12} + c$$

$$(ii) \int (5-9x)^6 dx = \frac{(5-9x)^7}{7(-9)} + c \quad \left[\begin{array}{l} \because \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c \\ \text{Here } a=-9, n=6 \end{array} \right]$$

$$= -\frac{1}{63}(5-9x)^7 + c$$

$$(iii) \int (9x+5)^{3/2} dx = \frac{(9x+5)^{5/2}}{\frac{5}{2} \times 9} + c \quad \left[\begin{array}{l} \because \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c \\ \text{Here } a=9, n=3/2 \end{array} \right]$$

$$= \frac{2}{45}(9x+5)^{5/2} + c$$

$$(iv) \int \sqrt[5]{8-3x} dx = \int (8-3x)^{1/5} dx = \frac{(8-3x)^{6/5}}{\frac{6}{5} \times (-3)} + c = \frac{-5}{18}(8-3x)^{6/5} + c$$

$$\left[\begin{array}{l} \because \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c \\ \text{Here } a=-3, n=1/5 \end{array} \right]$$

$$(v) \int \frac{(3+2x)^{7/2}}{\sqrt{3+2x}} dx = \int (3+2x)^{\frac{7}{2} - \frac{1}{2}} dx \quad \left[\because \frac{a^m}{a^n} = a^{m-n} \right]$$

$$= \int (3+2x)^3 dx$$

$$= \frac{(3+2x)^4}{4 \times 2} + c \quad \left[\begin{array}{l} \because \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c \\ \text{Here } a=2, n=3 \end{array} \right]$$

$$= \frac{1}{8}(3+2x)^4 + c$$

$$(vi) \int \frac{1}{(7x+2)^3} dx = \int (7x+2)^{-3} dx$$

$$= \frac{(7x+2)^{-2}}{-2 \times 7} + c \left[\begin{array}{l} \because \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c \\ \text{Here } a = 7, n = -3 \end{array} \right]$$

$$= -\frac{1}{14} (7x+2)^{-2} + c$$

$$(vii) \int \frac{1}{\sqrt{3x+5}} dx = \int (3x+5)^{-1/2} dx$$

$$= \frac{(3x+5)^{1/2}}{\frac{1}{2} \times 3} + c \left[\begin{array}{l} \because \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c \\ \text{Here } a = 3, n = -1/2 \end{array} \right]$$

$$= \frac{2}{3} \sqrt{3x+5} + c$$

$$(viii) \int e^{5 \log \sqrt{3x-5}} dx = \int e^{\log(3x-5)^{\frac{5}{2}}} dx = \int (3x-5)^{\frac{5}{2}} dx \left[\because a^{\log_a f(x)} = f(x) \right]$$

$$= \frac{(3x-5)^{7/2}}{(7/2)(3)} + c = \frac{2}{21} (3x-5)^{7/2} + c$$

$$(ix) \int a^{\log_a \sqrt{4x+5}} dx = \int \sqrt{4x+5} dx \left[\because a^{\log_a f(x)} = f(x) \right]$$

$$= \frac{(4x+5)^{3/2}}{(3/2) \times 4} + c = \frac{1}{6} (4x+5)^{3/2} + c$$

$$(x) \int a^{3x} dx = \frac{a^{3x}}{3 \log |a|} + c \left[\begin{array}{l} \because \int a^{mx} dx = \frac{a^{mx}}{m \log |a|} + c \\ \text{Here } a = a, m = 3 \end{array} \right]$$

$$(xi) \int e^{3x} dx = \frac{e^{3x}}{3} + c \left[\begin{array}{l} \because \int e^{ax} dx = \frac{e^{ax}}{a} + c \\ \text{Here } a = 3 \end{array} \right]$$

$$(xii) \int e^{5x+7} dx = \frac{e^{5x+7}}{5} + c \left[\begin{array}{l} \because \int e^{ax+b} dx = \frac{e^{ax+b}}{a} + c \\ \text{Here } a = 5, b = 7 \end{array} \right]$$

$$(xiii) \int a^{3-2x} dx = \frac{a^{3-2x}}{-2 \log |a|} + c \left[\begin{array}{l} \because \int a^{mx+n} dx = \frac{a^{mx+n}}{m \log |a|} + c \\ \text{Here } m = -2, n = 3 \end{array} \right]$$

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$$(xiv) \int (5e^{7x} + x^2)dx = \frac{5}{7}e^{7x} + \frac{x^3}{3} + c$$

$$(xv) \int (a^x + e^x + a^a + e^a + a^e)dx = \frac{a^x}{\log|a|} + e^x + a^a x + e^a x + a^e x + c$$

$$\left[\begin{array}{l} \text{Here } a^a, e^a, a^e \text{ all are constants} \\ \text{and if } k \text{ is constant then } \int k dx = kx + c \end{array} \right]$$

$$(xvi) \int 5^x 2^x dx = \int (5 \cdot 2)^x dx = \int 10^x dx = \frac{10^x}{\log 10} + c \left[\begin{array}{l} \because \int a^x dx = \frac{a^x}{\log|a|} + c \\ \text{Here } a = 10 \end{array} \right]$$

$$(xvii) \int \frac{2^x}{3^x} dx = \int (2/3)^x dx = \frac{(2/3)^x}{\log 2/3} + c \left[\begin{array}{l} \because \int a^x dx = \frac{a^x}{\log|a|} + c \\ \text{Here } a = 2/3 \end{array} \right]$$

$$= \frac{(2/3)^x}{\log 2 - \log 3} + c$$

$$(xviii) \int \frac{(a^x - b^x)^2}{a^x b^x} dx = \int \frac{a^{2x} + b^{2x} - 2a^x b^x}{a^x b^x} dx \quad [\because (a-b)^2 = a^2 + b^2 - 2ab]$$

$$= \int \left(\frac{a^{2x}}{a^x b^x} + \frac{b^{2x}}{a^x b^x} - \frac{2a^x b^x}{a^x b^x} \right) dx = \int \left(\frac{a^x}{b^x} + \frac{b^x}{a^x} - 2 \right) dx$$

$$= \int \left((a/b)^x + (b/a)^x - 2 \right) dx$$

$$= \frac{(a/b)^x}{\log|a/b|} + \frac{(b/a)^x}{\log|b/a|} - 2x + c \quad \left[\because \int m^x dx = \frac{m^x}{\log|m|} + c \right]$$

$$(xix) \int (e^{a \log x} + a^{x \log_a a} + a^{m \log_a a}) dx = \int e^{\log x^a} + a^{\log_a a^x} + a^{\log_a a^m} dx$$

$$= \int (x^a + a^x + a^m) dx \quad [\because a^{\log_a f(x)} = f(x)]$$

$$= \frac{x^{a+1}}{a+1} + \frac{a^x}{\log|a|} + a^m x + c$$

$$\left[\because a^m \text{ is a constant quantity} \right]$$

$$\begin{aligned}
 \text{(xx)} \quad & \int \left[\frac{x}{3} + (5x-3)^3 + x\sqrt{x} + \frac{1}{\sqrt{5+2x}} + a^a \right] dx \\
 &= \int \left[\frac{x}{3} + (5x-3)^3 + x^{3/2} + (5+2x)^{-1/2} + a^a \right] dx \\
 &= \frac{x^2}{3 \times 2} + \frac{(5x-3)^4}{4 \times 5} + \frac{x^{5/2}}{5/2} + \frac{(5+2x)^{1/2}}{2 \times 1/2} + a^a x + c \\
 &= \frac{x^2}{6} + \frac{(5x-3)^4}{20} + \frac{2}{5} x^{5/2} + (5+2x)^{1/2} + a^a x + c
 \end{aligned}$$

Example 3: Evaluate the following integrals:

$$\text{(i)} \int \frac{1}{x} dx \quad \text{(ii)} \int \frac{3}{x} dx \quad \text{(iii)} \int \frac{5}{x+1} dx \quad \text{(iv)} \int \frac{7}{5x+2} dx \quad \text{(v)} \int \frac{3}{9-2x} dx$$

Solution:

$$\text{(i)} \quad \int \frac{1}{x} dx = \log|x| + c \quad [\text{Using formula 3 of the table}]$$

$$\text{(ii)} \quad \int \frac{3}{x} dx = 3 \int \frac{1}{x} dx = 3 \log|x| + c \quad [\text{Using formula 3 of the table}]$$

$$\text{(iii)} \quad \int \frac{5}{x+1} dx = 5 \log|x+1| + c \quad [\text{Using formula 5 of the table}]$$

$$\begin{aligned}
 \text{(iv)} \quad \int \frac{7}{5x+2} dx &= \frac{7 \log|5x+2|}{5} + c \quad [\text{Using formula 5 of the table}] \\
 &= \frac{7}{5} \log|5x+2| + c
 \end{aligned}$$

$$\begin{aligned}
 \text{(v)} \quad \int \frac{3}{9-2x} dx &= \frac{3 \log|9-2x|}{(-2)} + c \quad [\text{Using formula 5 of the table}] \\
 &= -\frac{3}{2} \log|9-2x| + c
 \end{aligned}$$

Remark 2: In solving these examples you have noted that integration is in fact anti derivative of a function.

For example, consider (ix) part of Example 1

$$\text{Let } f(x) = x^3 + x + 5 \text{ then } \int f(x) dx = \frac{x^4}{4} + \frac{x^2}{2} + 5x + c \text{ (already calculated)}$$

Indefinite Integration

$$\text{Now, let } F(x) = \frac{x^4}{4} + \frac{x^2}{2} + 5x + c$$

Diff. w.r.t.x

$$\frac{d}{dx}(F(x)) = \frac{4x^3}{4} + \frac{2x}{2} + 5 + 0 = x^3 + x + 5$$

Thus, we note that if $\int f(x)dx = F(x)$ then $\frac{d}{dx}(F(x)) = f(x)$

i.e. integral $F(x)$ of $f(x)$ is indefinite because of the presence of arbitrary constant c .

In the next unit you will meet definite integral, where constant c will be cancel out.

13.4 INTEGRATION BY SUBSTITUTION

In section 13.3, we have taken into consideration the integrations for which a formula can directly be used. But sometimes integrand cannot be directly integrated using standard formula. In order to convert it into a form for which standard formula can be applied, we substitute some function in place of some other function and this technique of obtaining the integration is known as integration by substitution method.

Substitution Method

If integral is of the type $\int (f(x))^n f'(x) dx$ or $\int \frac{f'(x)}{(f(x))^n} dx$, then

Step I We substitute $f(x) = t$... (1)

Step II Differentiate on both sides of (1)

Step III Change the given integral in terms of t

Step IV After simplification, if necessary, we get one of the standard forms discussed in Sec. 13.3. Using appropriate formula we can obtain the integral of given integrand in terms of t .

Step V Replace t in terms of x , we get the desired result after simplification, if required.

Following example will explain the substitution method and the steps involved in it:

Example 4: Evaluate the following integrals:

- (i) $\int \frac{x^9}{x^{10}+1} dx$ (ii) $\int \frac{x^{n-1}}{x^n+a} dx$ (iii) $\int \frac{e^x}{e^x+5} dx$
- (iv) $\int \frac{1}{x \log x} dx$ (v) $\int \frac{2ax+b}{ax^2+bx+c} dx$ (vi) $\int \frac{8x^3+4x}{(x^4+x^2+1)^6} dx$
- (vii) $\int \frac{e^{2x}+e^{-2x}}{e^{2x}-e^{-2x}} dx$ (viii) $\int \frac{1}{x+\sqrt{x}} dx$ (ix) $\int (2ax+b)\sqrt{ax^2+bx+c} dx$
- (x) $\int \frac{2x}{(1+x^2)\log(1+x^2)} dx$

Solution:

(i) Let $I = \int \frac{x^9}{x^{10}+1} dx \quad \dots (1)$

Putting $x^{10}+1 = t$

Differentiating

$$10x^9 dx = dt \Rightarrow x^9 dx = \frac{dt}{10}$$

$\therefore (1)$ becomes

$$I = \int \frac{1}{t} \frac{dt}{10} = \frac{1}{10} \int \frac{dt}{t} = \frac{1}{10} \log|t| + c \quad \left[\because \int \frac{1}{y} dy = \log|y| + c \right]$$

$$= \frac{1}{10} \log|x^{10}+1| + c$$

[Replacing t in terms of x]

$$= \frac{1}{10} \log(x^{10}+1) + c$$

$\left[\because x^{10}+1 \text{ cannot be -ve for real } x \right]$

Alternatively: We can also put

$$x^{10} = t$$

Differentiating

$$10x^9 dx = dt \Rightarrow x^9 dx = \frac{dt}{10}$$

$\therefore (1)$ becomes

$$I = \int \frac{1}{t+1} \times \frac{dt}{10} = \frac{1}{10} \int \frac{1}{t+1} dt$$

$$= \frac{1}{10} \log|t+1| + c$$

$$\left[\because \int \frac{1}{x+a} dx = \log|x+a| + c \right]$$

Indefinite Integration

$$= \frac{1}{10} \log |x^{10} + 1| + c \quad [\text{Replacing } t \text{ in terms of } x]$$

$$= \frac{1}{10} \log(x^{10} + 1) + c \quad \left[\because x^{10} + 1 \text{ is always +ve for real } x \right]$$

(ii) Let $I = \int \frac{x^{n-1}}{x^n + a} dx \quad \dots (1)$

Putting $x^n + a = t$

Differentiating

$$n x^{n-1} dx = dt \Rightarrow x^{n-1} dx = \frac{dt}{n}$$

$\therefore$ (1) becomes

$$I = \frac{1}{n} \int \frac{dt}{t} = \frac{1}{n} \log|t| + c \quad \left[\because \int \frac{1}{x} dx = \log|x| + c \right]$$

$$= \frac{1}{n} \log|x^n + a| + c \quad [\text{Replacing } t \text{ in terms of } x]$$

(iii) Let $I = \int \frac{e^x}{e^x + 5} dx \quad \dots (1)$

Putting $e^x + 5 = t$

Differentiating

$$e^x dx = dt$$

$\therefore$ (1) becomes

$$I = \int \frac{dt}{t} = \log|t| + c = \log|e^x + 5| + c \quad \left[\because \int \frac{1}{x} dx = \log|x| + c \right]$$

(iv) Let $I = \int \frac{1}{x \log x} dx \quad \dots (1)$

Putting $\log x = t$

Differentiating

$$\frac{1}{x} dx = dt$$

$\therefore$ (1) becomes

$$I = \int \frac{dt}{t} = \log|t| + c \quad \left[\because \int \frac{1}{x} dx = \log|x| + c \right]$$

$$= \log|\log x| + c \quad [\text{Replacing } t \text{ in terms of } x]$$

(v) Let $I = \int \frac{2ax + b}{ax^2 + bx + c} dx \quad \dots (1)$

Putting $ax^2 + bx + c = t$

Differentiating

$$(2ax + b)dx = dt$$

$\therefore$ (1) becomes

$$I = \int \frac{dt}{t} = \log|t| + k = \log|ax^2 + bx + c| + k$$

where k is constant of integration

(vi) Let $I = \int \frac{8x^3 + 4x}{(x^4 + x^2 + 1)^6} dx = 2 \int \frac{4x^3 + 2x}{(x^4 + x^2 + 1)^6} dx \quad \dots (1)$

Putting $x^4 + x^2 + 1 = t$

Differentiating

$$(4x^3 + 2x)dx = dt$$

$\therefore$ (1) becomes

$$\begin{aligned} I &= 2 \int \frac{dt}{t^6} = 2 \int t^{-6} dt = 2 \times \frac{t^{-5}}{-5} + c \\ &= \frac{-2}{5} (x^4 + x^2 + 1)^{-5} + c \quad [\text{Replacing } t \text{ in terms of } x] \end{aligned}$$

(vii) Let $I = \int \frac{e^{2x} + e^{-2x}}{e^{2x} - e^{-2x}} dx \quad \dots (1)$

Putting $e^{2x} - e^{-2x} = t$

Differentiating

$$(2e^{2x} + 2e^{-2x})dx = dt \Rightarrow (e^{2x} + e^{-2x})dx = \frac{dt}{2}$$

$\therefore$ (1) becomes

$$\begin{aligned} I &= \frac{1}{2} \int \frac{dt}{t} = \frac{1}{2} \log|t| + c \quad \left[\because \int \frac{1}{x} dx = \log|x| + c \right] \\ &= \frac{1}{2} \log|e^{2x} - e^{-2x}| + c \quad [\text{Replacing } t \text{ in terms of } x] \end{aligned}$$

(viii) Let $I = \int \frac{1}{x + \sqrt{x}} dx = \int \frac{1}{\sqrt{x}(\sqrt{x} + 1)} dx \quad \dots (1)$

Putting $\sqrt{x} + 1 = t$

Indefinite Integration

Differentiating

$$\frac{1}{2\sqrt{x}} dx = dt \Rightarrow \frac{dx}{\sqrt{x}} = 2dt$$

$\therefore$ (1) becomes

$$I = 2 \int \frac{dt}{t} = 2 \log|t| + c = 2 \log|\sqrt{x} + 1| + c$$

$$(ix) \text{ Let } I = \int (2ax + b) \sqrt{ax^2 + bx + c} dx \quad \dots (1)$$

Putting $ax^2 + bx + c = t$

Differentiating

$$(2ax + b)dx = dt$$

$\therefore$ (1) becomes

$$I = \int \sqrt{t} dt = \frac{t^{3/2}}{3/2} + k = \frac{2}{3} (ax^2 + bx + c)^{3/2} + k,$$

where k is constant of integration

$$(x) \text{ Let } I = \int \frac{2x}{(1+x^2)\log(1+x^2)} dx \quad \dots (1)$$

Putting $\log(1+x^2) = t$

Differentiating

$$\frac{1}{1+x^2} 2x dx = dt$$

$\therefore$ (1) becomes

$$I = \int \frac{dt}{t} = \log|t| + c = \log|\log(1+x^2)| + c \text{ [Replacing t in terms of x]}$$

Check Your Progress – 1

1) Evaluate the following integrals:

$$(i) \int \left(x^2 + \frac{1}{x^2}\right)^2 dx \quad (ii) \int \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^3 dx \quad (iii) \int (\alpha - 3) dx$$

$$(iv) \int \sqrt{x} \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right) dx \quad (v) \int (x^a + 1)(x^b + 1) dx \quad (vi) \int \frac{x^m}{x^n} dx$$

$$(vii) \int \left(x + \frac{1}{x}\right) \left(x^3 + \frac{1}{x^3}\right) dx$$

2) Evaluate the following integral:

$$(i) \int \left(a^x + e^x a^x + \frac{x}{a} \right) dx \quad (ii) \int (3^{2 \log_3 x} + 3^{x \log_3 a} + a^{a \log_a x} + a^{a \log_a a}) dx$$

3) Evaluate the following integrals:

$$(i) \int \frac{2x+1}{(x^2+x+7)^5} dx \quad (ii) \int x\sqrt{x+ax} dx$$

$$(iii) \int \frac{x}{\sqrt{x+a}} dx \quad (iv) \int \frac{1}{(1+x) \log(1+x)} dx$$

13.5 INTEGRATION USING PARTIAL FRACTIONS

The integrand may be in the form that it can be integrated only after resolving it into partial fractions. Here, in this section, we are going to deal with integration of such functions:

First of all we discuss the process of resolving such functions into partial fractions:

Important steps for resolving into partial fractions are:

1. Check degree of numerator, if it is less than that of denominator, go to step 2 and if it is greater than or equal to that of denominator, then first divide the numerator by the denominator and then go to step 2.
2. We may have one of the following main types of functions which we will deal as discussed below:

Indefinite Integration

Type 1 Denominator involve all linear factors with exponent as unity.

e.g. $\frac{x+5}{(x-1)(x-2)(x-3)}$.

Step I Let $\frac{x+5}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3} \dots (1)$

Step II Equate each of the factors of denominator to zero.

i.e. $x-1=0 \Rightarrow x=1$, $x-2=0 \Rightarrow x=2$, $x-3=0 \Rightarrow x=3$

Step III Put $x = 1, 2, 3$ every where (in the given expression) but not in the factor from which it has come out,

$$A = \frac{1+5}{(1-2)(1-3)} = \frac{6}{2} = 3, \quad [\text{By putting } x = 1 \text{ in L.H.S. of (1)}]$$

Step IV $B = \frac{2+5}{(2-1)(2-3)} = \frac{7}{-1} = -7, \quad [\text{By putting } x = 2 \text{ in L.H.S. of (1)}]$

Step V and $C = \frac{3+5}{(3-1)(3-2)} = \frac{8}{2 \times 1} = 4 \quad [\text{By putting } x = 3 \text{ in L.H.S. of (1)}]$

Thus, we may write $\frac{x+5}{(x-1)(x-2)(x-3)} = \frac{3}{x-1} + \frac{-7}{x-2} + \frac{4}{x-3}$

R.H.S. is nothing but the partial fractions of the given expression. Here we note that integration of R.H.S. is directly available, as we will see in the Example 5 of this unit.

Type 2 Denominator involves all linear factors but some have 2, 3, etc. as exponents

e.g. $\frac{x^2 + x + 5}{(x+5)(x+2)^3}$

Step I Let $\frac{x^2 + x + 5}{(x+5)(x+2)^3} = \frac{A}{x+5} + \frac{B}{x+2} + \frac{C}{(x+2)^2} + \frac{D}{(x+2)^3}$

Multiply on both sides by denominator of L.H.S. in this case

by $(x+5)(x+2)^3$, we get

$$x^2 + x + 5 = A(x+2)^3 + B(x+5)(x+2)^2 + C(x+5)(x+2) + D(x+5) \dots (1)$$

Step II Equate each of the factors to zero.

i.e. $x+5=0 \Rightarrow x=-5$, $x+2=0 \Rightarrow x=-2$

Step III Put $x = -5$ in (1) we get value of A, as given below

$$(-5)^2 + (-5) + 5 = A(-5+2)^3 + B(0) + C(0) + D(0)$$

$$\Rightarrow 25 = -27A \Rightarrow \boxed{A = -\frac{25}{27}}$$

Step IV Put $x = -2$ in (1) we get value of D, as given below

$$(-2)^2 + (-2) + 5 = A(0) + B(0) + C(0) + D(-2+5)$$

$$\Rightarrow 7 = 3D \Rightarrow \boxed{D = \frac{7}{3}}$$

Step V In order to find the values of B, C we have to equate the coefficients of different powers of x on both sides of (1).

In present case equating coefficients of x^3 and constant terms, we get

$$0 = A + B \quad \dots (2)$$

$$5 = 8A + 20B + 10C + 5D \quad \dots (3)$$

By putting value of A from Step III and value of D from step IV in equations (2) and (3), we get.

$$0 = -\frac{25}{27} + B = 0 \Rightarrow \boxed{B = \frac{25}{27}}$$

$$5 = 8\left(-\frac{25}{27}\right) + 20B + 10C + 5\left(\frac{7}{3}\right)$$

$$\Rightarrow 10C = 5 + \frac{200}{27} - 20\left(\frac{25}{27}\right) - \frac{35}{3} \Rightarrow \boxed{C = -48}$$

Thus, we may write

$$\frac{x^2 + x + 5}{(x+5)(x+2)^3} = \frac{-25/27}{x+5} + \frac{25/27}{x+2} + \frac{-48}{(x+2)^2} + \frac{7/3}{(x+2)^3}$$

R.H.S. is nothing but the partial fractions of the given expression. Here we note that integration of R.H.S. is directly available, as we will see in Example 5 of this unit.

Type 3 Denominator involves quadratic expressions. We will not discuss the problems based on this type, because it will involve the integral formulae which are beyond our contents.

Type 4 Denominator involves higher powers of quadratic expressions.

This type is also not discussed here because it will involve the integral formulae which are beyond our contents.

Following example will illustrate how these types are used in evaluating integrals.

Example 5: Evaluate the following integrals:

$$(i) \int \frac{4x+1}{(x-1)(x-2)} dx \quad (ii) \int \frac{3x+4}{x^2-x-12} dx \quad (iii) \int \frac{8x}{(x+1)(x-3)^2} dx$$

$$(iv) \int \frac{x^2+x+2}{(x+2)(x+1)^3} dx \quad (v) \int \frac{x^2+1}{x^2-1} dx$$

Solution:

$$(i) \text{ Let } I = \int \frac{4x+1}{(x-1)(x-2)} dx$$

$$= \int \left[\frac{-5}{x-1} + \frac{9}{x-2} \right] dx$$

$$= -5 \log|x-1| + 9 \log|x-2| + c$$

Using type 1 procedure as already discussed. Put $x=1$ every where except in $x-1$ and $x=2$ every where except in $x-2$, we have

$$A = \frac{4.1+1}{1-2} = -5, B = \frac{4.2+1}{2-1} = 9$$

$$(ii) \text{ Let } I = \int \frac{3x+4}{x^2-x-12} dx = \int \frac{3x+4}{(x-4)(x+3)} dx$$

$$= \int \left(\frac{16/7}{x-4} + \frac{5/7}{x+3} \right) dx$$

[Using partial fractions as discussed in type 1]

$$= \frac{16}{7} \log|x-4| + \frac{5}{7} \log|x+3| + c$$

$$(iii) \text{ Let } I = \int \frac{8x}{(x+1)(x-3)^2} dx \quad \dots (1)$$

Let us first resolve into partial fractions

$$\text{Let } \frac{8x}{(x+1)(x-3)^2} = \frac{A}{x+1} + \frac{B}{x-3} + \frac{C}{(x-3)^2}$$

Multiplying on both sides by $(x+1)(x-3)^2$, we get

$$8x = A(x-3)^2 + B(x+1)(x-3) + C(x+1) \quad \dots (2)$$

Putting $x = -1$ in (2), we get $[\because x + 1 = 0 \text{ gives } x = -1]$

$$-8 = A(-1-3)^2 + B(0) + C(0) \Rightarrow -8 = 16A \Rightarrow \boxed{A = -\frac{1}{2}}$$

Putting $x = 3$ in (2), we get $[\because x - 3 = 0 \text{ gives } x = 3]$

$$24 = A(0) + B(0) + C(3+1) \Rightarrow 24 = 4C \Rightarrow \boxed{C = 6}$$

Comparing coefficient of x^2 on both sides of (2), we get

$$0 = A + B \Rightarrow B = -A \Rightarrow \boxed{B = \frac{1}{2}}$$

$$\therefore I = \int \left(\frac{-1/2}{x+1} + \frac{1/2}{x-3} + \frac{6}{(x-3)^2} \right) dx = \int \left[\frac{-1/2}{x+1} + \frac{1/2}{x-3} + 6(x-3)^{-2} \right] dx$$

$$= -\frac{1}{2} \log|x+1| + \frac{1}{2} \log|x-3| + 6 \frac{(x-3)^{-1}}{-1} + c$$

$$= -\frac{1}{2} \log|x+1| + \frac{1}{2} \log|x-3| - \frac{6}{x-3} + c$$

(iv) Let $I = \int \frac{x^2 + x + 2}{(x+2)(x+1)^3} dx$

Let us first resolve into partial fractions

$$\text{Let } \frac{x^2 + x + 2}{(x+2)(x+1)^3} = \frac{A}{x+2} + \frac{B}{x+1} + \frac{C}{(x+1)^2} + \frac{D}{(x+1)^3}$$

Multiplying on both sides by $(x+2)(x+1)^3$, we get

$$x^2 + x + 2 = A(x+1)^3 + B(x+2)(x+1)^2 + C(x+2)(x+1) + D(x+2) \dots (2)$$

Putting $x = -2$ in (2), we get $[\because x + 2 = 0 \text{ gives } x = -2]$

$$(-2)^2 + (-2) + 2 = A(-2+1)^2 + B(0) + C(0) + D(0)$$

$$\Rightarrow 4 = A \Rightarrow \boxed{A = 4}$$

Putting $x = -1$ in (2), we get $[\because x + 1 = 0 \text{ gives } x = -1]$

$$(-1)^2 + (-1) + 2 = A(0) + B(0) + C(0) + D(-1+2)$$

$$\Rightarrow 2 = D \Rightarrow \boxed{D = 2}$$

Comparing coefficients of x^3 and constant terms on both sides of (2), we get

$$0 = A + B \Rightarrow B = -A \Rightarrow \boxed{B = -4}$$

$$2 = A + 2B + 2C + 2D \Rightarrow 2C = 2 - A - 2B - 2D$$

$$\Rightarrow 2C = 2 - 4 + 8 - 4$$

$$\Rightarrow 2C = 2$$

$$\Rightarrow \boxed{C=1}$$

$$\begin{aligned}\therefore I &= \int \left[\frac{4}{x+2} + \frac{-4}{x+1} + \frac{1}{(x+1)^2} + \frac{2}{(x+1)^3} \right] dx \\ &= 4 \log|x+2| - 4 \log|x+1| + \frac{(x+1)^{-1}}{-1} + \frac{2(x+1)^{-2}}{-2} + c \\ &= 4 \log|x+2| - 4 \log|x+1| - \frac{1}{x+1} - \frac{1}{(x+1)^2} + c\end{aligned}$$

$$\begin{aligned}\text{(v) Let } I &= \int \frac{x^2+1}{x^2-1} dx = \int \left[1 + \frac{2}{x^2-1} \right] dx & \frac{x^2-1}{x^2-1} \sqrt{\frac{x^2+1}{x^2-1}} \\ &= \int \left[1 + \frac{2}{(x-1)(x+1)} \right] dx & \frac{-}{+} \frac{1}{2} \\ &= \int \left[1 + 2 \left(\frac{1/2}{x-1} + \frac{-1/2}{x+1} \right) \right] dx & \left[\begin{array}{l} \text{Using partial fractions} \\ \text{as discussed in type 1} \end{array} \right] \\ &= x + \log|x-1| - \log|x+1| + c \\ &= x + \log \left| \frac{x-1}{x+1} \right| + c & [\because \log m - \log n = \log \frac{m}{n}]\end{aligned}$$

13.6 INTEGRATION BY PARTS

If u and v are any two functions of a single variable x such that first derivatives of u and v w.r.t. x exist, then by product rule, we have

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

Integrating on both sides, we have

$$uv = \int u \frac{dv}{dx} dx + \int v \frac{du}{dx} dx$$

$$\Rightarrow \int u \frac{dv}{dx} dx = uv - \int \frac{du}{dx} v dx \quad \dots (1)$$

$$\text{Let } u = f(x) \text{ and } \frac{dv}{dx} = g(x) \quad \dots (2)$$

$$\Rightarrow \frac{du}{dx} = f'(x) \text{ and } v = \int g(x)dx \quad \dots (3)$$

Using (2) and (3) in (1), we get

$$\int f(x) g(x)dx = f(x) \int g(x)dx - \int \left[f'(x) \int g(x)dx \right] dx$$

$$\text{Or } \int I II \, dx = I \int II \, dx - \int \left[\frac{d}{dx}(I) \int II \, dx \right] dx \quad \dots (4)$$

where I = first function = f(x)

II = second function = g(x)

R.H.S. of equation (4) is known as integration by parts of L.H.S. of equation (4), where I, and II just indicate our choice between the product of two functions taking as first and second functions.

Remark 3:

- (i) In case of integration by parts, choice of first and second function is important as explained in part (i) of Example 6 given below.
- (ii) If in the product of two functions, one is polynomial function then we take polynomial function as first function.
- (iii) Integration by parts is one of the methods (techniques) of integration. It does not mean that integration of product of any two functions exists.

Example 6: Evaluate the following integrals:

$$(i) \int x e^x dx \quad (ii) \int x^2 e^{3x} dx \quad (iii) \int x^3 a^x dx \quad (iv) \int \log x dx$$

Solution:

$$(i) \text{ Let } I = \int x e^x dx$$

I II

Integrating by parts (taking x as first and e^x as second function)

$$I = x \int e^x dx - \int \left[\left(\frac{d}{dx}(x) \right) \left(\int e^x dx \right) \right] dx + c_1$$

where c_1 is constant of integration

$$= x e^x - \int (1)(e^x) dx + c_1 = x e^x - \int e^x dx + c_1 = x e^x - (e^x + c_2) + c_1$$

where c_2 is constant of integration

Indefinite Integration

$$= xe^x - e^x + c, \text{ where } c = c_1 - c_2$$

Let us see what happens if we integrate by parts by taking x as second and e^x as first function:

$$\begin{aligned} I &= e^x \int x dx - \int \left[\left(\frac{d}{dx}(e^x) \right) \left(\int x dx \right) \right] dx + c_1 \\ &= e^x \frac{x^2}{2} - \int \left[e^x \frac{x^2}{2} \right] dx + c_1 = \frac{x^2 e^x}{2} - \frac{1}{2} \int x^2 e^x dx + c_1 \end{aligned}$$

We see that integration becomes more complicated. So choice of first and second function is important.

Note: In future we will add c as constant of integration only once.

(ii) Let $I = \int x^2 e^{3x} dx$

Integrating by parts (taking x^2 as first and e^{3x} as second function)

$$\begin{aligned} I &= x^2 \left(\frac{e^{3x}}{3} \right) - \int (2x) \left(\frac{e^{3x}}{3} \right) dx + c \\ &\text{where } c \text{ is constant of integration} \\ &= \frac{x^2 e^{3x}}{3} - \frac{2}{3} \int x e^{3x} dx + c \end{aligned}$$

Again integrating by parts (taking x as first and e^{3x} as second function)

$$\begin{aligned} I &= \frac{x^2 e^{3x}}{3} - \frac{2}{3} \left[x \left(\frac{e^{3x}}{3} \right) - \int (1) \left(\frac{e^{3x}}{3} \right) dx \right] + c \\ &= \frac{x^2 e^{3x}}{3} - \frac{2}{3} \left[\frac{x e^{3x}}{3} - \frac{1}{3} \int e^{3x} dx \right] + c \\ &= \frac{x^2 e^{3x}}{3} - \frac{2}{3} \left(\frac{x e^{3x}}{3} - \frac{e^{3x}}{9} \right) + c = \frac{1}{3} x^2 e^{3x} - \frac{2}{9} x e^{3x} + \frac{2}{27} e^{3x} + c \end{aligned}$$

(iii) Let $I = \int x^3 a^x dx$

I II

Integrating by parts (taking x^3 as first and a^x as second function)

$$I = (x^3) \left(\frac{a^x}{\log a} \right) - \int (3x^2) \left(\frac{a^x}{\log a} \right) dx + c = \frac{x^3 a^x}{\log a} - \frac{3}{\log a} \int x^2 a^x dx + c$$

Again integrating by parts (taking x^2 as first and a^x as second function)

$$\begin{aligned} I &= \frac{x^3 a^x}{\log a} - \frac{3}{\log a} \left[(x^2) \left(\frac{a^x}{\log a} \right) - \int (2x) \left(\frac{a^x}{\log a} \right) dx \right] + c \\ &= \frac{x^3 a^x}{\log a} - \frac{3x^2 a^x}{(\log a)^2} + \frac{6}{(\log a)^2} \int x a^x dx + c \end{aligned}$$

I II

Again integrating by parts (taking x as first and a^x as second function)

$$\begin{aligned} &= \frac{x^3 a^x}{\log a} - \frac{3x^2 a^x}{(\log a)^2} + \frac{6}{(\log a)^2} \left[(x) \left(\frac{a^x}{\log a} \right) - \int (1) \left(\frac{a^x}{\log a} \right) dx \right] + c \\ &= \frac{x^3 a^x}{\log a} - \frac{3x^2 a^x}{(\log a)^2} + \frac{6}{(\log a)^2} \left[\frac{x a^x}{\log a} - \frac{a^x}{(\log a)^2} \right] + c \\ &= \frac{x^3 a^x}{\log a} - \frac{3x^2 a^x}{(\log a)^2} + \frac{6x a^x}{(\log a)^3} - \frac{6a^x}{(\log a)^4} + c \end{aligned}$$

$$(iv) \text{ Let } I = \int \log x \, dx = \int 1 \times \log x \, dx$$

II I

Integrating by parts (taking $\log x$ as first and 1 as second function)

$$I = (\log x)(x) - \int \left(\frac{1}{x} \right) (x) dx + c = x \log x - \int 1 dx + c = x \log x - x + c$$

Check Your Progress – 2

1) Evaluate the following integrals:

$$(i) \int \frac{3x+2}{(x-1)(x-2)(x-3)} dx \quad (ii) \int \frac{x^3+5x+1}{x^2-4} dx$$

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.....

.....

2) Evaluate the following integrals:

$$(i) \int x^2 e^{-x} dx \quad (ii) \int x^3 e^{x^2} dx$$

.....

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13.7 SUMMARY

This unit introduced integration as the inverse process of differentiation, defining it both as a method for finding antiderivatives and as a tool for calculating the area under a curve. It establishes essential terminology, such as the integrand, the constant of integration (C), and the distinction between indefinite and definite integrals, while providing a library of standard integrals for particular functions, such as polynomial, trigonometric, and exponential forms. In this unit, it shown how to find the integral of some functions such as constant, $x^n, \frac{1}{x^n}$, polynomial, $(ax+b)^n, \frac{1}{ax+b}, (ax+b)^n, \frac{1}{ax+b}, (ax+b)^n, \frac{1}{ax+b}$, and exponential whose integral are directly available. To handle more complex expressions, the unit details three core techniques: Integration by Substitution, which simplifies integrals by changing variables; Integration using Partial Fractions, which decomposes rational functions into simpler, integrable parts; and Integration by Parts, which uses the product rule in reverse to evaluate the integral of a product of two functions.

13.8 SOLUTIONS/ANSWERS TO CHECK YOUR PROGRESS

Check Your Progress – 1

$$1) (i) \int \left(x^2 + \frac{1}{x^2} \right)^2 dx = \int \left(x^4 + \frac{1}{x^4} + 2x^2 \cdot \frac{1}{x^2} \right) dx = \int \left(x^4 + \frac{1}{x^4} + 2 \right) dx$$

$$= \int (x^4 + x^{-4} + 2) dx$$

$$= \frac{x^5}{5} + \frac{x^{-3}}{-3} + 2x + c \quad \left[\begin{array}{l} \because \int x^n dx = \frac{x^{n+1}}{n+1} + c \\ \text{and } \int k dx = kx + c \end{array} \right]$$

$$= \frac{x^5}{5} - \frac{1}{3x^3} + 2x + c$$

$$(ii) \int \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^3 dx = \int \left[x^{3/2} + \frac{1}{x^{3/2}} + 3\sqrt{x} \cdot \frac{1}{\sqrt{x}} \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right) \right] dx$$

$$= \int (x^{3/2} + x^{-3/2} + 3x^{1/2} + 3x^{-1/2}) dx$$

$$= \frac{x^{5/2}}{5/2} + \frac{x^{-1/2}}{-1/2} + \frac{3x^{3/2}}{3/2} + \frac{3x^{1/2}}{1/2} + c$$

$$= \frac{2}{5} x^{5/2} - \frac{2}{\sqrt{x}} + 2x^{3/2} + 6\sqrt{x} + c \quad \left[\because \int x^n dx = \frac{x^{n+1}}{n+1} + c \right]$$

$$(iii) \int (\alpha - 3) dx = (\alpha - 3)x + c \quad [\because \alpha - 3 \text{ is a constant}]$$

$$(iv) \int \sqrt{x} \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right) dx = \int (x + 1) dx = \frac{x^2}{2} + x + c$$

$$(v) \int (x^a + 1)(x^b + 1) dx = \int (x^{a+b} + x^a + x^b + 1) dx + c$$

$$= \frac{x^{a+b+1}}{a+b+1} + \frac{x^{a+1}}{a+1} + \frac{x^{b+1}}{b+1} + x + c,$$

$$\left[\because \int x^n dx = \frac{x^{n+1}}{n+1} \text{ and } \int k dx = kx, \text{ where } k \text{ is constant} \right]$$

where $a \neq -1$, $b \neq -1$, $a + b \neq -1$

Indefinite Integration

$$(vi) \int \frac{x^m}{x^n} dx = \int x^{m-n} dx = \frac{x^{m-n+1}}{m-n+1} + c, m-n \neq -1 \quad \left[\because \int x^n dx = \frac{x^{n+1}}{n+1} \right]$$

$$\begin{aligned}(vii) \int \left(x + \frac{1}{x} \right) \left(x^3 + \frac{1}{x^3} \right) dx &= \int \left(x^4 + \frac{1}{x^2} + x^2 + \frac{1}{x^4} \right) dx \\&= \int (x^4 + x^{-2} + x^2 + x^{-4}) dx \\&= \frac{x^5}{5} + \frac{x^{-1}}{-1} + \frac{x^3}{3} + \frac{x^{-3}}{-3} + c \quad \left[\because \int x^n dx = \frac{x^{n+1}}{n+1} \right] \\&= \frac{x^5}{5} - \frac{1}{x} + \frac{x^3}{3} - \frac{1}{3x^3} + c\end{aligned}$$

$$2) (i) \int \left(a^x + e^x a^x + \frac{x}{a} \right) dx = \int \left(a^x + (ea)^x + \frac{x}{a} \right) dx = \frac{a^x}{\log a} + \frac{(ea)^x}{\log ea} + \frac{x^2}{2a} + c$$

$$\begin{aligned}(ii) \int \left(3^{2 \log_3 x} + 3^{x \log_3 a} + a^{a \log_a x} + a^{a \log_a a} \right) dx \\&= \int \left(3^{\log_3 x^2} + 3^{\log_3 a^x} + a^{\log_a x^a} + a^{\log_a a^a} \right) dx \\&= \int \left(x^2 + a^x + x^a + a^a \right) dx \quad [\because a^{\log_a f(x)} = f(x)] \\&= \frac{x^3}{3} + \frac{a^x}{\log |a|} + \frac{x^{a+1}}{a+1} + a^a x + c\end{aligned}$$

$$3) (i) \text{ Let } I = \int \frac{2x+1}{(x^2+x+7)^5} dx \quad \dots (1)$$

$$\text{Putting } x^2 + x + 7 = t$$

Differentiating

$$(2x+1)dx = dt$$

$\therefore$ (1) becomes

$$I = \int \frac{dt}{t^5} = \int t^{-5} dt = \frac{t^{-4}}{-4} + c = \frac{-1}{4t^4} + c = -\frac{1}{4(x^2+x+7)^4} + c$$

$$(ii) \text{ Let } I = \int x\sqrt{x+a} dx \quad \dots (1)$$

$$\text{Putting } \sqrt{x+a} = t \Rightarrow x+a = t^2$$

Differentiating

$$dx = 2t dt$$

$\therefore$ (1) becomes

$$I = \int (t^2 - a)t(2t)dt = 2 \int (t^4 - at^2)dt = 2 \left(\frac{t^5}{5} - \frac{at^3}{3} \right) + c$$

$$= 2 \left[\frac{(x+a)^{5/2}}{5} - \frac{a}{3}(x+a)^{3/2} \right] + c$$

(iii) Let $I = \int \frac{x}{\sqrt{x+a}} dx \quad \dots (1)$

Putting $\sqrt{x+a} = t \Rightarrow x+a = t^2$

Differentiating

$$dx = 2t dt$$

$\therefore$ (1) becomes

$$I = \int \frac{(t^2 - a)}{t} 2t dt = 2 \int (t^2 - a) dt = 2 \left(\frac{t^3}{3} - at \right) + c$$

$$= 2 \left(\frac{(x+a)^{3/2}}{3} - a\sqrt{x+a} \right) + c$$

(iv) Let $I = \int \frac{1}{(1+x)\log(1+x)} dx \quad \dots (1)$

Putting $\log(1+x) = t$

Differentiating

$$\frac{1}{1+x} dx = dt$$

$\therefore$ (1) becomes

$$I = \int \frac{dt}{t} = \log|t| + c = \log|\log(1+x)| + c$$

Check Your Progress – 2

1) (i) Let $I = \int \frac{3x+2}{(x-1)(x-2)(x-3)} dx = \int \left[\frac{5/2}{x-1} + \frac{-8}{x-2} + \frac{11/2}{x-3} \right] dx$

$$\left[\begin{array}{l} \text{Using partial fractions as discussed in type 1, we get} \\ A = \frac{3.1+2}{(1-2)(1-3)} = \frac{5}{2}, B = \frac{3.2+2}{(2-1)(2-3)} = -8, C = \frac{3.3+2}{(3-1)(3-2)} = \frac{11}{2} \end{array} \right]$$

Indefinite Integration

$$= \frac{5}{2} \log|x-1| - 8 \log|x-2| + \frac{11}{2} \log|x-3| + c$$

$$(ii) \text{ Let } I = \int \frac{x^3 + 5x + 1}{x^2 - 4} dx \quad \dots (1)$$

Dividing numerator by denominator, we can write (1) as

$$I = \int \left[x + \frac{9x+1}{x^2-4} \right] dx = \int \left[x + \frac{9x+1}{(x-2)(x+2)} \right] dx \quad \begin{array}{r} x^2-4 \overline{) x^3+5x+1} \\ \underline{x^3-4x} \\ 9x+1 \end{array}$$

$$I = \int \left[x + \frac{19/4}{x-2} + \frac{17/4}{x+2} \right] dx \quad \left[\begin{array}{l} \text{Using partial fractions} \\ \text{as discussed in type 1} \end{array} \right]$$

$$= \frac{x^2}{2} + \frac{19}{4} \log|x-2| + \frac{17}{4} \log|x+2| + c \quad \left[\because \int \frac{1}{x+a} dx = \log|x+a| \right]$$

$$= \frac{x^2}{2} + \frac{1}{4} (19 \log|x-2| + 17 \log|x+2|) + c$$

$$= \frac{x^2}{2} + \frac{1}{4} (\log|x-2|^{19} + \log|x+2|^{17}) + c \quad [\because n \log m = \log m^n]$$

$$= \frac{x^2}{2} + \frac{1}{4} \log|(x-2)^{19}(x+2)^{17}| + c \quad [\because \log m + \log n = \log mn]$$

$$2) (i) \text{ Let } I = \int x^2 e^{-x} dx$$

I II

Integrating by parts (taking x^2 as first and e^{-x} as second function)

$$I = (x^2) \left(\frac{e^{-x}}{-1} \right) - \int (2x) \left(\frac{e^{-x}}{-1} \right) dx + c$$

where c is constant of integration

$$= -x^2 e^{-x} + 2 \int x e^{-x} dx + c$$

I II

Integrating by parts (taking x as first and e^{-x} as second function)

$$I = -x^2 e^{-x} + 2 \left[(x) \left(\frac{e^{-x}}{-1} \right) - \int (1) \left(\frac{e^{-x}}{-1} \right) dx \right] + c$$

$$= -x^2 e^{-x} + 2[-x e^{-x} + \int e^{-x} dx] + c$$

$$= -x^2 e^{-x} + 2 \left[-x e^{-x} + \left(\frac{e^{-x}}{-1} \right) \right] + c$$

$$= -x^2 e^{-x} - 2x e^{-x} - 2e^{-x} + c$$

(ii) Let $I = \int x x^2 e^{x^2} dx \dots (1)$

Putting $x^2 = t$

Differentiating

$$2x dx = dt \Rightarrow x dx = \frac{dt}{2}$$

$\therefore$ (1) becomes

$$I = \frac{1}{2} \int t e^t dt$$

I II

Integrating by parts (taking t as first and e^t as second function)

$$\begin{aligned} I &= \frac{1}{2} \left[(t)(e^t) - \int (1)(e^t) dt \right] + c = \frac{1}{2} (te^t - e^t) + c = \frac{1}{2} (x^2 e^{x^2} - e^{x^2}) + c \\ &= \frac{1}{2} (x^2 - 1) e^{x^2} + c \end{aligned}$$

13.9 REFERENCES/FURTHER READINGS

1. Mathematics Class XI By RD Sharma , Dhanpat Rai Publication , 2011
2. Mathematics Class XII Part – 2 By Ravi Prakash , GRB Books, Second Edition 2025
3. BCS-012 : Basic Mathematics :
https://onlinecourses.swayam2.ac.in/nou26_ma07/preview (January 2026 session)

UNIT 14 DEFINITE INTEGRATION

Structure

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14.0 INTRODUCTION

In Units 12 and 13 we have discussed concept of differentiation and concept of indefinite integral. But on many occasions, we are interested in finding out the probability of a continuous random variable in certain limits, this job is done by using the concept of definite integral.

This unit discusses about definite integral, evaluation of definite integral of some commonly used functions with the help of large number of examples. Properties of definite integral and how these are used have been also discussed in this unit with the help of number of examples.

14.1 OBJECTIVES

After completing this unit, you should be able to:

- define definite integration and give its geometrical meaning;
 - evaluate the integration of some commonly used functions;
 - explain the properties of the definite integrations; and
 - evaluate the integrations using properties of definite integrations.
-

14.2 MEANING AND GEOMETRICAL INTERPRETATION

Notation and Definition

You have already studied that if

$$\frac{d}{dx}(F(x)) = f(x), \text{ then } \int f(x)dx = F(x) + c$$

where c is arbitrary constant and hence value of $\int f(x)dx$ is indefinite.

Here, we are going to discuss definite integrals.

Definite integral of a function $f(x)$ within the limits

$$a < x < b \text{ or } a \leq x < b \text{ or } a < x \leq b \text{ or } a \leq x \leq b \quad \dots (1)$$

is denoted by $\int_a^b f(x)dx$

and is defined as

$$\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a) \quad \dots (2) \quad \left[\begin{array}{l} \text{By fundamental theorem} \\ \text{of integral calculus} \end{array} \right]$$

Fundamental theorem of Integral Calculus:

If f is integrable on $[a, b]$ and F is such that

$$\frac{d}{dx}(F(x)) = f(x), \quad x \in [a, b]$$

then

$$\int_a^b f(x)dx = F(b) - F(a)$$

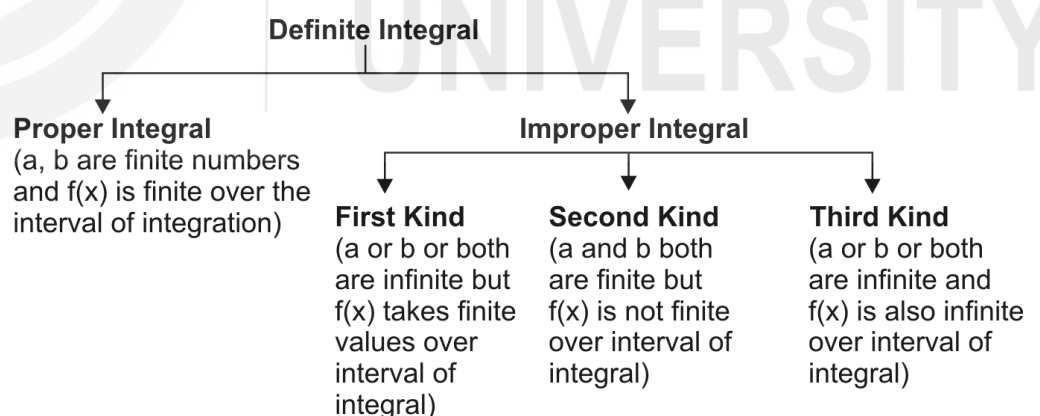
, where a is called lower limit, b is called upper limit and either of the intervals in (1) to be used is known as interval of integration.

For example, $\int_2^5 (x+3)dx = \left[\frac{x^2}{2} + 3x + c \right]_2^5 = \frac{(5)^2}{2} + 3(5) + c - \left[\frac{2^2}{2} + 3 \times 2 + c \right]$
 $= 25/2 + 15 - 2 - 6 = 39/2$

The integral $\int_a^b f(x)dx$ being definite integral, so it will have a definite value,

but equation (2) suggests that value of L.H.S. of (2) depends on a , b and $f(x)$.

Following diagram shows that definite integral have different names based on the role of a , b and $f(x)$.



Here we shall discuss proper integrals and only first kind of improper integrals as these will be applicable later on in the subsequent courses.

Note: The notation $[F(x)]_a^b$ means that function $F(x)$ is to be evaluated at top and bottom limits and then subtract. Some authors use American text book

notation $|F(x)|_a^b$ instead of $[F(x)]_a^b$. But here we will use square bracket notation.

Geometrical Interpretation

The definite integral $\int_a^b f(x)dx$ represents the area bounded by the function

$y = f(x)$, x-axis and between the lines $x = a$, $x = b$ as shown by the shaded region in the following Fig. 14.1

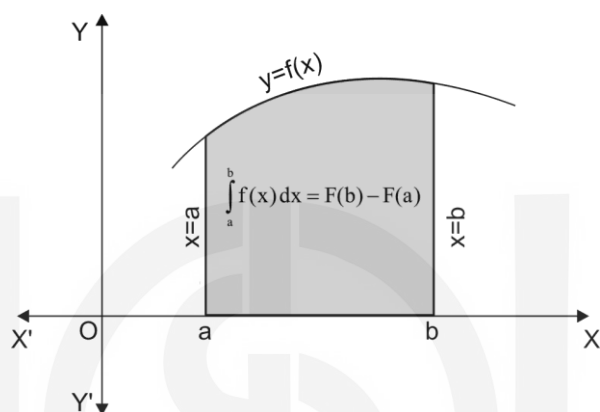


Fig. 14.1: Geometrical Interpretation of definite integral

Remark 1: In solving numerical problems, generally c is not used.

For example, if $f(x) = x^2 + x + 6$ then $\int f(x)dx = \frac{x^3}{3} + \frac{x^2}{2} + 6x + c = F(x)$

We will not write it as

$$\begin{aligned} \int_1^2 f(x)dx &= [F(x)]_1^2 = \left[\frac{x^3}{3} + \frac{x^2}{2} + 6x + c \right]_1^2 \\ &= \frac{2^3}{3} + \frac{2^2}{2} + 6 \times 2 + c - \left(\frac{1^3}{3} + \frac{1^2}{2} + 6 \times 1 + c \right) \\ &= \frac{8}{3} + 2 + 12 + c - \frac{1}{3} - \frac{1}{2} - 6 - c = \frac{8}{3} + 8 - \frac{1}{3} - \frac{1}{2} \\ &= \frac{16 + 48 - 2 - 3}{6} = \frac{59}{6} \end{aligned}$$

But throughout the unit, we will write it as

$$\int_1^2 f(x)dx = \int_1^2 (x^2 + x + 6)dx = \left[\frac{x^3}{3} + \frac{x^2}{2} + 6x \right]_1^2$$

$$= \frac{8}{3} + \frac{4}{3} + 12 - \left(\frac{1}{3} + \frac{1}{2} + 6 \right) = \frac{59}{6}$$

14.3 DEFINITE INTEGRAL OF SOME COMMONLY USED FUNCTIONS

Let us here consider some examples of definite integrals based on the formulae of indefinite integral already discussed in Unit 13 of this course.

Example 1: Evaluate the following integrals:

(i) $\int_2^6 8dx$

(ii) $\int_0^2 (x^2 + 1)dx$

(iii) $\int_2^3 (3x + 4)dx$

(iv) $\int_2^5 4x^3 dx$

(v) $\int_2^3 x^a dx$

(vi) $\int_2^4 \sqrt{18x - 24} dx$

(vii) $\int_1^2 (x + 1)(x^2 - 1)dx$

(viii) $\int_2^3 \frac{x^3 + 5}{x^2} dx$

(ix) $\int_2^5 \frac{1}{2x - 3} dx$

(x) $\int_2^3 \frac{4}{x} dx$

(xi) $\int_1^4 2^x dx$

(xii) $\int_0^2 e^{3x} dx$

(xiii) $\int_0^2 3^{4x+1} dx$

(xiv) $\int_1^3 e^{2x+5} dx$

Solution:

(i) $\int_2^6 8dx = [8x]_2^6 = 8 \times 6 - 8 \times 2 = 48 - 16 = 32$

(ii) $\int_0^2 (x^2 + 1)dx = \left[\frac{x^3}{3} + x \right]_0^2 = \frac{8}{3} + 2 - 0 - 0 = \frac{14}{3}$

(iii) $\int_2^3 (3x + 4)dx = \left[\frac{3x^2}{2} + 4x \right]_2^3 = \frac{27}{2} + 12 - 6 - 8 = \frac{27}{2} - 2 = \frac{23}{2}$

(iv) $\int_2^5 4x^3 dx = \left[\frac{4x^4}{4} \right]_2^5 = [x^4]_2^5 = 625 - 16 = 609$

(v) $\int_2^3 x^a dx = \left[\frac{x^{a+1}}{a+1} \right]_2^3 = \frac{1}{a+1} (3^{a+1} - 2^{a+1})$

$$\begin{aligned}
 \text{(vi)} \quad \int_2^4 \sqrt{18x-24} \, dx &= \left[\frac{(18x-24)^{3/2}}{\frac{3}{2} \times 18} \right]_2^4 \quad \left[\because \int (ax+b)^n \, dx = \frac{(ax+b)^{n+1}}{a \times (n+1)} \right] \\
 &= \frac{1}{27} \left[(72-24)^{3/2} - (36-24)^{3/2} \right] = \frac{1}{27} \left[48^{3/2} - (12)^{3/2} \right] \\
 &= \frac{1}{27} \left[48\sqrt{48} - 12\sqrt{12} \right] = \frac{1}{27} \left[48 \times 4\sqrt{3} - 12 \times 2\sqrt{3} \right] \\
 &= \frac{24}{27} \left[8\sqrt{3} - \sqrt{3} \right] = \frac{8}{9} (7\sqrt{3}) = \frac{56\sqrt{3}}{9}
 \end{aligned}$$

$$\begin{aligned}
 \text{(vii)} \quad \int_1^2 (x+1)(x^2-1) \, dx &= \int_1^2 (x^3 - x + x^2 - 1) \, dx \\
 &= \left[\frac{x^4}{4} - \frac{x^2}{2} + \frac{x^3}{3} - x \right]_1^2 \\
 &= \frac{16}{4} - \frac{4}{2} + \frac{8}{3} - 2 - \left(\frac{1}{4} - \frac{1}{2} + \frac{1}{3} - 1 \right) \\
 &= 4 - 2 + \frac{8}{3} - 2 - \left(\frac{3-6+4-12}{12} \right) \\
 &= \frac{8}{3} + \frac{11}{12} = \frac{32+11}{12} = \frac{43}{12}
 \end{aligned}$$

$$\begin{aligned}
 \text{(viii)} \quad \int_2^3 \frac{x^3+5}{x^2} \, dx &= \int_2^3 \left(\frac{x^3}{x^2} + \frac{5}{x^2} \right) \, dx = \int_2^3 (x + 5x^{-2}) \, dx = \left[\frac{x^2}{2} + \frac{5x^{-1}}{-1} \right]_2^3 \\
 &= \left[\frac{x^2}{2} - \frac{5}{x} \right]_2^3 = \frac{9}{2} - \frac{5}{3} - \left(\frac{4}{2} - \frac{5}{2} \right) \\
 &= \frac{27-10}{6} - \frac{4-5}{2} = \frac{17}{6} + \frac{1}{2} = \frac{17+3}{6} = \frac{20}{6} = \frac{10}{3}
 \end{aligned}$$

$$\text{(ix)} \quad \int_2^5 \frac{1}{2x-3} \, dx = \left[\frac{1}{2} \log |2x-3| \right]_2^5 = \frac{1}{2} [\log 7 - \log 1] = \frac{1}{2} \log 7 \quad \text{as } \log 1 = 0$$

$$\text{(x)} \quad \int_2^3 \frac{4}{x} \, dx = 4 \int_2^3 \frac{1}{x} \, dx = 4 [\log |x|]_2^3 = 4(\log 3 - \log 2) = 4 \log \frac{3}{2}$$

$$\text{(xi)} \quad \int_1^4 2^x \, dx = \left[\frac{2^x}{\log 2} \right]_1^4 = \frac{1}{\log 2} [2^4 - 2^1] = \frac{14}{\log 2}$$

Definite Integration

$$(xii) \int_0^2 e^{3x} dx = \left[\frac{e^{3x}}{3} \right]_0^2 = \frac{1}{3}(e^6 - e^0) = \frac{e^6 - 1}{3}$$

$$(xiii) \int_0^2 3^{4x+1} dx = \left[\frac{3^{4x+1}}{4 \log 3} \right]_0^2 = \frac{1}{4 \log 3} [3^9 - 3^1] = \frac{1}{4 \log 3} (19683 - 3) = \frac{19680}{4 \log 3}$$

$$(xiv) \int_1^3 e^{2x+5} dx = \left[\frac{e^{2x+5}}{2} \right]_1^3 = \frac{1}{2}(e^{11} - e^7)$$

Example 2: Evaluate the following integrals:

$$(i) \int_1^6 x\sqrt{x+3} dx$$

$$(ii) \int_0^1 \frac{2x+3}{x^2+3x+5} dx$$

$$(iii) \int_0^2 \frac{2x+7}{(x-3)(x+1)(x-4)} dx \quad (iv) \int_0^3 \frac{x-5}{(x+1)(x+2)^2} dx$$

Solution:

$$(i) \text{ Let } I = \int_1^6 x\sqrt{x+3} dx \quad \dots (1)$$

$$\text{Putting } \sqrt{x+3} = t \Rightarrow x+3 = t^2$$

Differentiating

$$dx = 2t dt$$

Also when $x = 1$, $t = 2$ and when $x = 6$, $t = 3$

$\therefore$ (1) becomes

$$I = \int_2^3 (t^2 - 3)t(2t) dt = 2 \int_2^3 (t^4 - 3t^2) dt = 2 \left[\frac{t^5}{5} - t^3 \right]_2^3$$

$$= 2 \left[\frac{243}{5} - 27 - \left(\frac{32}{5} - 8 \right) \right] = 2 \left(\frac{243}{5} - \frac{32}{5} - 19 \right)$$

$$= 2 \left(\frac{243 - 32 - 95}{5} \right) = 2 \left(\frac{116}{5} \right) = \frac{232}{5}$$

$$(ii) \text{ Let } I = \int_0^1 \frac{2x+3}{x^2+3x+5} dx \quad \dots (1)$$

$$\text{Putting } x^2 + 3x + 5 = t$$

Differentiating

$$(2x+3)dx = dt$$

Also when $x = 0$, $t = 5$ and when $x = 1$, $t = 9$

∴ (1) becomes

$$I = \int_5^9 \frac{dt}{t} = [\log|t|]_5^9 = \log 9 - \log 5 = \log \frac{9}{5}$$

$$(iii) \text{ Let } I = \int_0^2 \frac{2x+7}{(x-3)(x+1)(x-4)} dx = \int_0^2 \left(\frac{-13/4}{x-3} + \frac{1/4}{x+1} + \frac{3}{x-4} \right) dx$$

$$\left[\begin{array}{l} \text{Using partial fractions as discussed in type 1, we get} \\ A = \frac{2 \times 3 + 7}{(3+1)(3-4)} = -\frac{13}{4}, B = \frac{2(-1)+7}{(-1-3)(-1-4)} = \frac{1}{4}, C = \frac{2 \times 4 + 7}{(4-3)(4+1)} = 3 \end{array} \right]$$

$$I = -\frac{13}{4} [\log|x-3|]_0^2 + \frac{1}{4} [\log|x+1|]_0^2 + 3 [\log|x-4|]_0^2$$

$$= -\frac{13}{4} (\log 1 - \log 3) + \frac{1}{4} (\log 3 - \log 1) + 3 (\log 2 - \log 4)$$

$$= -\frac{13}{4} (0 - \log 3) + \frac{1}{4} (\log 3 - 0) + 3 (\log 2 - \log 2^2) \quad \text{as } \log 1 = 0$$

$$= \frac{13}{4} \log 3 + \frac{1}{4} \log 3 + 3 (\log 2 - 2 \log 2) \quad [\because \log m^n = n \log m]$$

$$= \frac{1}{4} (13 \log 3 + \log 3) + 3 (-\log 2)$$

$$= \frac{14 \log 3}{4} - 3 \log 2 = \frac{7}{2} \log 3 - 3 \log 2$$

$$(iv) \text{ Let } I = \int_0^3 \frac{x-5}{(x+1)(x+2)^2} dx \quad \dots (1)$$

First we resolve into partial fractions

$$\text{Let } \frac{x-5}{(x+1)(x+2)^2} = \frac{A}{x+1} + \frac{B}{x+2} + \frac{C}{(x+2)^2}$$

Multiply on both sides by $(x+1)(x+2)^2$, we get

$$x-5 = A(x+2)^2 + B(x+1)(x+2) + C(x+1) \quad \dots (2)$$

Putting $x = -1$ in (2), we get $[\because x+1=0 \text{ gives } x=-1]$

$$-6 = A(-1+2)^2 + B(0) + C(0) \Rightarrow -6 = A \Rightarrow \boxed{A = -6}$$

Putting $x = -2$ in (2), we get $[\because x+2=0 \text{ gives } x=-2]$

$$-7 = A(0) + B(0) + C(-2+1) \Rightarrow -7 = -C \Rightarrow \boxed{C = 7}$$

Comparing coefficient of x^2 on both sides of (2), we get

$$0 = A + B \Rightarrow B = -A \Rightarrow \boxed{B = 6}$$

$$\begin{aligned} \therefore I &= \int_0^3 \left[\frac{-6}{x+1} + \frac{6}{x+2} + \frac{7}{(x+2)^2} \right] dx \\ &= -6 \left[\log|x+1| \right]_0^3 + 6 \left[\log|x+2| \right]_0^3 + 7 \left[\frac{(x+2)^{-1}}{-1} \right]_0^3 \\ &= -6(\log 4 - \log 1) + 6(\log 5 - \log 2) - 7 \left(\frac{1}{5} - \frac{1}{2} \right) \\ &= -6 \log 2^2 + 6 \log 5 - 6 \log 2 - 7 \left(\frac{2-5}{10} \right) \quad \text{as } \log 1 = 0 \\ &= -12 \log 2 + 6 \log 5 - 6 \log 2 + \frac{21}{10} \quad [\because \log m^n = n \log m] \\ &= 6 \log 5 - 18 \log 2 + \frac{21}{10} \end{aligned}$$

Check Your Progress – 1

1) Evaluate the following integrals:

(i) $\int_2^4 x^2 dx$

(ii) $\int_0^2 \left(x - \frac{5}{2} \right) dx$

2) Evaluate the following integrals:

(i) $\int_2^5 \frac{x}{x^2+3} dx$ (ii) $\int_0^2 \frac{e^x}{3+e^{2x}} dx$ (iii) $\int_0^1 \frac{2x^2+1}{(x-4)(x-2)^3} dx$

14.4 ELEMENTARY PROPERTIES OF DEFINITE INTEGRAL

Here, first we list some properties and then we will use these properties to evaluate some integrals.

$$\mathbf{P\ 1:} \quad \int_a^b f(x)dx = \int_a^b f(t)dt \quad (\text{Change of variable property})$$

$$\mathbf{P\ 2:} \quad \int_a^b f(x)dx = -\int_b^a f(x)dx \quad (\text{Interchange of limits property})$$

$$\mathbf{P\ 3:} \quad \int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx, \quad a < c < b$$

In general

We can introduce any number of points between a and b

$$\text{e.g. } \int_a^b f(x)dx = \int_a^{c_1} f(x)dx + \int_{c_1}^{c_2} f(x)dx + \dots + \int_{c_{n-1}}^{c_n} f(x)dx + \int_{c_n}^b f(x)dx$$

$$\text{where, } a < c_1 < c_2 < \dots < c_{n-1} < c_n < b$$

$$\mathbf{P\ 4:} \quad \int_{-a}^a f(x)dx = \begin{cases} 2\int_0^a f(x)dx, & \text{if } f(x) \text{ is an even function} \\ 0, & \text{if } f(x) \text{ is an odd function} \end{cases}$$

$$\mathbf{P\ 5:} \quad \int_a^b f(x)dx = \int_a^b f(a+b-x)dx$$

$$\text{In particular, } \int_0^a f(x)dx = \int_0^a f(a-x)dx$$

$$\mathbf{P\ 6:} \quad \int_0^{2a} f(x)dx = \int_0^a f(x)dx + \int_0^a f(2a-x)dx$$

$$\mathbf{P7:} \quad \int_0^{2a} f(x)dx = \begin{cases} 2\int_0^a f(x)dx, & \text{if } f(2a-x) = f(x) \\ 0, & \text{if } f(2a-x) = -f(x) \end{cases}$$

Proof:

P 1: Let $\int f(x)dx = F(x)$, so $\int f(t)dt = F(t)$

Now, by fundamental theorem of integral calculus

$$\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a) \quad \dots (1)$$

$$\text{and } \int_a^b f(t)dt = [F(t)]_a^b = F(b) - F(a) \quad \dots (2)$$

From (1) and (2)

$$\int_a^b f(x)dx = \int_a^b f(t)dt$$

P 2: Let $\int f(x)dx = F(x)$

$\therefore$ by fundamental theorem of integral calculus

$$\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a) \quad \dots (1)$$

$$\text{and } \int_b^a f(x)dx = [F(x)]_b^a = F(a) - F(b) = -[F(b) - F(a)] \quad \dots (2)$$

From (1) and (2)

$$\int_a^b f(x)dx = -\int_b^a f(x)dx$$

P 3: Let $\int f(x)dx = F(x)$

$\therefore$ by fundamental theorem of integral calculus

$$\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a) \quad \dots (1)$$

$$\begin{aligned} \text{and } \int_a^c f(x)dx + \int_c^b f(x)dx &= [F(x)]_a^c + [F(x)]_c^b = (F(c) - F(a)) + (F(b) - F(c)) \\ &= F(b) - F(a) \quad \dots (2) \end{aligned}$$

From (1) and (2)

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$$

P 4: Using property 3, we have

$$\int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx \quad \dots (1)$$

$$\text{Let } I = \int_{-a}^0 f(x) dx$$

Putting $x = -t$

Differentiating

$$dx = -dt$$

Also, when $x = -a$, $t = a$ and when $x = 0$, $t = 0$

$$\therefore I = -\int_a^0 f(-t) dt = \int_0^a f(-x) dx \quad \dots (2) \quad [\text{Using properties 1 and 2}]$$

Using (2) in (1), we get

$$\begin{aligned} \int_{-a}^a f(x) dx &= \int_0^a f(-x) dx + \int_0^a f(x) dx \quad \dots (3) \\ &= \begin{cases} \int_0^a f(x) dx + \int_0^a f(x) dx, & \text{if } f \text{ is even} \\ -\int_0^a f(x) dx + \int_0^a f(x) dx, & \text{if } f \text{ is odd} \end{cases} \\ &= \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f \text{ is even function} \\ 0, & \text{if } f \text{ is an odd function} \end{cases} \end{aligned}$$

$$\text{P 5: Let } I = \int_a^b f(x) dx \quad \dots (1)$$

R.H.S. suggests that we should put

$$x = a + b - t$$

Differentiating

$$dx = -dt$$

Also, when $x = a \Rightarrow t = b$ and when $x = b \Rightarrow t = a$

$\therefore$ (1) becomes

$$I = \int_b^a f(a + b - t)(-dt) = -\int_b^a f(a + b - t) dt$$

Definite Integration

$$= \int_a^b f(a+b-t)dt \quad \text{[Using property 2]}$$

$$= \int_a^b (a+b-x)dx \quad \text{[Using property 1]}$$

In particular

If we put $a = 0, b = a$ in this result, then

$$\int_0^a f(x)dx = \int_0^a f(a-x)dx$$

$$\begin{aligned} \mathbf{P\ 6:} \quad \int_0^{2a} f(x)dx &= \int_0^a f(x)dx + \int_a^{2a} f(x)dx \quad \text{[Using property 3]} \\ &= I_1 + I_2 \quad \dots (1) \end{aligned}$$

$$I_2 = \int_a^{2a} f(x)dx$$

Putting $x = 2a - t$

Differentiating

$$dx = -dt$$

Also, when $x = a, t = a$ and when $x = 2a, t = 0$

$$\begin{aligned} \therefore I_2 &= \int_a^0 f(2a-t)(-dt) = \int_0^a f(2a-t)dt \quad \text{[Using property 2]} \\ &= \int_0^a (2a-x)dx \quad \dots (2) \quad \text{[Using property 1]} \end{aligned}$$

Using (2) in (1), we get

$$\int_0^{2a} f(x)dx = \int_0^a f(x)dx + \int_0^a f(2a-x)dx$$

P7: From property 6

$$\begin{aligned} \int_0^{2a} f(x)dx &= \int_0^a f(x)dx + \int_0^a f(2a-x)dx \\ &= \begin{cases} \int_0^a f(x)dx + \int_0^a f(x)dx, & \text{if } f(2a-x) = f(x) \\ \int_0^a f(x)dx - \int_0^a f(x)dx, & \text{if } f(2a-x) = -f(x) \end{cases} \end{aligned}$$

$$= \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f(2a-x) = f(x) \\ 0, & \text{if } f(2a-x) = -f(x) \end{cases}$$

14.5 EXAMPLES BASED ON PROPERTIES OF DEFINITE INTEGRAL

In this section, you will see how the properties of definite integral, discussed in previous Sec. are used and save lot of calculation work.

Example 3: Evaluate the following integrals:

(i) $\int_1^4 |x-2| dx$ (ii) $\int_0^2 |2x-3| dx$ (iii) $\int_0^3 f(x) dx$, where $f(x) = \begin{cases} x+1, & 0 \leq x < 1 \\ 2x+3, & 1 \leq x < 3 \end{cases}$

(iv) $\int_{-99}^{99} (x^3 + x + e^x - e^{-x}) dx$ (v) $\int_{-2}^2 \log\left(\frac{5-x}{5+x}\right) dx$ (vi) $\int_{-3}^3 2^{|2x|} dx$

(vii) $\int_a^b \frac{f(x)}{f(x) + f(a+b-x)} dx$ (viii) $\int_2^3 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{5-x}} dx$ (ix) $\int_2^7 \frac{\sqrt[5]{x+1}}{\sqrt[5]{x+1} + \sqrt[5]{10-x}} dx$

Solution:

(i) Let $I = \int_1^4 |x-2| dx = \int_1^2 |x-2| dx + \int_2^4 |x-2| dx$ [By P3]

$$I = \int_1^2 -(x-2) dx + \int_2^4 (x-2) dx \quad \left[\begin{array}{l} \because \text{for } 1 \leq x < 2, x-2 < 0 \text{ so} \\ |x-2| = -(x-2) \\ \text{and for } 2 \leq x \leq 4, x-2 \geq 0 \text{ so} \\ |x-2| = x-2 \end{array} \right]$$

$$= -\left[\frac{x^2}{2} - 2x \right]_1^2 + \left[\frac{x^2}{2} - 2x \right]_2^4 = -\left(\frac{4}{2} - 4 - \left(\frac{1}{2} - 2 \right) \right) + \left(\frac{16}{2} - 8 - \left(\frac{4}{2} - 4 \right) \right)$$

$$= -\left(2 - 4 - \frac{1}{2} + 2 \right) + (8 - 8 - 2 + 4) = \frac{1}{2} + 2 = \frac{5}{2}$$

(ii) $\int_0^2 |2x-3| dx = \int_0^{3/2} |2x-3| dx + \int_{3/2}^2 |2x-3| dx$ [By P 3]

Definite Integration

$$= \int_0^{3/2} -(2x-3)dx + \int_{3/2}^2 (2x-3)dx$$

$$\left[\begin{array}{l} \because \text{for } 0 \leq x < 3/2 \Rightarrow 2x-3 < 0 \text{ so } |2x-3| = -(2x-3) \text{ and} \\ \text{for } 3/2 \leq x \leq 2 \Rightarrow 2x-3 \geq 0 \text{ so } |2x-3| = 2x-3 \end{array} \right]$$

$$= -\left[x^2 - 3x\right]_0^{3/2} + \left[x^2 - 3x\right]_{3/2}^2$$

$$= -\left(\frac{9}{4} - \frac{9}{2} - 0 + 0\right) + \left(4 - 6 - \frac{9}{4} + \frac{9}{2}\right)$$

$$= -\left(-\frac{9}{4}\right) - 2 + \frac{9}{4} = \frac{9}{4} - 2 + \frac{9}{4} = \frac{9-8+9}{4} = \frac{10}{4} = \frac{5}{2}$$

$$(iii) \text{ Let } I = \int_0^3 f(x)dx, \text{ where } f(x) = \begin{cases} x+1, & 0 \leq x < 1 \\ 2x+3, & 1 \leq x < 3 \end{cases} \quad \dots (1)$$

$$\text{Now, } I = \int_0^1 f(x)dx + \int_1^3 f(x)dx \quad [\text{Using property 3}]$$

$$= \int_0^1 (x+1)dx + \int_1^3 (2x+3)dx \quad [\text{Using (1)}]$$

$$= \left[\frac{x^2}{2} + x\right]_0^1 + \left[x^2 + 3x\right]_1^3$$

$$= \frac{1}{2} + 1 - 0 - 0 + (9 + 9 - 1 - 3) = \frac{3}{2} + 14 = \frac{31}{2}$$

$$(iv) \text{ Let } I = \int_{-99}^{99} (x^3 + x + e^x - e^{-x})dx$$

$$\text{Let } f(x) = x^3 + x + e^x - e^{-x}$$

$$\therefore f(-x) = (-x)^3 + (-x) + e^{-x} - e^{-(-x)} = -x^3 - x + e^{-x} - e^x$$

$$= -(x^3 + x + e^x - e^{-x}) = -f(x)$$

$\Rightarrow f(x)$ is an odd function

$$\therefore I = \int_{-99}^{99} (x^3 + x + e^x + e^{-x})dx = 0 \quad [\text{By property 4}]$$

$$(v) \text{ Let } I = \int_{-2}^2 \log\left(\frac{5-x}{5+x}\right)dx$$

$$\text{Let } f(x) = \log\left(\frac{5-x}{5+x}\right)$$

$$\begin{aligned}\therefore f(-x) &= \log\left(\frac{5-(-x)}{5+(-x)}\right) = \log\left(\frac{5+x}{5-x}\right) = \log\left(\frac{5-x}{5+x}\right)^{-1} \\ &= -\log\left(\frac{5-x}{5+x}\right) = -f(x) \quad [\because \log m^n = n \log m]\end{aligned}$$

$\Rightarrow f(x)$ is an odd function

$$\therefore I = \int_{-2}^2 \log\left(\frac{5-x}{5+x}\right) dx = 0 \quad [\text{By property 4}]$$

$$\begin{aligned}\text{(vi)} \quad \int_{-3}^3 2^{|2x|} dx &= \int_{-3}^0 e^{|2x|} dx + \int_0^3 e^{|2x|} dx \\ &= \int_{-3}^0 e^{-2x} dx + \int_0^3 e^{2x} dx \left[\begin{array}{l} \because \text{for } -3 < x < 0, 2x < 0 \text{ so } |2x| = -2x \text{ and} \\ \text{for } 0 < x < 3, 2x > 0 \text{ so } |2x| = 2x \end{array} \right] \\ &= \left[\frac{e^{-2x}}{-2} \right]_{-3}^0 + \left[\frac{e^{2x}}{2} \right]_0^3 = -\frac{1}{2}(1 - e^6) + \frac{1}{2}(e^6 - 1) \\ &= \frac{1}{2}(-1 + e^6 + e^6 - 1) = \frac{2e^6 - 2}{2} = e^6 - 1\end{aligned}$$

$$\begin{aligned}\text{(vii)} \quad \text{Let } I &= \int_a^b \frac{f(x)}{f(x) + f(a+b-x)} dx \quad \dots (1) \\ &= \int_a^b \frac{f(a+b-x)}{f(a+b-x) + f(a+b-(a+b-x))} dx \quad [\text{Using property 5}] \\ I &= \int_a^b \frac{f(a+b-x)}{f(a+b-x) + f(x)} dx \quad \dots (2)\end{aligned}$$

(1) + (2) gives

$$\begin{aligned}2I &= \int_a^b \frac{f(x) + f(a+b-x)}{f(x) + f(a+b-x)} dx = \int_a^b 1 dx = [x]_a^b = b - a \\ \Rightarrow I &= \frac{b-a}{2}\end{aligned}$$

$$\begin{aligned}\text{(viii)} \quad \text{Let } I &= \int_2^3 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{5-x}} dx \quad \dots (1) \\ &= \int_2^3 \frac{\sqrt{5-x}}{\sqrt{5-x} + \sqrt{5-(5-x)}} dx \quad [\text{Using property 5}]\end{aligned}$$

$$I = \int_2^3 \frac{\sqrt{5-x}}{\sqrt{5-x} + \sqrt{x}} dx \quad \dots (2)$$

(1) + (2) gives

$$2I = \int_2^3 \frac{\sqrt{x} + \sqrt{5-x}}{\sqrt{x} + \sqrt{5-x}} dx = \int_2^3 1 dx = [x]_2^3 = 3 - 2 = 1$$

$$\Rightarrow I = 1/2$$

$$(ix) \text{ Let } I = \int_2^7 \frac{\sqrt[5]{x+1}}{\sqrt[5]{x+1} + \sqrt[5]{10-x}} dx \quad \dots (1)$$

$$= \int_2^7 \frac{\sqrt[5]{9-x+1}}{\sqrt[5]{9-x+1} + \sqrt[5]{10-(9-x)}} dx \quad [\text{Using property 5}]$$

$$I = \int_2^7 \frac{\sqrt[5]{10-x}}{\sqrt[5]{10-x} + \sqrt[5]{1+x}} dx \quad \dots (2)$$

(1) + (2) gives

$$2I = \int_2^7 \frac{\sqrt[5]{x+1} + \sqrt[5]{10-x}}{\sqrt[5]{10-x} + \sqrt[5]{1+x}} dx = \int_2^7 1 dx = [x]_2^7 = 7 - 2 = 5$$

$$\Rightarrow I = 5/2$$

Check Your Progress – 2

1) Evaluate the following integrals:

$$(i) \int_0^5 |x^2 - 3x + 2| dx \quad (ii) \int_1^5 (|x-1| + |x+2| + |x-3|) dx$$

$$(iii) \int_{-1}^2 f(x) dx, \text{ where } f(x) = \begin{cases} 4-3x, & -1 \leq x < 1 \\ 2x+1, & 1 \leq x \leq 2 \end{cases}$$

$$(iv) \int_{-77}^{77} x^5 \sqrt{a^2 - x^2} dx \quad (v) \int_{-1}^1 \frac{x^5 + x^7}{4 - x^4} dx \quad (vi) \int_0^2 x(2-x)^{11/2} dx \quad (vii) \int_{-3}^3 4x|x| dx$$

$$(viii) \int_1^5 f(x) dx, \text{ where } f(x) = \begin{cases} 1, & 1 \leq x < 2 \\ 2, & 2 \leq x < 3 \\ 3, & 3 \leq x < 4 \\ 4, & 4 \leq x < 5 \\ 5, & x \geq 5 \end{cases}$$

14.6 SUMMARY

This unit introduced the definite integral, defining it as the net signed area between a function's graph and the x-axis over a specific interval $[a,b]$. Unlike indefinite integrals, the definite integral results in a unique numerical value, geometrically. This unit also elaborated on the fundamental theorem of Integral Calculus. This unit provided evaluations for commonly used functions—such as polynomials, trigonometric, and exponential forms—by applying the fundamental theorem of calculus. The unit also demonstrated how to find a definite integral using substitution and partial fractions. A significant portion of the unit is dedicated to the elementary properties of definite integrals, including linearity, interval switching, and some properties, such as the change of variable property.

14.7 SOLUTIONS/ANSWERS TO CHECK YOUR PROGRESS

Check Your Progress – 1

$$1) \text{ (i) } \int_2^4 x^2 dx = \left[\frac{x^3}{3} \right]_2^4 = \frac{1}{3}(64 - 8) = \frac{56}{3}$$

$$\text{(ii) } \int_0^2 \left(x - \frac{5}{2} \right) dx = \left[\frac{x^2}{2} - \frac{5}{2}x \right]_0^2 = \frac{4}{2} - \frac{5}{2} \times 2 - 0 - 0 = 2 - 5 = -3$$

$$2) \text{ (i) Let } I = \int_2^5 \frac{x}{x^2 + 3} dx \quad \dots (1)$$

Putting $x^2 = t$

Differentiating

$$2x dx = dt \Rightarrow x dx = dt/2$$

Also, when $x = 2$, $t = 4$ and when $x = 5$, $t = 25$

$$\begin{aligned}\therefore I &= \frac{1}{2} \int_4^{25} \frac{dt}{t+3} = \frac{1}{2} [\log|t+3|]_4^{25} = \frac{1}{2} (\log 28 - \log 7) = \frac{1}{2} \log \left(\frac{28}{7} \right) \\ &= \frac{1}{2} \log 4 = \frac{1}{2} \log 2^2 = \frac{1}{2} \times 2 \log 2 = \log 2\end{aligned}$$

(ii) Let $I = \int_0^2 \frac{e^{2x}}{3+e^{2x}} dx \quad \dots(1)$

Putting $e^{2x} = t$

Differentiating

$$2e^{2x} dx = dt \Rightarrow e^{2x} dx = \frac{dt}{2}$$

Also, when $x = 0, t = 1$ and when $x = 2, t = e^4$

$$\therefore I = \frac{1}{2} \int_1^{e^4} \frac{dt}{3+t} = \frac{1}{2} [\log|3+t|]_1^{e^4} = \frac{1}{2} [\log(3+e^4) - \log 4] = \frac{1}{2} \log \left(\frac{3+e^4}{4} \right)$$

(iii) $\int_0^1 \frac{2x^2 + 1}{(x-4)(x-2)^3} dx$

Let us first resolve into partial fractions

$$\text{Let } \frac{2x^2 + 1}{(x-4)(x-2)^3} = \frac{A}{x-4} + \frac{B}{x-2} + \frac{C}{(x-2)^2} + \frac{D}{(x-2)^3}$$

Multiplying on both sides by $(x-4)(x-2)^3$, we get

$$2x^2 + 1 = A(x-2)^3 + B(x-4)(x-2)^2 + C(x-4)(x-2) + D(x-4) \quad \dots(2)$$

Putting $x = 4$ in (2) we get $[\because x-4=0 \text{ gives } x=4]$

$$33 = A(4-2)^2 + B(0) + C(0) + D(0) \Rightarrow 33 = 4A \Rightarrow \boxed{A = \frac{33}{4}}$$

Putting $x = 2$ in (2), we get $[\because x-2=0 \text{ gives } x=2]$

$$9 = A(0) + B(0) + C(0) + D(2-4) \Rightarrow 9 = -2D \Rightarrow \boxed{D = -\frac{9}{2}}$$

Comparing coefficients of x^3 and constant terms on both sides of (2), we get

$$0 = A + B \Rightarrow B = -A \Rightarrow \boxed{B = -\frac{33}{4}}$$

$$1 = -8A - 16B + 8C - 4D \quad \dots (3)$$

Putting values of A, B and D in (3), we get

$$1 = -8 \times \frac{33}{4} - 16 \left(-\frac{33}{4} \right) + 8C - 4 \left(-\frac{9}{2} \right)$$

$$1 = -66 + 132 + 8C + 18 \Rightarrow 8C = -83 \Rightarrow \boxed{C = \frac{-83}{8}}$$

$$\therefore I = \int_0^1 \left[\frac{33/4}{x-4} + \frac{-33/4}{x-2} + \frac{-83/8}{(x-2)^2} + \frac{-9/2}{(x-2)^3} \right] dx$$

$$= \frac{33}{4} [\log|x-4|]_0^1 - \frac{33}{4} [\log|x-2|]_0^1 - \frac{83}{8} \left[\frac{(x-2)^{-1}}{-1} \right]_0^1$$

$$- \frac{9}{2} \left[\frac{(x-2)^{-2}}{-2} \right]_0^1$$

$$= \frac{33}{4} (\log|1-4| - \log|0-4|) - \frac{33}{4} (\log|1-2| - \log|0-2|)$$

$$+ \frac{83}{8} \left[\frac{1}{1-2} - \frac{1}{0-2} \right] + \frac{9}{4} \left[\frac{1}{(1-2)^2} - \frac{1}{(0-2)^2} \right]$$

$$= \frac{33}{4} (\log 3 - \log 4) - \frac{33}{4} (\log 1 - \log 2) + \frac{83}{8} \left(-1 + \frac{1}{2} \right) + \frac{9}{4} \left(1 - \frac{1}{4} \right)$$

$$= \frac{33}{4} (\log 3 - \log 2^2) - \frac{33}{4} (0 - \log 2) - \frac{83}{16} + \frac{27}{16} \quad \text{as } \log 1 = 0$$

$$= \frac{33}{4} \log 3 - \frac{33}{2} \log 2 + \frac{33}{4} \log 2 - \frac{56}{16} \quad [\text{as } \log m^n = n \log m]$$

$$= \frac{33}{4} \log 3 - \frac{33}{4} \log 2 - \frac{56}{16}$$

Check Your Progress – 2

1) (i) Let $I = \int_0^5 |x^2 - 3x + 2| dx = \int_0^5 |(x-2)(x-1)| dx$

$$= \int_0^1 |(x-2)(x-1)| dx + \int_1^2 |(x-2)(x-1)| dx + \int_2^5 |(x-2)(x-1)| dx$$

Definite Integration

$$= \int_0^1 (x-2)(x-1)dx + \int_1^2 -(x-2)(x-1)dx + \int_2^5 (x-2)(x-1)dx$$

$$\left[\begin{array}{l} \because \text{for } 0 \leq x < 1, (x-2)(x-1) > 0 \text{ so } |(x-2)(x-1)| = (x-2)(x-1) \\ \text{for } 1 \leq x < 2, (x-2)(x-1) \leq 0 \text{ so } |(x-2)(x-1)| = -(x-2)(x-1) \text{ and} \\ \text{for } 2 \leq x < 5, (x-2)(x-1) \geq 0 \text{ so } |(x-2)(x-1)| = (x-2)(x-1) \end{array} \right]$$

$$= \int_0^1 (x^2 - 3x + 2)dx - \int_1^2 (x^2 - 3x + 2)dx + \int_2^5 (x^2 - 3x + 2)dx$$

$$= \left[\frac{x^3}{3} - \frac{3x^2}{2} + 2x \right]_0^1 - \left[\frac{x^3}{3} - \frac{3x^2}{2} + 2x \right]_1^2 + \left[\frac{x^3}{3} - \frac{3x^2}{2} + 2x \right]_2^5$$

$$= \left[\frac{1}{3} - \frac{3}{2} + 2 - (0 - 0 + 0) \right] - \left[\left(\frac{8}{3} - 6 + 4 \right) - \left(\frac{1}{3} - \frac{3}{2} + 2 \right) \right]$$

$$+ \left[\left(\frac{125}{3} - \frac{75}{2} + 10 \right) - \left(\frac{8}{3} - 6 + 4 \right) \right]$$

$$= \frac{2-9+12}{6} - \left(\frac{8}{3} - 2 - \frac{1}{3} + \frac{3}{2} - 2 \right) + \left(\frac{125}{3} - \frac{75}{2} + 10 - \frac{8}{3} + 2 \right)$$

$$= \frac{5}{6} - \frac{16-12-2+9-12}{6} + \frac{250-225+60-16+12}{6}$$

$$= \frac{5}{6} - \frac{-1}{6} + \frac{81}{6} = \frac{5+1+81}{6} = \frac{87}{6} = \frac{29}{2}$$

$$(ii) \text{ Let } I = \int_1^5 (|x-1| + |x+2| + |x-3|)dx$$

$$= \int_1^5 |x-1|dx + \int_1^5 |x+2|dx + \int_1^5 |x-3|dx$$

$$= \int_1^5 (x-1)dx + \int_1^5 (x+2)dx + \int_1^3 |x-3|dx + \int_3^5 |x-3|dx$$

$$\left[\begin{array}{l} \because \text{for } 1 < x < 5, x-1 > 0 \\ \text{so } |x-1| = x-1 \end{array} \right]$$

$$= \left[\frac{x^2}{2} - x \right]_1^5 + \left[\frac{x^2}{2} + 2x \right]_1^5 + \int_1^3 -(x-3)dx + \int_3^5 (x-3)dx$$

$$\left[\begin{array}{l} \because \text{for } 1 < x < 3, x - 3 < 0 \text{ so } |x - 3| = -(x - 3) \text{ and} \\ \text{for } 3 \leq x \leq 5, x - 3 \geq 0 \text{ so } |x - 3| = x - 3 \end{array} \right]$$

$$\begin{aligned} &= \left(\frac{25}{2} - 5 - \frac{1}{2} + 1 \right) + \left(\frac{25}{2} + 10 - \frac{1}{2} - 2 \right) - \left[\frac{x^2}{2} - 3x \right]_1^3 + \left[\frac{x^2}{2} - 3x \right]_3^5 \\ &= \frac{25 - 10 - 1 + 2}{2} + \frac{25 + 20 - 1 - 4}{2} - \left(\frac{9}{2} - 9 - \frac{1}{2} + 3 \right) + \left(\frac{25}{2} - 15 - \frac{9}{2} + 9 \right) \\ &= \frac{16}{2} + \frac{40}{2} - \left(\frac{9 - 18 - 1 + 6}{2} \right) + \left(\frac{25 - 30 - 9 + 18}{2} \right) \\ &= 8 + 20 - \frac{-4}{2} + \frac{4}{2} = 28 + 2 + 2 = 32 \end{aligned}$$

(iii) Let $I = \int_{-1}^2 f(x) dx$,

where $f(x) = \begin{cases} 4 - 3x, & -1 \leq x < 1 \\ 2x + 1, & 1 \leq x \leq 2 \end{cases} \quad \dots (1)$

Now,

$$\begin{aligned} I &= \int_{-1}^1 f(x) dx + \int_1^2 f(x) dx \quad [\text{Using property (2)}] \\ &= \int_{-1}^1 (4 - 3x) dx + \int_1^2 (2x + 1) dx \quad [\text{Using (1)}] \\ &= \left[4x - \frac{3x^2}{2} \right]_{-1}^1 + \left[x^2 + x \right]_1^2 = 4 - \frac{3}{2} - \left(-4 - \frac{3}{2} \right) + 4 + 2 - 1 - 1 \\ &= \frac{8 - 3}{2} - \left(\frac{-8 - 3}{2} \right) + 4 = \frac{5}{2} + \frac{11}{2} + 4 = \frac{5 + 11 + 8}{2} = \frac{24}{2} = 12 \end{aligned}$$

(iv) Let $I = \int_{-77}^{77} x^5 \sqrt{a^2 - x^2} dx$

Let $f(x) = x^5 \sqrt{a^2 - x^2}$

$$\therefore f(-x) = (-x)^5 \sqrt{a^2 - (-x)^2} = -x^5 \sqrt{a^2 - x^2} = -f(x)$$

$\Rightarrow f(x)$ is an odd function.

$$\therefore I = \int_{-77}^{77} x^5 \sqrt{a^2 - x^2} dx = 0$$

[By property 4]

$$(v) \int_{-1}^1 \frac{x^5 + x^7}{4 - x^4} dx$$

$$\text{Let } f(x) = \frac{x^5 + x^7}{4 - x^4}$$

$$f(-x) = \frac{(-x)^5 + (-x)^7}{4 - (-x)^4} = \frac{-x^5 - x^7}{4 - x^4} = -\frac{x^5 + x^7}{4 - x^4} = -f(x)$$

$\Rightarrow f(x)$ is an odd function.

$$\therefore \int_{-1}^1 \frac{x^5 + x^7}{4 - x^4} dx = 0$$

[By property 4]

$$(vi) \text{ Let } I = \int_0^2 x(2-x)^{11/2} dx = \int_0^2 (2-x)(2-(2-x))^{11/2} dx \quad [\text{Using property 5}]$$

$$= \int_0^2 (2-x)(x)^{11/2} dx = \int_0^2 (2x^{11/2} - x^{13/2}) dx$$

$$= \left[2 \frac{x^{13/2}}{13/2} - \frac{x^{15/2}}{15/2} \right]_0^2 = \frac{4}{13} \times 2^{13/2} - \frac{2}{15} \times (2)^{15/2} - 0 + 0$$

$$= \frac{4}{13} \times 2^6 \times \sqrt{2} - \frac{2}{15} \times 2^7 \times \sqrt{2} = 2^8 \left(\frac{1}{13} - \frac{1}{15} \right) \sqrt{2}$$

$$= 2^8 \left(\frac{15-13}{13 \times 15} \right) \sqrt{2} = \frac{2^8 \times 2}{195} \sqrt{2} = \frac{512}{195} \sqrt{2}$$

$$(vii) \text{ Let } I = \int_{-3}^3 4x|x| dx$$

$$\text{Let } f(x) = 4x|x|$$

$$f(-x) = 4(-x)|-x| = -4x|(-1)x| = -4x|-1| \times |x| = -4x|x| = -f(x)$$

$\Rightarrow f(x)$ is an odd function.

$$\therefore I = \int_{-3}^3 4x|x| dx = 0$$

[Using property 4]

$$(viii) \int_1^5 f(x) dx,$$

$$\text{where } f(x) = \begin{cases} 1, & 1 \leq x < 2 \\ 2, & 2 \leq x < 3 \\ 3, & 3 \leq x < 4 \\ 4, & 4 \leq x < 5 \\ 5, & x \geq 5 \end{cases} \quad \dots (1)$$

$$\text{Now, } \int_1^5 f(x) dx = \int_1^2 f(x) dx + \int_2^3 f(x) dx + \int_3^4 f(x) dx + \int_4^5 f(x) dx$$

[Using property 2]

$$= \int_1^2 1 dx + \int_2^3 2 dx + \int_3^4 3 dx + \int_4^5 4 dx \quad [\text{Using (1)}]$$

$$= [x]_1^2 + [2x]_2^3 + [3x]_3^4 + [4x]_4^5$$

$$= (2 - 1) + (6 - 4) + (12 - 9) + (20 - 16)$$

$$= 1 + 2 + 3 + 4 = 10$$

14.8 REFERENCES/FURTHER READINGS

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